EXAM PACK

EEE455

VLSI I



Exam Pack

Session 2019–20 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination-2023*

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Question Paper Solutions 2019–20 pages 3--70 **68 pages**

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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2019–20

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PAST PAPER SOLUTIONS

EEE455

VLSI I

Question Paper Solutions

Session 2019–20

4th Year 1st Semester Final Examination-2023

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part - A · (Answer any two of the following questions)				
Q1	(a) Etching Chemistries and the Effect of H2 and O2			
	(b) RCA Clean	6	11 min	☐
	(c) Photolithography and Its Major Steps	8	14 min	
	(d) DIBL, Subthreshold Leakage and Temperature	5	9 min	
Q2	(a) Long-Channel I-V Equations of the MOSFET	8	14 min	
	(b) Delays of a NAND3 Driving Four NAND3 Gates	7	13 min	☐
	(c) Moore's Law and the Problems of Scaling	6	11 min	
	(d) Cascode Voltage Switch Logic	4	7 min	
Q3	(a) DC Characteristics of the CMOS Inverter	7	13 min	
	(b) Minimum Delay of the Path A to B in Figure 3.1	6	11 min	☐
	(c) Decoder: Best Number of Stages and Gate Sizes	7	13 min	
	(d) Input Ordering Delay Effect	5	9 min	
Part - B · (Answer any two of the following questions)				
Q4	(a) Scaling and How the Parameters Scale	5	9 min	
	(b) Glitches and Their Power	6	11 min	
	(c) Dynamic Power of a Synthesized Chip	4	7 min	☐
	(d) Crosstalk: Noise Cases and Cures	7	13 min	
	(e) Sheet Resistance and the Effect of Interconnect	3	5 min	
Q5	(a) Min-Delay Constraint and Hold Time	6	11 min	
	(b) Cycle Time and Hold Check of the Itanium 2 Bypass Path			☐
	(c) F = AB + CD with NAND Gates and with a Compound Gate	9	16 min	
	(d) Stack Effect and Leakage of Two OFF Transistors	6	11 min	
Q6	(a) Principle of Dynamic Logic	6	11 min	
	(b) Stick Diagram of f = (A + B)(C + D)	6	11 min	☐
	(c) SRAM Cell and Its Modes of Operation	6	11 min	
	(d) Pseudo-nMOS ROM for the Data of Figure 6.1	7	13 min	

Formulas follow Weste and Harris, CMOS VLSI Design (delay, logical effort, power, leakage, interconnect, timing, memories), Kang and Leblebici, CMOS Digital Integrated Circuits (long-channel MOSFET and inverter DC equations), and Plummer, Deal and Griffin, Silicon VLSI Technology (etching, RCA clean, photolithography).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination-2023
Course Code: EEE 455 Session: 2019-20
Course Title: VLSI I

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

(Answer any two of the following questions)

1. (a) Briefly discuss the chemistries involved in in [*sic*: should be "involved in"] etching. Also explain the effect of adding H_2 and O_2 on etching rate. 6
   (b) What is RCA clean? What are the steps? What contaminants can be removed in this process? 6
   (c) Define photolithography. briefly describe the major steps involved in photolithography. 8
   (d) Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities? 5

2. (a) Find the equations and I-V characteristics curve of MOSFET current using long-channel model. 8
   (b) Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance. 7
   (c) State Moore's law. Discuss the problems that arise due to MOSFET scaling. 6
   (d) Explain the principle of cascode voltage switch logic. 4

3. (a) Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter. 7
   (b) Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay. 6
   (c) Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size. 7
   (d) Explain input ordering delay effect. 5

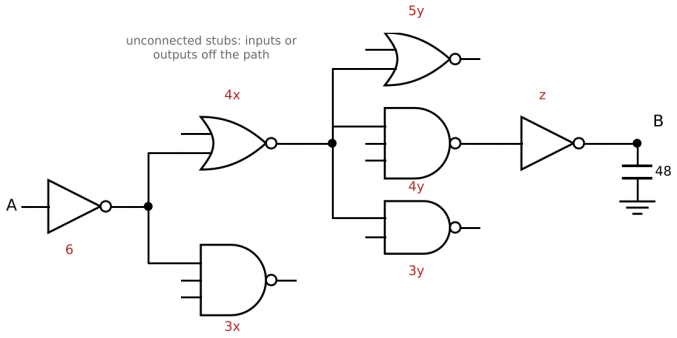


Figure 3.1 of the paper (redrawn)

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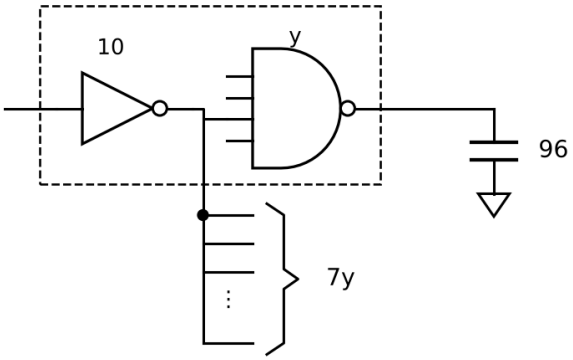


Figure 3.2 of the paper (redrawn from the hand-drawn sketch on the paper)

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Part - B

(Answer any two of the following questions)

4. (a) What is scaling? Explain how different parameters of transistors are scaled? 5

   (b) Define glitch? How does it increase power consumption? How it can be reduced? 6

   (c) You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at $V_{DD} = 0.9$ V. 4

   (d) Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced? 7

   (e) Define sheet resistance. How does interconnect affect the performance of a system? 3

5. (a) What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution? 6

   (b) The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time T_c at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures? 4

   (c) Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer. 9

   (d) Define stack effect. Find subthreshold leakage current for two series OFF transistor. 6

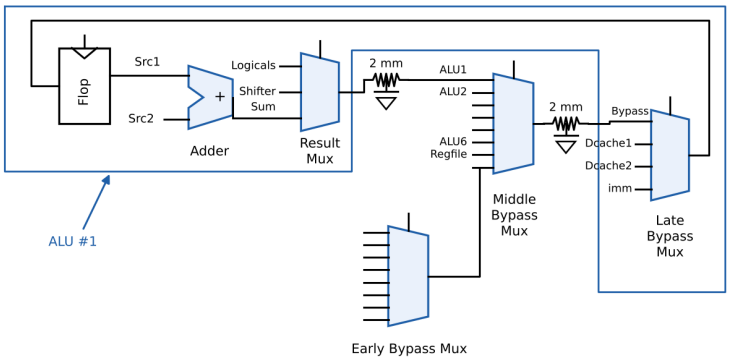


Figure 5.1 of the paper: the Itanium 2 ALU self-bypass path (redrawn)

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Elements	Propagation delay	Contamination delay
Adder	590 ps	100 ps
Result Mux	60 ps	35 ps
Early Bypass Mux	110 ps	95 ps
Middle Bypass Mux	80 ps	55 ps
Late Bypass Mux	70 ps	45 ps
2-mm Wire	100 ps	65 ps

Table 1: Combinational Logic Delays (Table 5.1 of the paper)

6. (a) Explain the principle of dynamic logic circuit. 6

   (b) Draw the stick diagram of the following Boolean function: $f = (A+B).(C+D)$ with their symbolic diagram and truth table. 6

   (c) Draw logic diagram of a SRAM cell and explain its different modes of operation. 6

   (d) Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation. 7

```
word0: 010101
word1: 011001
word2: 100101
word3: 101010
```

Figure 6.1 of the paper

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1. (a) Briefly discuss the chemistries involved in in [*sic*: should be "involved in"] etching. Also explain the effect of adding H_2 and O_2 on etching rate. Marks: 6
   (b) What is RCA clean? What are the steps? What contaminants can be removed in this process? Marks: 6
   (c) Define photolithography. briefly describe the major steps involved in photolithography. Marks: 8
   (d) Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities? Marks: 5

1.(a) ⚠ Note on the paper: The phrase "involved in in etching" has a doubled "in"; the question means the chemistries involved in etching.

1. Theory

1. What is etching? (Core Idea)

A chip is built layer by layer. After a layer (oxide, nitride, polysilicon, metal) is grown or deposited on the whole wafer, photolithography covers the parts to keep with a patterned **photoresist**. **Etching** then removes the layer wherever it is *not* covered. The resist is the stencil; etching is the cutting.

Two numbers describe an etch:

- **Etch rate:** how fast the layer is removed (nm/min).
- **Selectivity:** the etch rate of the layer we want to remove divided by the etch rate of the layer below it (or of the resist). A selectivity of 20 means that while 200 nm of oxide is removed, only 10 nm of the silicon below is lost.

A third property is the **shape** of the cut. An etch that removes material equally in all directions is **isotropic**; it eats sideways under the resist (**undercut**). An etch that removes material only downwards is **anisotropic**; it gives vertical walls.

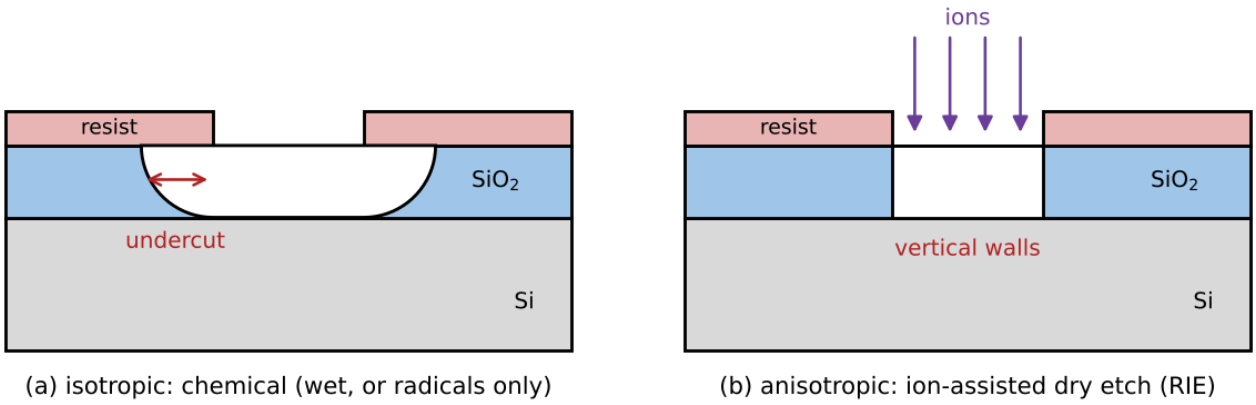


Figure 1: Etch profiles: (a) an isotropic etch undercuts the resist; (b) an anisotropic, ion-assisted dry etch gives vertical walls

Reading the figure. In (a) the opening under the resist has grown sideways by about the depth of the etch (red arrow), so a line drawn on the mask becomes narrower on the wafer. In (b) ions arrive straight down (purple arrows) and the walls stay vertical, so the pattern is copied exactly. Small modern features need (b).

2. Wet etching: liquid chemistry

The wafer is dipped in a liquid that reacts with the layer and turns it into something that dissolves. The reaction goes in all directions, so wet etching is **isotropic**, but it is very **selective** and cheap. The standard reactions:

Layer	Etchant	Reaction
SiO_2	HF, usually buffered with NH_4F (BHF)	$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$
Si (poly or crystal)	HNO_3 + HF (+ acetic acid)	HNO_3 oxidises Si to SiO_2; HF dissolves the oxide
Si (anisotropic, along crystal planes)	KOH	$Si + 2OH^- + 2H_2O \rightarrow SiO_2(OH)_2^{2-} + 2H_2$
Si_3N_4	hot H_3PO_4 (about 160 °C)	slow, selective to SiO_2
Al	H_3PO_4 + HNO_3 + acetic acid	HNO_3 oxidises Al; H_3PO_4 dissolves Al_2O_3

Table 2: Common wet etches

3. Dry (plasma) etching: gas chemistry

The wafer sits in a low-pressure chamber with a gas such as CF_4. A radio-frequency voltage turns the gas into a **plasma**: free electrons hit the molecules and break them into reactive pieces (**radicals** such as F and CF_3) and **ions**.

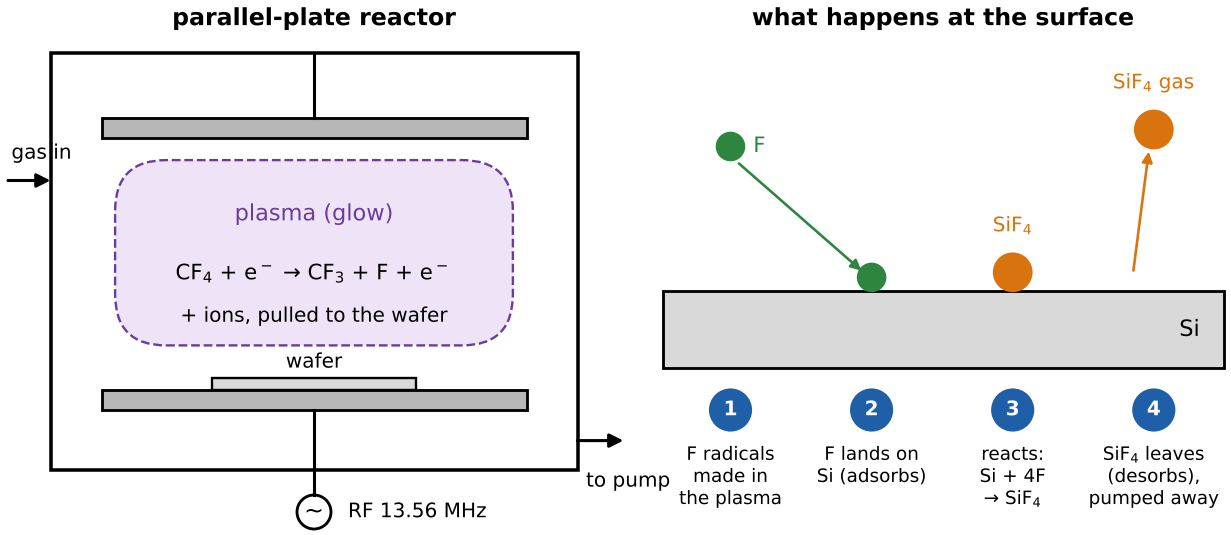


Figure 2: A parallel-plate plasma etcher and the four steps of the chemical etch at the silicon surface

Reading the figure. On the left, gas enters, the RF supply on the lower electrode keeps a plasma glowing above the wafer, and the products are pumped away. On the right, the chemical part of the etch takes four steps: (1) the plasma makes F radicals; (2) an F atom lands on the silicon; (3) four F atoms join one Si atom to form SiF_4; (4) SiF_4 is a gas, so it leaves the surface and is pumped out. An etch works only if step 4 works: the product must be **volatile**.

The main reactions in a CF_4 plasma:

$$CF_4 + e^- \rightarrow CF_3 + F + e^-$$

$$Si + 4F \rightarrow SiF_4 \uparrow \qquad SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$$

The F radicals alone etch isotropically. The ions, accelerated by the electric field above the wafer, hit the surface **vertically**; they break bonds and knock off protective films only on the bottom of the opening, so the etch goes mainly downwards. This **ion-assisted** etch (reactive ion etching, RIE) is what makes vertical walls. The choice of gas follows the volatile product:

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Layer	Typical gases	Volatile product
Si, poly-Si	CF_4/O_2, SF_6, Cl_2, HBr	SiF_4, $SiCl_4$, $SiBr_4$
SiO_2	CF_4/H_2, CHF_3, C_2F_6	SiF_4, CO, CO_2
Si_3N_4	CF_4/O_2, CHF_3	SiF_4, N_2
Al	Cl_2, BCl_3	$AlCl_3$
Photoresist	O_2	CO, CO_2, H_2O

Table 3: Common dry-etch gases

4. Why do O_2 and H_2 change the rate?

The etch rate of silicon depends mainly on **how many free F atoms** are in the plasma. In pure CF_4, many of the F atoms made by the plasma join a CF_3 fragment again ($F + CF_3 \rightarrow CF_4$), so few are left. Anything that removes the CF_x fragments raises the F count; anything that removes F lowers it.

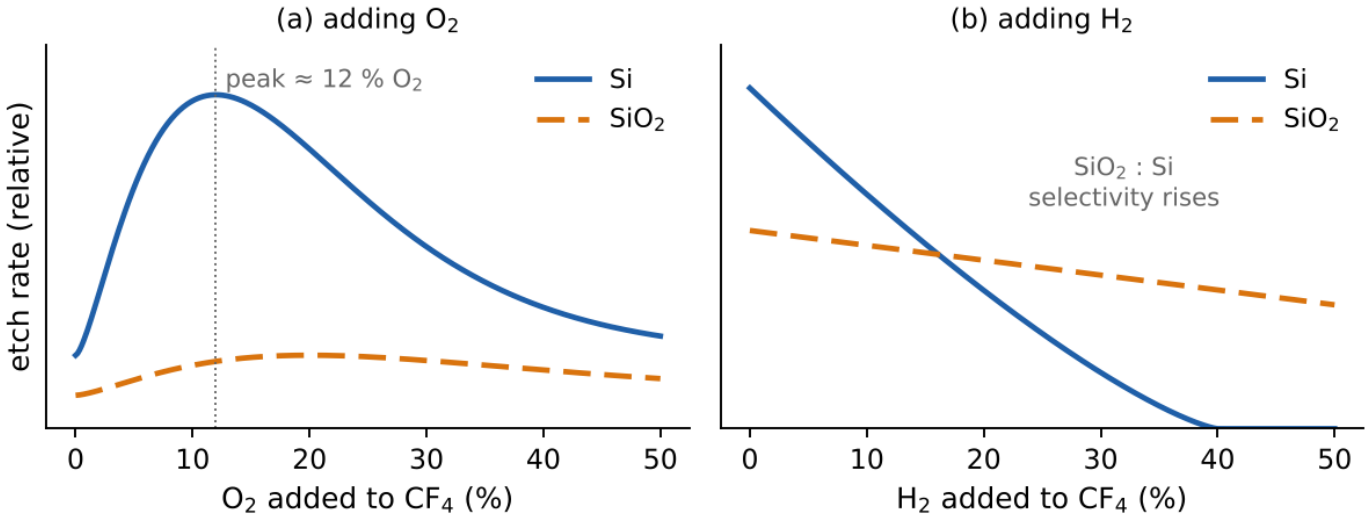


Figure 3: Etch rates of Si and SiO2 in a CF4 plasma: (a) with O2 added, (b) with H2 added (shapes after the classic measurements reported in Plummer; the numbers are relative)

Reading the figure. In (a), the silicon etch rate (blue) rises steeply as O_2 is added, peaks near 12 % O_2 and then falls; the oxide rate (orange) changes much less, with a small peak near 20 %. In (b), the silicon rate falls quickly as H_2 is added and is almost zero at about 40 % H_2, while the oxide rate stays nearly the same. So H_2 makes the etch **selective**: it removes oxide but stops on silicon.

2. Answer

Etching removes a layer where the photoresist does not protect it. It is done in a liquid (**wet**) or in a plasma (**dry**).

Wet chemistries (isotropic, highly selective):

- SiO_2 in hydrofluoric acid: $SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$ (buffered with NH_4F for a steady rate).
- Si in HNO_3 + HF: HNO_3 oxidises the silicon, HF dissolves the oxide; KOH etches Si along crystal planes.
- Si_3N_4 in hot H_3PO_4; Al in H_3PO_4 + HNO_3 + acetic acid.

Dry (plasma) chemistries: an RF discharge breaks the gas into radicals and ions, $CF_4 + e^- \rightarrow CF_3 + F + e^-$. The radicals react with the surface to form a **volatile** product that is pumped away:

$$Si + 4F \rightarrow SiF_4 \uparrow, \qquad SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$$

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Fluorine gases (CF_4, SF_6, CHF_3) etch Si, SiO_2 and Si_3N_4; chlorine gases (Cl_2, BCl_3) etch Si and Al ($AlCl_3$); O_2 removes photoresist (CO_2, H_2O). Ions striking vertically make the etch anisotropic (RIE).

Effect of adding O_2:

- O atoms react with the CF_x fragments ($O + CF_3 \rightarrow COF_2 + F$), so fewer F atoms recombine into CF_4 and more free F is left. The Si etch rate **rises**, to a maximum at about 12 % O_2 (the SiO_2 rate peaks later, near 20 %).
- Beyond the peak, the extra O_2 only **dilutes** the F, and O atoms adsorb on the silicon and oxidise it, blocking the F. The rate **falls**.

Effect of adding H_2:

- H atoms capture F: $H + F \rightarrow HF$. Free F drops, so the **Si etch rate falls** quickly (almost zero at about 40 % H_2).
- The plasma becomes rich in carbon, which forms a thin fluorocarbon polymer on the silicon and stops it. On SiO_2, the oxygen released from the oxide removes the carbon as CO and CO_2, so the **SiO_2 rate stays almost constant**.
- Result: the **SiO_2:Si selectivity rises** strongly (to 10 or more), which is used to etch contact holes in oxide that must stop on silicon.

Summary: O_2 in moderation increases the Si etch rate (more free F); H_2 decreases the Si etch rate (F tied up as HF, polymer on Si) and makes the oxide etch selective.

3. Keyword Explanation

1. Radical

↩ back

A neutral fragment of a molecule with an unpaired electron, such as F or CF_3. It is chemically very active, so it does most of the chemical etching.

2. Selectivity

↩ back

The ratio of the etch rates of two materials, for example SiO_2:Si = 20:1. High selectivity lets an etch stop on the layer below.

3. Anisotropic etch

↩ back

An etch that removes material in one direction only (downwards), giving vertical walls with no undercut.

4. Notes and Remarks

- The curves in the figure follow the shapes of the classic measurements by Mogab et al. (O_2 in CF_4) and Ephrath (H_2 in CF_4) as shown in Plummer, Deal and Griffin; the exact percentages depend on the reactor, so quote them as "about".
- The same idea explains why CHF_3 (which already contains H) is the usual gas for selective oxide etching, and why O_2 is added to CF_4 for fast silicon and nitride etching.

1.(b) What is RCA clean? What are the steps? What contaminants can be removed in this process? Marks: 6

1. Theory

1. Why must a wafer be cleaned? (Core Idea)

A transistor's gate oxide is only a few nanometres thick, and its junctions are only a fraction of a micrometre deep. A single dust particle can break a line; a trace of sodium in the oxide moves the threshold voltage; a metal atom such as iron in the silicon makes leakage current. So before every high-temperature step (oxidation, diffusion, deposition) the wafer is cleaned.

The dirt falls into four groups:

- **Particles:** dust, silicon chips, flakes from equipment.
- **Organic films:** photoresist residue, oils, skin grease.

- **Metal ions:** alkali (Na^+, K^+) and heavy metals (Fe, Cu, Ni, Au, Al, Zn).
- **Native oxide:** the thin oxide that grows by itself in air.

2. The idea of the RCA clean

In 1965 Werner Kern at RCA (Radio Corporation of America) found a sequence of hot **hydrogen peroxide** solutions, one basic and one acidic, that removes all four groups without attacking the silicon. H_2O_2 is a strong oxidiser: it burns organics and turns metals into ions. The base (NH_4OH) lifts particles; the acid (HCl) keeps metal ions dissolved so they do not settle back. It is still the standard clean.

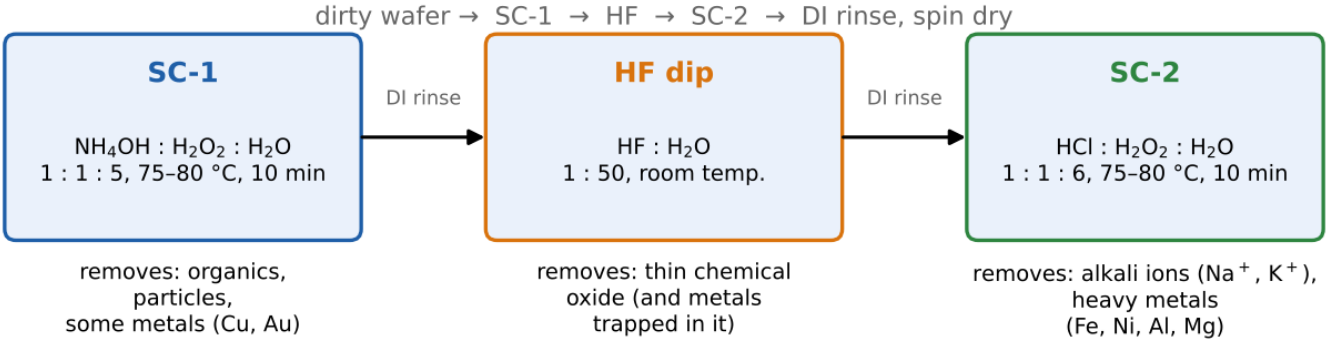


Figure 4: The RCA clean: SC-1, HF dip and SC-2, each followed by a rinse in deionised water

Reading the figure. The wafer passes from left to right. Each box gives the mixture (by volume) and its temperature; below it is what that step removes. Between steps the wafer is rinsed in ultrapure (deionised, DI) water so one chemical does not carry into the next.

2. Answer

RCA clean is the standard wet chemical cleaning of silicon wafers (developed at RCA by W. Kern, 1965), used before oxidation and other high-temperature steps. It uses hot hydrogen-peroxide solutions to remove particles, organics and metal contamination without etching the silicon.

Steps:

1. **SC-1 (Standard Clean 1):** $NH_4OH : H_2O_2 : H_2O = 1 : 1 : 5$, at 75–80 °C for about 10 min. H_2O_2 oxidises organic films; the alkaline solution slightly etches and lifts particles, and keeps them away from the surface by charging them. It also removes some metals (Cu, Au, Ag, Zn) by forming ammonia complexes. It leaves a thin chemical oxide.
2. **DI water rinse.**
3. **HF dip (optional):** $HF : H_2O = 1 : 50$ at room temperature for a few seconds, which removes the thin oxide from SC-1 together with the metals trapped in it, leaving bare silicon.
4. **DI water rinse.**
5. **SC-2 (Standard Clean 2):** $HCl : H_2O_2 : H_2O = 1 : 1 : 6$, at 75–80 °C for about 10 min. It dissolves alkali ions and heavy metals (Na, K, Fe, Al, Mg, Ni) as soluble chlorides and keeps them from depositing again.
6. **DI water rinse and spin dry.**

(Many fabs add a **piranha** clean, $H_2SO_4 : H_2O_2 = 4 : 1$ at about 120 °C, first, when there is a lot of photoresist or other organic residue.)

Contaminants removed:

Contaminant	Removed by
Organic films (resist residue, oils)	SC-1 (and piranha)
Particles	SC-1
Metals such as Cu, Au, Ag, Zn	SC-1 (ammonia complexes)
Thin native or chemical oxide	HF dip
Alkali ions (Na^+, K^+) and heavy metals (Fe, Ni, Al, Mg)	SC-2

Table 4: What each step of the RCA clean removes

3. Keyword Explanation

1. Deionised (DI) water

↩ back

Water with almost all ions removed (resistivity about 18 MΩ·cm), so rinsing adds no new contamination.

2. Mobile ion contamination

↩ back

Alkali ions such as Na^+ that drift in the gate oxide under an electric field and shift the threshold voltage; SC-2 removes them.

4. Notes and Remarks

- Modern fabs often use more dilute versions of SC-1 and SC-2 (for example 1 : 1 : 50) with megasonic agitation, which removes particles with less roughening of the silicon; the chemistry is the same.

1.(c) Define photolithography. briefly describe the major steps involved in photolithography. Marks: 8

1. Theory

1. What is photolithography? (Core Idea)

Every layer of a chip has a pattern: where the gates go, where the contacts go, where the wires run. The pattern is drawn on a **mask** (a glass plate with an opaque chromium pattern). **Photolithography** copies the mask's pattern onto the wafer using light, much like printing a photograph from a negative:

- The wafer is coated with **photoresist**, a light-sensitive polymer.
- Light shines through the mask. Where the mask is clear, the resist is exposed; where it is chromium, the resist stays dark.
- A developer dissolves either the exposed resist (**positive resist**) or the unexposed resist (**negative resist**).

The resist left on the wafer then protects the layer below during etching or ion implantation. A modern chip needs 30 or more such steps.

2. Positive and negative resist

- **Positive resist:** light breaks its molecules (or, in modern resists, makes them soluble), so the **exposed** areas dissolve. The resist pattern is the same as the mask's opaque pattern. It gives better resolution and is used for almost all modern work.
- **Negative resist:** light cross-links it, so the **exposed** areas stay and the dark areas dissolve. The pattern is the reverse of the mask. It tends to swell in the developer, which limits resolution.

3. What limits how small a line can be?

The smallest line a projection system can print is about

$$W_{min} = k_1 \frac{\lambda}{NA}$$

where

- λ is the wavelength of the light (436 nm g-line, 365 nm i-line, 248 nm KrF, 193 nm ArF, 13.5 nm EUV)
- NA is the numerical aperture of the lens (up to about 0.93 in air, 1.35 with water immersion)
- k_1 is a process factor (about 0.25 to 0.8)

This is why the industry moved to shorter and shorter wavelengths.

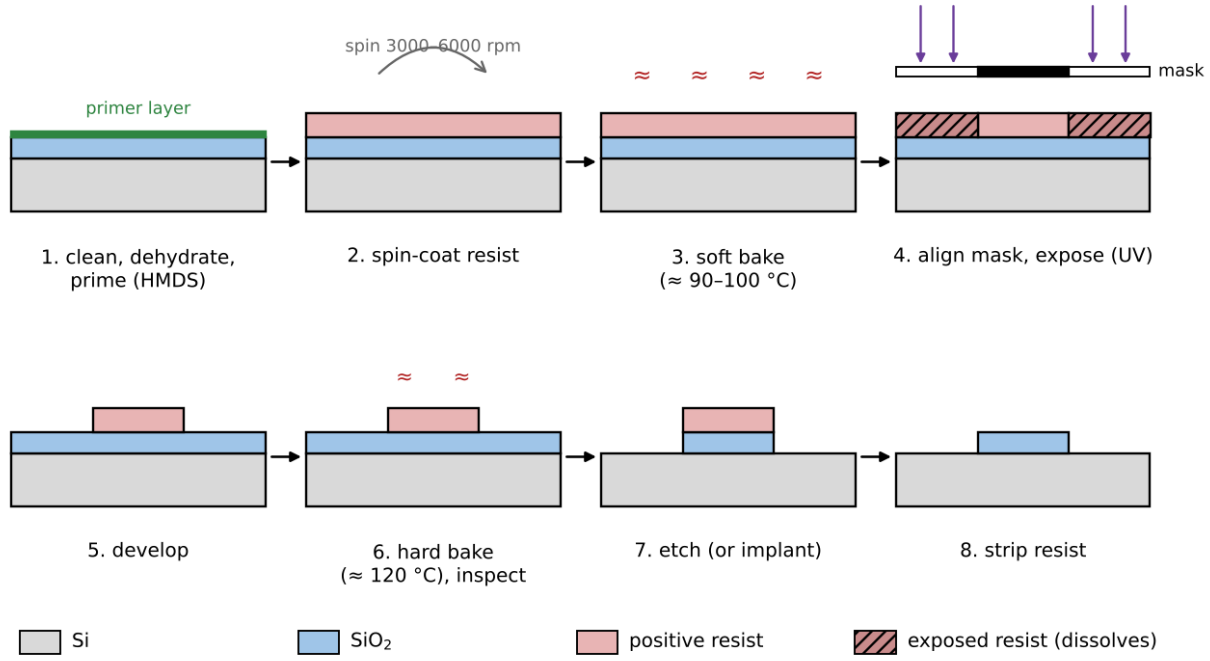


Figure 5: The steps of photolithography with a positive resist, followed by etching and resist strip

Reading the figure. Follow the panels from 1 to 8. After cleaning and priming (1), the resist is spun on (2) and dried (3). The mask lets UV light through only at its clear ends (4), so the hatched resist there is exposed. The developer removes the exposed resist (5), the remaining resist is hardened (6), the oxide not covered by resist is etched (7), and finally the resist is removed (8), leaving the oxide in the shape of the mask's dark area.

2. Answer

Photolithography is the process of transferring a geometric pattern from a mask to a light-sensitive layer (photoresist) on the wafer, by exposing the resist to light through the mask and then developing it. The patterned resist then masks the wafer for etching, ion implantation or lift-off.

Major steps:

1. **Wafer cleaning and preparation:** the wafer is cleaned (RCA clean, see 1(b)), baked at about 150–200 °C to drive off moisture (**dehydration bake**), and primed with HMDS (hexamethyldisilazane), which makes the surface water-repellent so the resist sticks.
2. **Resist coating (spin coating):** a few drops of liquid resist are put on the wafer, which is spun at about 3000–6000 rpm for 30–60 s, giving a uniform film about 0.5–2 µm thick. Thickness falls with spin speed.
3. **Soft bake (pre-bake):** about 90–100 °C for 1 min on a hot plate. It drives the solvent out of the resist, improves adhesion and makes the film stable for exposure.

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4. **Alignment and exposure:** the mask is aligned to the patterns already on the wafer (alignment marks), then the resist is exposed to UV light through the mask. Exposure tools: **contact** (mask touches the resist; good resolution but damages the mask), **proximity** (small gap; less damage, poorer resolution) and **projection** (a lens images the mask, usually reduced 4×, step-and-repeat; used today).
5. **Post-exposure bake (when needed):** about 110 °C; it smooths standing-wave ripples and, in chemically amplified DUV resists, completes the reaction started by the light.
6. **Development:** the wafer is sprayed or dipped in developer (an alkaline solution, such as TMAH, for positive resist). The exposed regions of a positive resist dissolve (the unexposed regions of a negative resist), leaving the pattern.
7. **Hard bake (post-bake):** about 120–150 °C. It removes the remaining solvent and hardens the resist so it survives etching or implantation.
8. **Inspection:** the pattern is checked under a microscope or by SEM for line width (critical dimension), alignment and defects; a bad layer is stripped and redone.

After lithography the wafer is **etched** (or implanted) through the resist openings, and the resist is then **stripped** in an O_2 plasma (ashing) or a solvent.

3. Keyword Explanation

1. Mask (reticle)

↩ back

A quartz plate with a chromium pattern for one layer; a reticle holds the pattern for one or a few chips, enlarged 4× for a reduction stepper.

2. HMDS

↩ back

Hexamethyldisilazane, an adhesion promoter that replaces the water-loving OH groups on the oxide surface so the resist does not lift off in the developer.

3. Critical dimension (CD)

↩ back

The width of the smallest line in a layer, measured after development to check the process.

4. Notes and Remarks

- The 2020–21 paper asked for the narrower task of patterning SiO_2 by photolithography; see 5(d) of the 2020–21 paper for extra reading.

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1.(d) Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities? Marks: 5

1. Theory

1. The ideal switch and the real one (Core Idea)

In the ideal (long-channel) picture, an nMOS transistor is a perfect switch: no current at all when $V_{GS} < V_t$, and a current set only by V_{GS} (not by V_{DS}) in saturation. Real transistors, especially short ones, leak when they should be off, and their threshold depends on the drain voltage. These are two of the main **non-idealities**.

2. The barrier picture

Think of the electrons in the source as water behind a dam. The p-type region under the gate is a hill of potential energy (a **barrier**) between the source and the drain. The gate voltage sets the height of the hill: a high V_{GS} lowers it and electrons flow over (the transistor is on); at $V_{GS} = 0$ the hill is high and almost no electrons get over.

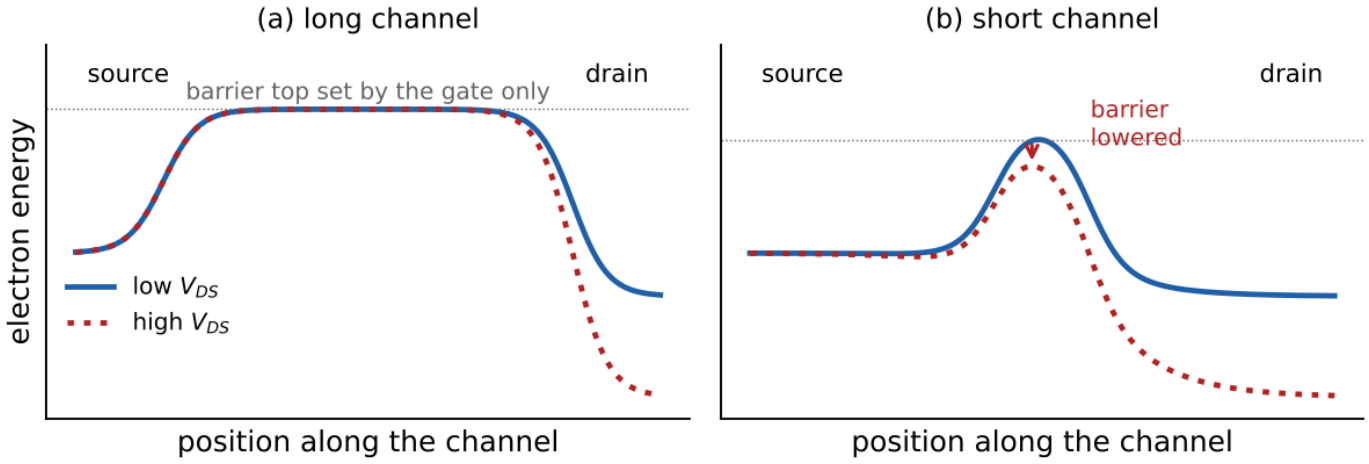


Figure 6: Electron energy along the channel from source to drain: (a) in a long channel the drain does not change the barrier top; (b) in a short channel a high drain voltage lowers it

Reading the figure. In (a) the barrier is wide; raising V_{DS} (red) only pulls down the drain end, and the top, which decides how many electrons can cross, stays where the gate put it. In (b) the channel is so short that the drain's field reaches the top of the barrier; at high V_{DS} the top is lower (red arrow), so more electrons cross. That is **drain-induced barrier lowering**.

3. Current below threshold

Electrons in the source have a spread of energies (a Boltzmann distribution). Even when the barrier is above most of them, a few have enough energy to cross. Their number falls by a factor e for every kT/q (25.9 mV at 300 K) that the barrier rises. So below threshold the current is **exponential** in V_{GS}:

$$I_{ds} = I_{ds0}\, e^{\frac{V_{gs}-V_t}{nv_T}} \left(1 - e^{-\frac{V_{ds}}{v_T}}\right)$$

where

- $v_T = kT/q$ is the thermal voltage
- n (about 1.3–1.5) accounts for the gate controlling the surface only partly, through the oxide and depletion capacitances
- I_{ds0} is the current at threshold

The **subthreshold slope** $S = nv_T \ln 10$ is the change in V_{GS} for a tenfold change in current: at least 60 mV/decade at room temperature, typically 80–100 mV/decade.

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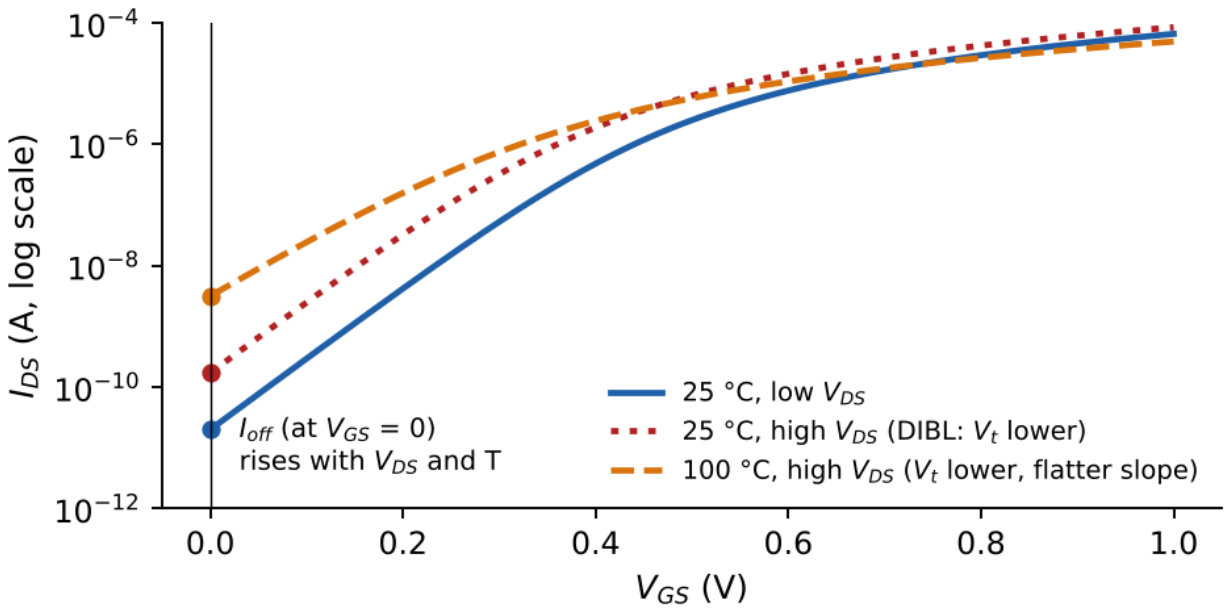


Figure 7: Drain current on a log scale against the gate voltage: the straight-line part is the subthreshold region; a high VDS (DIBL) and a high temperature both raise the off current

Reading the figure. On a log scale the subthreshold current is a straight line, rising by ten for every S volts. The dots at $V_{GS} = 0$ are the **off currents**. A higher V_{DS} (red) shifts the curve left because DIBL lowers V_t, so I_{off} rises about tenfold here. At 100 °C (orange), V_t is lower again and the line is flatter (larger S), so I_{off} rises further, while the on current at high V_{GS} actually falls a little (lower mobility).

2. Answer

(i) Drain-induced barrier lowering (DIBL). In a short-channel MOSFET the drain voltage, as well as the gate, lowers the potential barrier between source and drain, because the drain's depletion region and field reach the source end of the channel. A lower barrier means a lower threshold voltage:

$$V_t = V_{t0} - \eta\, V_{ds}$$

where η is the DIBL coefficient (typically about 0.1, i.e. 100 mV per volt of V_{ds}). Effects: the threshold falls as V_{ds} rises, the saturation current keeps rising with V_{ds} (low output resistance, poor gain), and the off current at $V_{ds} = V_{DD}$ is much larger than at small V_{ds}.

(ii) Subthreshold leakage. For $V_{gs} < V_t$ the transistor is not completely off: electrons with enough thermal energy diffuse over the barrier, and the current falls only exponentially below threshold:

$$I_{ds} = I_{ds0}\, e^{\frac{V_{gs} - V_t + \eta V_{ds}}{n v_T}} \left(1 - e^{-\frac{V_{ds}}{v_T}}\right)$$

with subthreshold slope $S = n v_T \ln 10 \approx$ 60–100 mV/decade. The off current ($V_{gs} = 0$) is $I_{off} \propto 10^{-(V_t - \eta V_{DD})/S}$: a lower V_t (from scaling or DIBL) raises leakage exponentially. It is the main static power in modern chips and it slowly drains dynamic nodes and memory cells.

Effect of temperature:

- **Subthreshold leakage rises strongly with temperature.** $v_T = kT/q$ rises, so S grows (flatter turn-off), and V_t falls by about 1–2 mV/K. Both raise I_{off} exponentially; leakage can grow ten to a hundred times between room temperature and 100 °C.
- **DIBL:** the coefficient η is set mainly by the geometry (channel length, oxide thickness, junction depth), so it changes little with temperature. But the threshold it lowers is already lower at high temperature, so the combined off current at high V_{DS} and high temperature is the worst case.
- The on current falls slightly at high temperature (mobility falls), so a hot chip is both slower and leakier.

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3. Keyword Explanation

1. Threshold voltage V_t

↩ back

The gate voltage at which a conducting channel (inversion layer) forms under the gate.

2. Subthreshold slope S

↩ back

The gate-voltage change needed to change the subthreshold current by a factor of ten; smaller is better.

4. Notes and Remarks

- **Why 60 mV/decade at least:** with $n = 1$, $S = v_T \ln 10 = 25.9 \times 2.303 = 59.6$ mV at 300 K. No ordinary MOSFET can turn off faster, which is why V_t (and so V_{DD}) cannot be scaled down freely; see 2(c).
- The stack of two OFF transistors reduces subthreshold leakage; see 5(d).

5. Final Concise Answer

(a) Etching removes unprotected material. Wet: $SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$; Si in HNO_3 + HF or KOH; Si_3N_4 in hot H_3PO_4 (isotropic, selective). Dry: the plasma makes radicals and ions ($CF_4 + e^- \rightarrow CF_3 + F + e^-$), which form volatile products ($Si + 4F \rightarrow SiF_4$; $Al + Cl \rightarrow AlCl_3$; resist + $O \rightarrow CO_2$); ions make it anisotropic. Adding O_2 removes CF_x so more F is free: the Si rate rises to a peak near 12 % O_2, then falls by dilution and surface oxidation. Adding H_2 ties F up as HF and deposits polymer on Si: the Si rate falls (near zero at about 40 % H_2) while SiO_2 etches on, so SiO_2:Si selectivity rises.

(b) RCA clean: the standard wet clean of silicon wafers. SC-1 (NH_4OH:H_2O_2:H_2O = 1:1:5, 75–80 °C) removes organics, particles and some metals; HF dip (1:50) removes the oxide; SC-2 (HCl:H_2O_2:H_2O = 1:1:6, 75–80 °C) removes alkali ions and heavy metals; DI rinses between, then dry.

(c) Photolithography transfers a mask pattern into photoresist with light. Steps: clean, dehydration bake and HMDS prime; spin coat; soft bake; align and expose (contact, proximity or projection); post-exposure bake; develop; hard bake; inspect; then etch or implant and strip the resist.

(d) DIBL: the drain field lowers the source barrier in a short channel, $V_t = V_{t0} - \eta V_{ds}$. Subthreshold leakage: below V_t the current falls only exponentially, $I \propto e^{(V_{gs}-V_t)/(nv_T)}$, $S \approx$ 60–100 mV/decade. Higher temperature raises v_T and lowers V_t, so leakage rises exponentially; η changes little, but its effect adds to the lower V_t.

2. (a) Find the equations and I-V characteristics curve of MOSFET current using long-channel model. Marks: 8
   (b) Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance. Marks: 7
   (c) State Moore's law. Discuss the problems that arise due to MOSFET scaling. Marks: 6
   (d) Explain the principle of cascode voltage switch logic. Marks: 4

2.(a) Find the equations and I-V characteristics curve of MOSFET current using long-channel model. Marks: 8

1. Theory

1. The MOSFET as a voltage-controlled switch (Core Idea)

An nMOS transistor has two n^+ regions, the **source** and the **drain**, in a p-type **body**. Above the body between them sits the **gate**, separated by a thin oxide. With no gate voltage, the source and drain are two back-to-back pn junctions, so no current flows. A positive gate voltage pulls electrons to the surface under the oxide. Once V_{GS} exceeds the **threshold voltage** V_t, there are enough of them to form a thin n-type **channel** joining source and drain, and a current flows when the drain is made positive.

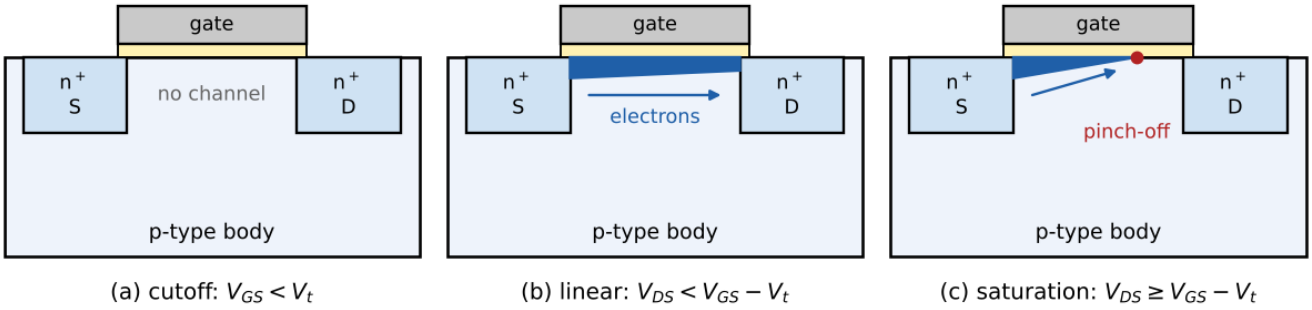


Figure 8: An nMOS transistor in its three regions: (a) no channel below threshold; (b) a channel from source to drain, thinner at the drain end; (c) the channel pinched off at the drain end

Reading the figure. In (a) the gate voltage is below V_t: no channel, no current. In (b) the channel (blue) runs all the way; it is thinner at the drain end, because the gate-to-channel voltage there is only $V_{GS} - V_{DS}$. In (c) V_{DS} is so large that the gate-to-channel voltage at the drain end falls to V_t: the channel disappears there (**pinch-off**), and electrons are swept across the short depleted gap by the strong field. Raising V_{DS} further changes almost nothing, so the current **saturates**.

2. The two ingredients: charge and speed

The current is (charge per unit length of channel) × (speed of that charge). The **long-channel model** (Shockley) assumes:

- The channel charge at any point is set by the local gate-to-channel voltage, like a parallel-plate capacitor: $Q(x) = C_{ox}W\,[V_{GS} - V(x) - V_t]$, where $V(x)$ is the channel voltage at distance x from the source and $C_{ox} = \varepsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area.
- The electrons drift with speed proportional to the field: $v = \mu_n E = \mu_n\, dV/dx$ (the channel is long, so the field is small and the speed never saturates).

3. The derivation

The current is the same at every point of the channel:

$$I_{DS} = Q(x)\, v(x) = \mu_n C_{ox} W \left[V_{GS} - V(x) - V_t\right] \frac{dV}{dx}$$

Multiply by dx and integrate from the source ($x = 0$, $V = 0$) to the drain ($x = L$, $V = V_{DS}$):

$$I_{DS} \int_0^L dx = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_t - V)\, dV$$

$$I_{DS}\, L = \mu_n C_{ox} W \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2}\right]$$

which gives the linear-region equation. It holds while the channel reaches the drain, i.e. while $V_{GS} - V_{DS} > V_t$. At $V_{DS} = V_{DSAT} = V_{GS} - V_t$ the channel pinches off and the current stops growing; putting $V_{DS} = V_{GS} - V_t$ in the equation gives the saturation current.

2. Answer

Assumptions (long-channel, Shockley model): gradual channel (the channel charge is set by the local gate-to-channel voltage), constant mobility (no velocity saturation), no channel-length modulation; $\beta = \mu_n C_{ox} \dfrac{W}{L}$, $C_{ox} = \dfrac{\varepsilon_{ox}}{t_{ox}}$.

Set-up. Take an nMOS transistor with x measured along the channel, from the source ($x = 0$) to the drain ($x = L$). Let $V(x)$ be the channel voltage at x, so $V(0) = 0$ and $V(L) = V_{DS}$.

Step 1: Charge in the channel

The gate, the oxide and the channel form a capacitor with C_{ox} per unit area. At point x the voltage between gate and channel is $V_{GS} - V(x)$. Only the part above the threshold V_t creates free electrons, so the charge per unit area is $C_{ox}(V_{GS} - V_t - V)$. For a channel of width W, the charge per unit length is

$$Q = C_{ox} W (V_{GS} - V_t - V)$$

Step 2: Speed of the electrons

The field along the channel is dV/dx, and with constant mobility the electrons move at

$$v = \mu_n \frac{dV}{dx}$$

Step 3: Current

Current is charge per unit length times speed:

$$I_{DS} = Qv = \mu_n C_{ox} W (V_{GS} - V_t - V) \frac{dV}{dx}$$

Step 4: Integrate along the channel

The same current flows through every point. Move dx to the left and integrate both sides from source to drain:

$$I_{DS} \int_0^L dx = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_t - V)\, dV$$

The left side is $I_{DS}L$. On the right, $V_{GS} - V_t$ is a constant:

$$I_{DS} L = \mu_n C_{ox} W \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2}\right]$$

Step 5: The linear region

Divide by L and use $\beta = \mu_n C_{ox} W/L$:

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$$I_{DS} = \beta \left(V_{GS} - V_t - \frac{V_{DS}}{2} \right) V_{DS}$$

This holds while a channel exists all the way to the drain.

Step 6: The saturation region

The channel charge at the drain, $C_{ox}W(V_{GS} - V_t - V_{DS})$, reaches zero at $V_{DS} = V_{DSAT} = V_{GS} - V_t$. The channel then pinches off there. Put $V_{DS} = V_{GS} - V_t$ into Step 5:

$$I_{DS} = \beta \left(V_{GS} - V_t - \frac{V_{GS} - V_t}{2} \right) (V_{GS} - V_t) = \frac{\beta}{2}(V_{GS} - V_t)^2$$

A larger V_{DS} only drops across the short pinched-off part next to the drain, so the voltage along the channel, and with it the current, stays the same.

Step 7: Cutoff

For $V_{GS} < V_t$ there is no channel, so $Q = 0$ and $I_{DS} = 0$.

Steps to remember

Channel charge $C_{ox}W(V_{GS} - V_t - V) \to$ speed $\mu_n dV/dx \to I = Qv \to$ integrate source to drain $\to$ linear-region law $\to$ put $V_{DS} = V_{GS} - V_t$ for saturation $\to$ no channel below V_t.

The equations (nMOS):

$$I_{DS} = \begin{cases} 0, & V_{GS} < V_t \quad \text{(cutoff)} \\ \beta \left(V_{GS} - V_t - \dfrac{V_{DS}}{2} \right) V_{DS}, & V_{DS} < V_{GS} - V_t \quad \text{(linear)} \\ \dfrac{\beta}{2} (V_{GS} - V_t)^2, & V_{DS} \geq V_{GS} - V_t \quad \text{(saturation)} \end{cases}$$

- **Cutoff:** no channel, no current.
- **Linear (triode):** for small V_{DS}, $I_{DS} \approx \beta(V_{GS} - V_t)V_{DS}$: the transistor is a resistor $R = 1/[\beta(V_{GS} - V_t)]$ controlled by the gate.
- **Saturation:** at $V_{DS} = V_{DSAT} = V_{GS} - V_t$ the channel pinches off at the drain; the current stays at $\frac{\beta}{2}(V_{GS} - V_t)^2$, independent of V_{DS} (square law). With channel-length modulation, $I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2(1 + \lambda V_{DS})$.
- **pMOS:** the same equations with all voltages and the current reversed in sign and μ_p ($\approx \mu_n/2$ to $\mu_n/3$) in place of μ_n.

I-V characteristics (example: $\beta = 200\ \mu\text{A/V}^2$, $V_t = 0.7$ V):

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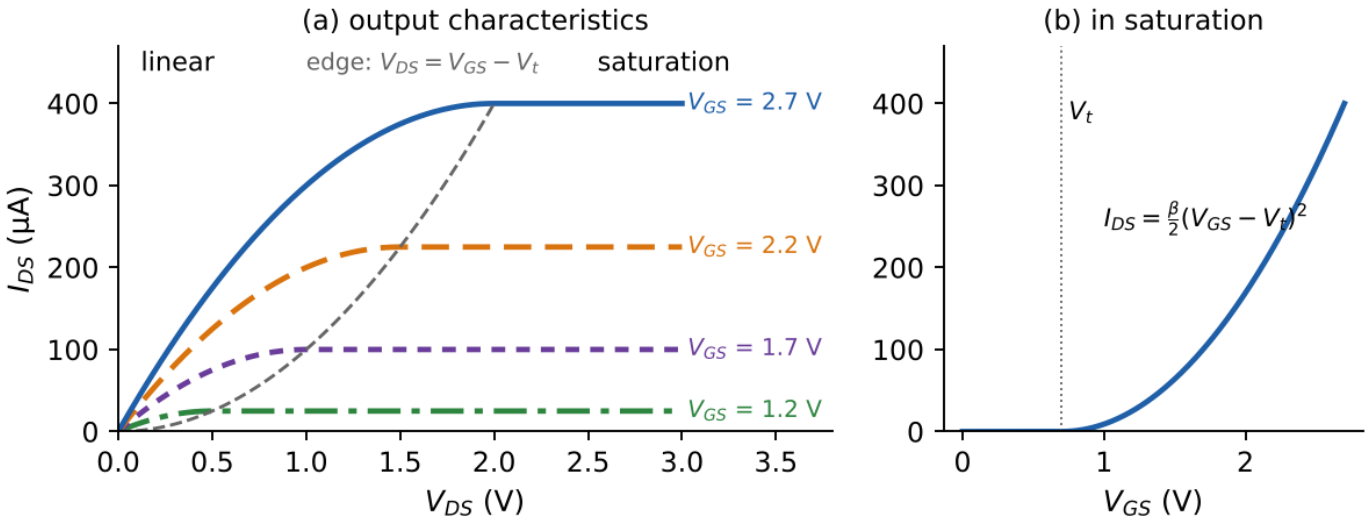


Figure 9: Long-channel I-V curves: (a) IDS against VDS for four gate voltages, with the linear/saturation edge dashed; (b) the saturation current against VGS (square law)

In (a) each curve rises as a parabola in the linear region and becomes flat at $V_{DS} = V_{GS} - V_t$ (dashed edge). Equal steps of V_{GS} give growing steps of current (25, 100, 225, 400 µA for $V_{GS} - V_t = 0.5$, 1, 1.5, 2 V), because the saturation current goes as the square of $V_{GS} - V_t$, as plotted in (b).

3. Keyword Explanation

1. Pinch-off

↩ back

The point at which the gate-to-channel voltage at the drain end falls to V_t, so the inversion layer vanishes there; beyond it the current saturates.

2. Transconductance parameter β

↩ back

$\beta = \mu C_{ox} W/L$ (A/V^2); a wider or shorter transistor carries more current.

4. Notes and Remarks

- Kang and Leblebici write the same equations as $I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$ and $I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_T)^2$; they are identical to the ones above.
- In short-channel transistors the carriers reach their saturation velocity and the current grows nearly **linearly** with $V_{GS} - V_t$ instead of as its square (the alpha-power model, $I \propto (V_{GS} - V_t)^\alpha$, $\alpha \approx 1.3$).

2.(b) Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance. Marks: 7

1. Theory

1. A transistor as a resistor and a capacitor (Core Idea)

For timing, Weste and Harris replace each ON transistor by a resistor and each transistor terminal by a capacitor:

- A **unit nMOS** (width 1) has resistance R. A transistor k times as wide has R/k. A pMOS has about twice the resistance of an nMOS of the same width (holes are slower), so a unit-resistance pMOS needs width 2.
- A transistor of width k has **gate capacitance** kC and **diffusion capacitance** of about kC on its source and on its drain.

The delay of a node is then estimated with the **Elmore delay**: the sum, over every capacitor, of the capacitor times the resistance between it and the source of the current:

$$t_{pd} = \sum_i R_{i,\text{to source}}\, C_i$$

2. Sizing a NAND3 for equal pull-up and pull-down resistance

A NAND3 has three nMOS in **series** (output low only when all inputs are 1) and three pMOS in **parallel**. For the pull-down to equal a unit inverter's R, each series nMOS must be 3 wide ($R/3$ each, $3 \times R/3 = R$). Each pMOS alone must pull up with R, so each is 2 wide. One input therefore drives $3 + 2 = 5$ units of gate capacitance: $C_{in} = 5C$.

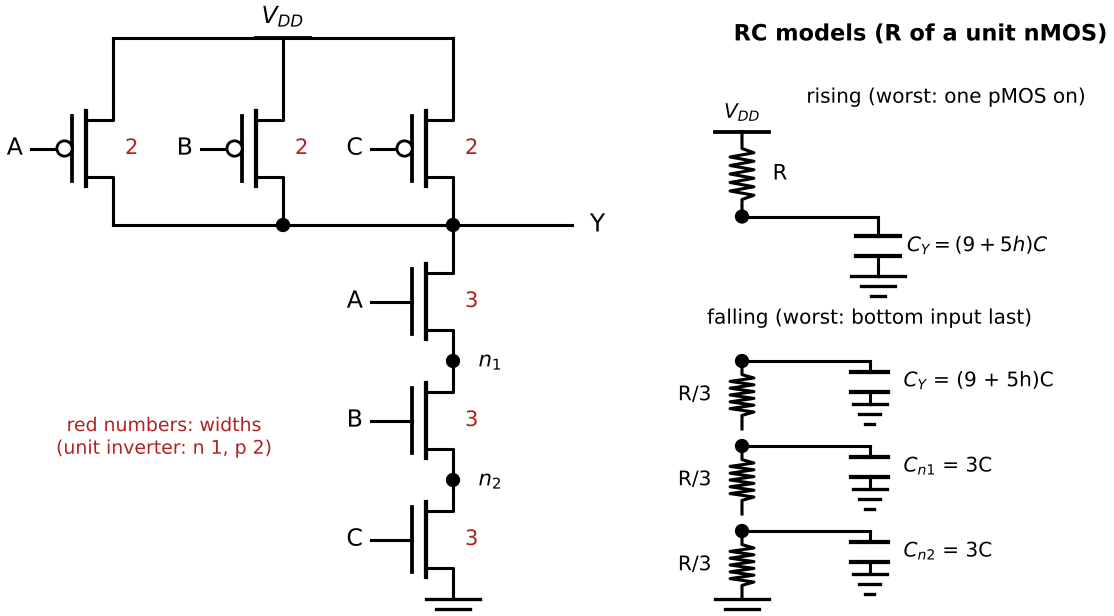


Figure 10: Unit NAND3 (widths in red) and its RC models for the worst rising and falling transitions

Reading the figure. On the left is the NAND3 with its widths. On the right are the RC models. Rising: in the worst case only one pMOS turns on, so the output charges through R. Falling: the three nMOS (each $R/3$) are in series; the internal nodes n_1 and n_2 also hold charge that must flow through the transistors below them.

3. Propagation and contamination delay

- **Propagation delay t_{pd}:** the **longest** time from an input change to the output settling (the worst case). It sets the speed of the circuit.
- **Contamination delay t_{cd}:** the **shortest** time from an input change to the output starting to change (the best case). It matters for hold-time checks; see 5(a).

2. Answer

Given: NAND3 sized for equal pull-up and pull-down resistance (nMOS width 3, pMOS width 2), driving $h = 4$ identical NAND3 gates. Unit transistor resistance R, capacitance C per unit width; the two series nMOS share their middle diffusion, so each internal node has $3C$.

Capacitances:

- Load: $h \times C_{in} = 4 \times 5C = 20C$
- Output diffusion: three pMOS drains $(3 \times 2C)$ + top nMOS drain $(3C) = 9C$
- $C_Y = 9C + 20C = 29C = (9 + 5h)C$; $C_{n1} = C_{n2} = 3C$

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Step 1: Rising propagation delay (worst case)

Only one pMOS is ON (one input falls, the other two stay 1); it charges C_Y through R. The internal nodes, if charged, do not slow it.

$$t_{pdr} = R\,(9 + 5h)C = R \times 29C$$

$$\boxed{t_{pdr} = 29\,RC}$$

Step 2: Falling propagation delay (worst case)

The bottom input (C) rises last, so n_1, n_2 and Y were all charged high and must discharge through the stack. Elmore delay:

$$t_{pdf} = \frac{R}{3}(3C) + \frac{2R}{3}(3C) + \frac{3R}{3}(29C) = RC + 2RC + 29RC$$

$$\boxed{t_{pdf} = 32\,RC = (12 + 5h)RC}$$

Step 3: Rising contamination delay (best case)

All three inputs fall together, so all three pMOS are ON in parallel ($R/3$):

$$t_{cdr} = \frac{R}{3}(29C) = \left(3 + \frac{5h}{3}\right) RC$$

$$\boxed{t_{cdr} = 9.67\,RC}$$

Step 4: Falling contamination delay (best case)

The top input (A) rises last: n_1 and n_2 were already discharged through the lower transistors, so only C_Y discharges through the three series nMOS ($3 \times R/3 = R$):

$$t_{cdf} = R(29C) = (9 + 5h)RC$$

$$\boxed{t_{cdf} = 29\,RC}$$

Final Answer: with $\tau = 3RC$:

Delay	In RC	In τ
t_{pdr}	29 RC	9.67 τ
t_{pdf}	32 RC	10.67 τ
t_{cdr}	9.67 RC	3.22 τ
t_{cdf}	29 RC	9.67 τ

Table 5: Delays of the NAND3 driving four NAND3 gates

3. Keyword Explanation

1. Elmore delay

↩ back

A first-order estimate of the delay of an RC tree: $\sum_i R_i C_i$, where R_i is the resistance shared between the path to node i and the path to the output.

2. τ

↩ back

$\tau = 3RC$, the delay of a unit inverter driving an identical inverter with no parasitic capacitance (Weste and Harris).

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4. Notes and Remarks

- **Check with logical effort:** NAND3 has $g = 5/3$ and $p = 3$, so $d = gh + p = \frac{5}{3} \times 4 + 3 = 9.67\tau$, the same as t_{pdr} above. The falling delay is a little larger because of the internal-node charge.
- If the internal nodes are counted as $6C$ each (two separate, uncontacted diffusions instead of one shared region), $t_{pdf} = 2RC + 4RC + 29RC = 35RC$; the other three delays do not change.

2.(c) State Moore's law. Discuss the problems that arise due to MOSFET scaling. Marks: 6

1. Theory

1. Moore's law (Core Idea)

In 1965 Gordon Moore (later a founder of Intel) noticed that the number of transistors on the most economical chip had doubled every year, and predicted it would go on doing so. In 1975 he revised the rate to a doubling about **every two years**. The industry then used the prediction as a target, and it held for about five decades.

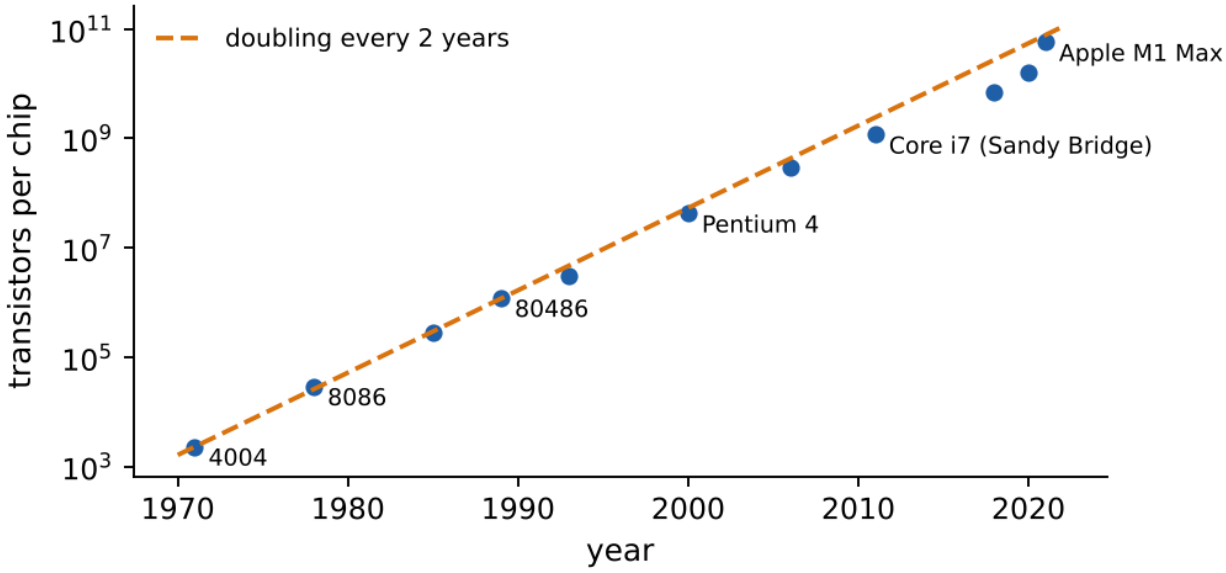


Figure 11: Transistors per chip against year, on a log scale, with the line for doubling every two years

Reading the figure. On a log scale, steady doubling is a straight line (dashed). The chips follow it from the Intel 4004 (2300 transistors, 1971) to the Apple M1 Max (57 billion, 2021): about 25 doublings in 50 years.

2. How the doubling was achieved

The main way is **scaling**: making every transistor smaller by a factor S (about 1.4 per generation, so the area halves). Under ideal constant-field scaling, the transistor gets faster and uses less power at the same time; see 4(a). But some quantities refuse to scale, and that causes the problems below.

2. Answer

Moore's law: the number of transistors on an integrated circuit doubles about every two years (Moore, 1965: every year; revised in 1975 to every two years), at roughly constant cost per chip.

Problems that arise from MOSFET scaling:

1. **Short-channel effects:** as L shrinks, the drain's field reaches the source. The threshold falls with shorter L (V_t **roll-off**) and with higher V_{DS} (**DIBL**), and in the extreme the source and drain depletion regions touch (**punch-through**), so the gate loses control.
2. **Subthreshold leakage:** V_{DD} must fall to keep the field constant, so V_t must fall too. Because the subthreshold slope cannot go below about 60 mV/decade, every 100 mV less of V_t raises the

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off current about ten to twenty times. Static power becomes a large part of the total.

3. **Gate leakage:** an oxide only a few atoms thick (about 1–2 nm) lets electrons **tunnel** through it; this needed high-k dielectrics (HfO_2) and metal gates.
4. **Velocity saturation and mobility degradation:** high fields make carriers reach a maximum speed, so the current no longer improves as much as the square law promises.
5. **Interconnect:** wires get thinner, so their resistance per length grows, while local wires do not get shorter; **wire RC delay**, crosstalk and electromigration grow and global wires become slower than gates.
6. **Power density and heat:** since about 2005 V_{DD} has stopped scaling (end of Dennard scaling), so power per area rises; clock frequencies stopped rising and chips went multi-core.
7. **Variability and reliability:** a channel contains only tens of dopant atoms, so random dopant fluctuation and line-edge roughness make V_t vary from transistor to transistor; hot carriers, oxide breakdown and electromigration shorten the lifetime.
8. **Lithography and cost:** printing features far smaller than the light's wavelength needs immersion, multiple patterning and EUV, and the cost of a fab and of a mask set rises sharply.

3. Keyword Explanation

1. Dennard scaling

↩ back

Constant-field scaling (Dennard, 1974): all dimensions and voltages shrink by S, so power density stays constant while speed rises.

2. Punch-through

↩ back

Merging of the source and drain depletion regions in a very short channel, giving a large current that the gate cannot turn off.

4. Notes and Remarks

- The industry answered these problems with new structures rather than plain shrinking: strained silicon (2003), high-k/metal gate (2007), FinFETs (2011) and gate-all-around nanosheets (2022 onwards), which wrap the gate around the channel to restore its control.

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2.(d) Explain the principle of cascode voltage switch logic. Marks: 4

1. Theory

1. Why another logic family? (Core Idea)

A static CMOS gate needs both an nMOS network and a pMOS network for every function, and the pMOS network is slow (holes are slow) and has a large input capacitance. **Cascode voltage switch logic (CVSL)**, also called differential cascode voltage switch logic (DCVSL), replaces the pMOS network with just **two cross-coupled pMOS transistors**, and computes both the output and its complement with two nMOS networks.

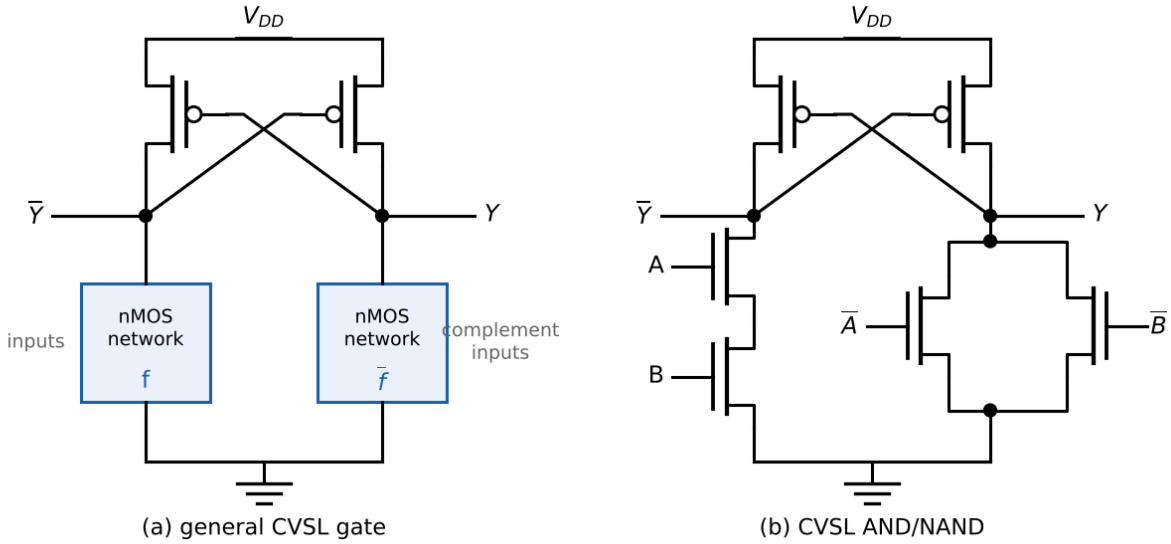


Figure 12: (a) The general CVSL gate: two complementary nMOS networks under two cross-coupled pMOS; (b) a CVSL AND/NAND gate

Reading the figure. In (a), the left network implements f and pulls the left node ($\overline{Y}$) low when $f = 1$; the right network implements $\overline{f}$ and pulls the right node (Y) low when $f = 0$. Each pMOS gate is wired to the opposite node. In (b), the left network is A in series with B (conducts when $AB = 1$), and the right network is $\overline{A}$ in parallel with $\overline{B}$ (conducts when $AB = 0$), so $Y = AB$ and $\overline{Y} = \overline{AB}$.

2. Answer

Principle: a CVSL gate is a **dual-rail** gate: it takes the true and complement inputs and produces both Y and $\overline{Y}$. It has two complementary nMOS pull-down networks, f and $\overline{f}$, and two cross-coupled pMOS pull-ups.

Operation (for $f = 1$):

1. The f network conducts and starts pulling $\overline{Y}$ low; the $\overline{f}$ network is off.
2. As $\overline{Y}$ falls, it turns ON the pMOS whose gate it drives, and that pMOS pulls Y up to V_{DD}.
3. Y high turns OFF the pMOS above $\overline{Y}$, so $\overline{Y}$ falls fully to 0 with no fight. The latch-like feedback makes the outputs full-swing: $Y = 1$, $\overline{Y} = 0$.

For $f = 0$ the roles swap. Exactly one network conducts at any time, so there is **no static current** path once switching is over.

Example

(Figure b): AND/NAND with A–B in series on the $\overline{Y}$ side and $\overline{A} \parallel \overline{B}$ on the Y side.

Features: both polarities are available without inverters; the inputs see only nMOS (small input capacitance, fast); the pMOS pull-ups are small. **Drawbacks:** about twice the nMOS count, dual-rail

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wiring, and during switching the two halves briefly fight (a ratioed contest), which costs some delay and short-circuit power.

3. Notes and Remarks

- CVSL is the static ancestor of dual-rail domino logic, used where both polarities are needed, for example in XOR-heavy adders.

4. Final Concise Answer

(a) Long-channel (Shockley) model with $\beta = \mu C_{ox} W/L$: from $I = Qv$, $Q = C_{ox}W(V_{GS} - V_t - V)$ and $v = \mu\, dV/dx$: $I_{DS} = 0$ (cutoff, $V_{GS} < V_t$); $I_{DS} = \beta(V_{GS} - V_t - V_{DS}/2)V_{DS}$ (linear, $V_{DS} < V_{GS} - V_t$); $I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2$ (saturation). Curves: parabolas that flatten at $V_{DS} = V_{GS} - V_t$; the saturation current grows as $(V_{GS} - V_t)^2$.

(b) Unit NAND3 (n 3, p 2), $C_{in} = 5C$, $C_Y = 9C + 20C = 29C$, internal nodes $3C$: $t_{pdr} = 29RC$, $t_{pdf} = 32RC$, $t_{cdr} = 9.67RC$, $t_{cdf} = 29RC$ (9.67, 10.67, 3.22, 9.67 τ).

(c) Moore's law: transistor count per chip doubles about every two years. Scaling problems: short-channel effects (DIBL, V_t roll-off, punch-through), subthreshold and gate-tunnelling leakage, velocity saturation, interconnect RC delay, power density (end of Dennard scaling), variability and reliability, lithography cost.

(d) CVSL: dual-rail logic with complementary nMOS networks f and $\overline{f}$ under two cross-coupled pMOS; the conducting network pulls one output low, which turns on the pMOS of the other output and pulls it high; full swing, no static current, both polarities, small input capacitance.

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3. (a) Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter. Marks: 7

   (b) Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay. Marks: 6

   (c) Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size. Marks: 7

   (d) Explain input ordering delay effect. Marks: 5

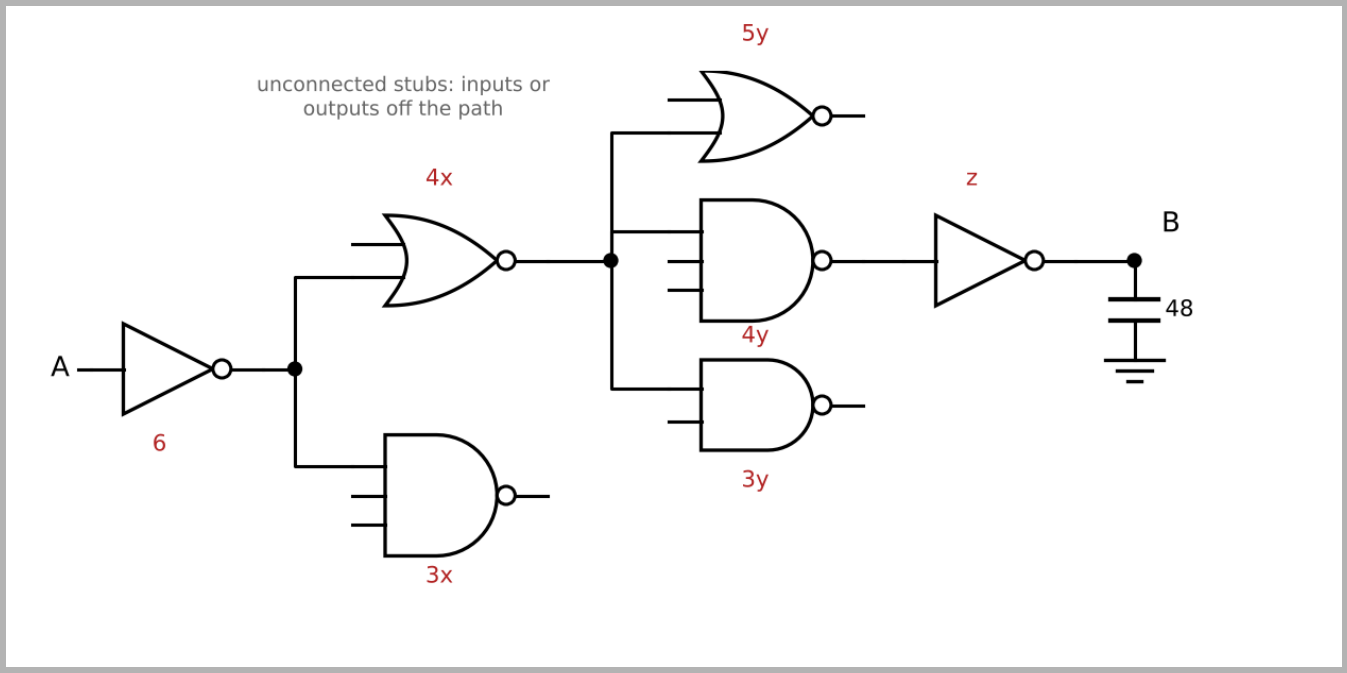


Figure 13: Figure 3.1 of the paper (redrawn)

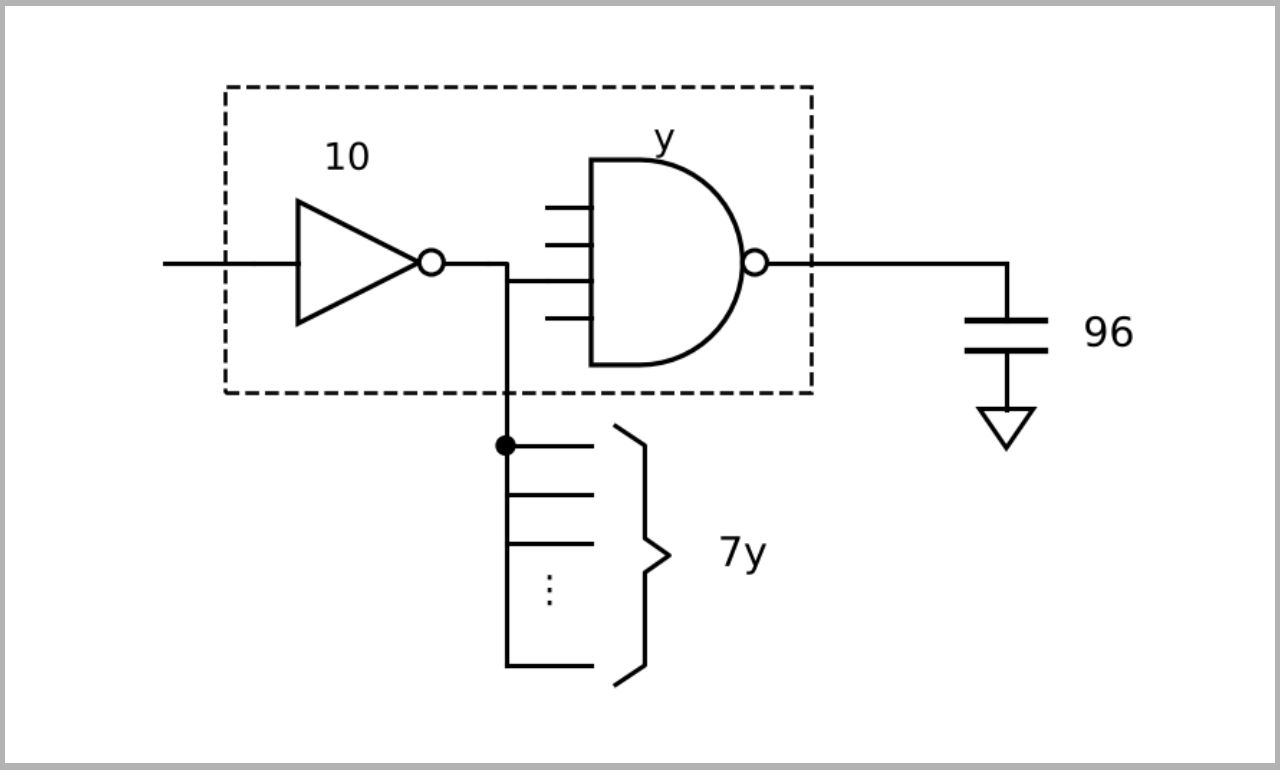


Figure 14: Figure 3.2 of the paper (redrawn from the hand-drawn sketch on the paper)

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3.(a) Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter. Marks: 7

1. Theory

1. The CMOS inverter (Core Idea)

A CMOS inverter is one pMOS from V_{DD} to the output and one nMOS from the output to ground, with both gates tied to the input. When the input is low, the pMOS is ON and the nMOS OFF, so the output is pulled to V_{DD}; when the input is high, the reverse. Its **DC characteristic**, or **voltage transfer characteristic (VTC)**, is the graph of V_{out} against V_{in} when the input changes slowly.

2. The key fact: the two currents are equal

With no load current (the next gate draws no DC current), whatever current flows down through the pMOS must flow on through the nMOS:

$$I_{Dn} = -I_{Dp} = |I_{Dp}|$$

So for each V_{in} the output settles where the nMOS current curve and the pMOS current curve cross. We draw both, as functions of V_{out}, using the long-channel equations of 2(a):

- nMOS: $V_{GSn} = V_{in}$, $V_{DSn} = V_{out}$.
- pMOS: $V_{SGp} = V_{DD} - V_{in}$, $V_{SDp} = V_{DD} - V_{out}$; its curve is the nMOS shape mirrored about the middle of the V_{out} axis.

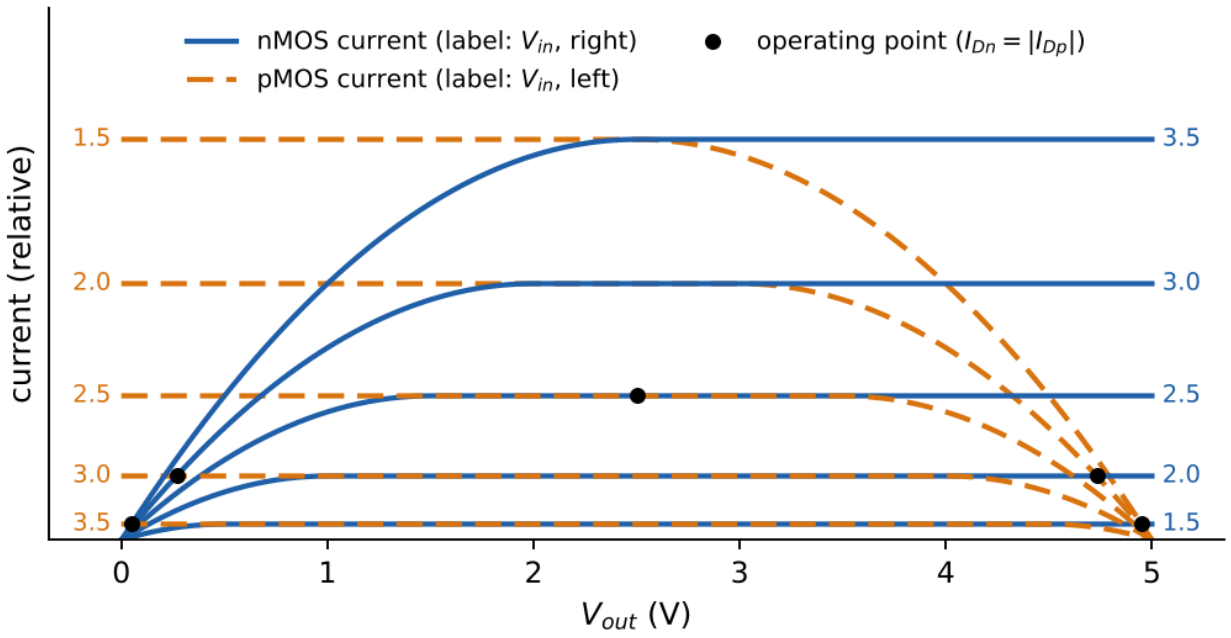


Figure 15: nMOS (blue) and pMOS (orange) currents against Vout for Vin from 1.5 V to 3.5 V (VDD = 5 V, Vt = 1 V, equal beta); each black dot is where the two currents are equal

Reading the figure. For $V_{in} = 1.5$ V the nMOS curve is low (it is barely on) and the pMOS curve is high, so they cross near $V_{out} = 5$ V: output high. As V_{in} rises, the nMOS curves rise and the pMOS curves fall, so the crossing moves left. At $V_{in} = 2.5$ V both are equal and flat (both saturated) and the crossing is at the middle. At 3.5 V the crossing is near 0 V: output low. Plotting each dot's V_{out} against its V_{in} gives the VTC.

2. Answer

Method: for each input, set $I_{Dn}(V_{in}, V_{out}) = |I_{Dp}|(V_{in}, V_{out})$ and solve for V_{out} (load-line intersection of the I-V curves above). With $V_{tn} = |V_{tp}| = V_t$:

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Region	V_{in}	nMOS	pMOS	V_{out}
A	$0 \le V_{in} < V_{tn}$	cutoff	linear	V_{DD}
B	$V_{tn} \le V_{in} < V_M$	saturation	linear	slightly below V_{DD}
C	$V_{in} = V_M$	saturation	saturation	falls steeply
D	$V_M < V_{in} \le V_{DD} - \lvert V_{tp}\rvert$	linear	saturation	slightly above 0
E	$V_{in} > V_{DD} - \lvert V_{tp}\rvert$	linear	cutoff	0

Table 6: Operating regions of the CMOS inverter

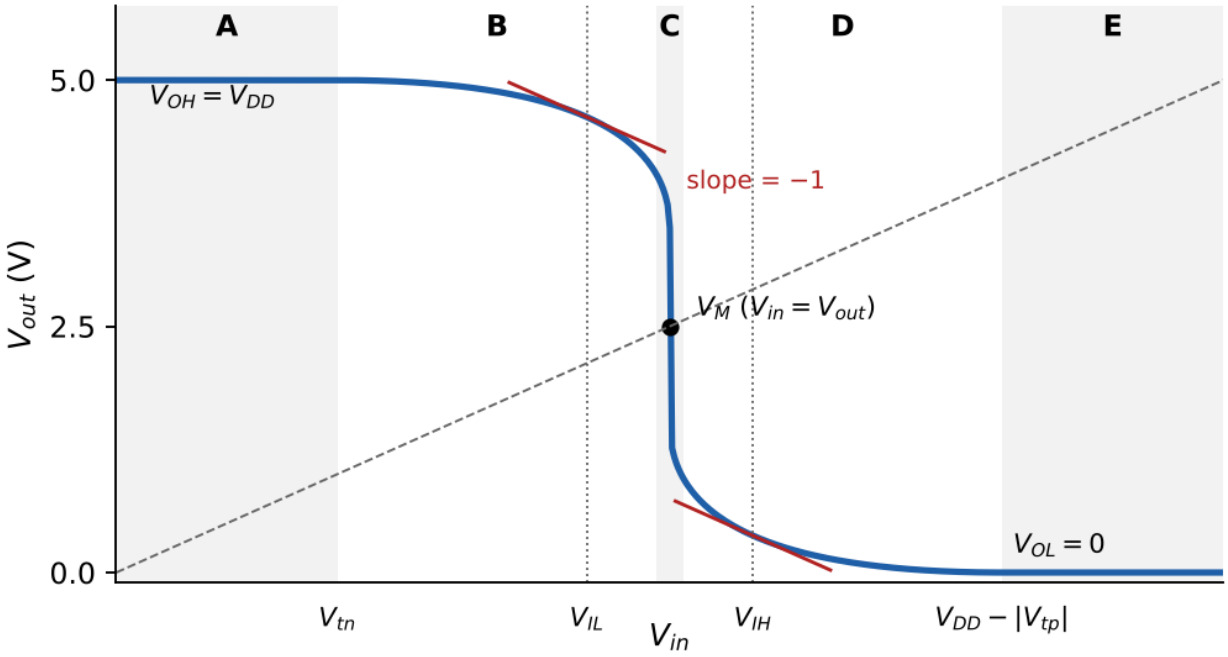


Figure 16: VTC of the CMOS inverter with its five regions (VDD = 5 V, Vt = 1 V, beta_n = beta_p)

Output levels: in A one transistor is off and no current flows, so $V_{OH} = V_{DD}$; in E, $V_{OL} = 0$. The inverter is **ratioless**: full swing whatever the sizes.

Derivation of the curve (symmetric inverter: $\beta_n = \beta_p = \beta$, $V_{tn} = |V_{tp}| = V_t$; long-channel square-law currents).

Step 1 (Basic): The two current laws

For a MOSFET with gate overdrive $V_{GT} = V_{GS} - V_t$: in saturation ($V_{DS} \ge V_{GT}$), $I_D = \frac{\beta}{2} V_{GT}^2$; in the linear region ($V_{DS} < V_{GT}$), $I_D = \beta \left(V_{GT} V_{DS} - \frac{V_{DS}^2}{2} \right)$. For the nMOS, $V_{GS} = V_{in}$ and $V_{DS} = V_{out}$; for the pMOS, $|V_{GS}| = V_{DD} - V_{in}$ and $|V_{DS}| = V_{DD} - V_{out}$.

Step 2: Region B (nMOS saturated, pMOS linear)

Set the currents equal, and write $x = V_{DD} - V_{out}$ for the pMOS drop:

$$\frac{\beta}{2}(V_{in} - V_t)^2 = \beta \left[(V_{DD} - V_{in} - V_t)\, x - \frac{x^2}{2} \right] \;\Rightarrow\; x^2 - 2(V_{DD} - V_{in} - V_t)\, x + (V_{in} - V_t)^2 = 0$$

The pMOS is linear, so x is the smaller root:

$$x = (V_{DD} - V_{in} - V_t) - \sqrt{(V_{DD} - V_{in} - V_t)^2 - (V_{in} - V_t)^2}$$

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The difference of squares factors as $[(V_{DD}-V_{in}-V_t)-(V_{in}-V_t)][(V_{DD}-V_{in}-V_t)+(V_{in}-V_t)] = (V_{DD}-2V_{in})(V_{DD}-2V_t)$. With $V_{out}=V_{DD}-x$:

$$V_{out} = V_{in} + V_t + \sqrt{(V_{DD}-2V_{in})(V_{DD}-2V_t)} \quad \text{(B)}$$

Step 3: Region D (nMOS linear, pMOS saturated)

$$\beta\left[(V_{in}-V_t)V_{out} - \frac{V_{out}^2}{2}\right] = \frac{\beta}{2}(V_{DD}-V_{in}-V_t)^2 \;\Rightarrow\; V_{out}^2 - 2(V_{in}-V_t)V_{out} + (V_{DD}-V_{in}-V_t)^2 = 0$$

The nMOS is linear, so V_{out} is the smaller root; the same factoring gives $(V_{in}-V_t)^2-(V_{DD}-V_{in}-V_t)^2 = (2V_{in}-V_{DD})(V_{DD}-2V_t)$:

$$V_{out} = V_{in} - V_t - \sqrt{(2V_{in}-V_{DD})(V_{DD}-2V_t)} \quad \text{(D)}$$

Step 4: Switching threshold V_M (region C)

At $V_{in}=V_{out}=V_M$ both transistors are saturated:

$$\frac{\beta_n}{2}(V_M-V_{tn})^2 = \frac{\beta_p}{2}(V_{DD}-V_M-|V_{tp}|)^2$$

Take the square root of both sides and let $r=\sqrt{\beta_p/\beta_n}$:

$$V_M - V_{tn} = r\,(V_{DD}-V_M-|V_{tp}|) \;\Rightarrow\; V_M = \frac{V_{tn}+r\,(V_{DD}-|V_{tp}|)}{1+r}$$

For $\beta_n=\beta_p$ ($W_p\approx 2W_n$) and $V_{tn}=|V_{tp}|$: $V_M = V_{DD}/2$.

Step 5: Noise-margin points (slope -1)

V_{IL} lies in region B. Differentiate (B): $\dfrac{dV_{out}}{dV_{in}} = 1 - \dfrac{V_{DD}-2V_t}{\sqrt{(V_{DD}-2V_{in})(V_{DD}-2V_t)}} = 1-\sqrt{\dfrac{V_{DD}-2V_t}{V_{DD}-2V_{in}}}$.
Setting it to -1 gives $V_{DD}-2V_t = 4(V_{DD}-2V_{in})$, so

$$V_{IL} = \frac{3V_{DD}+2V_t}{8}$$

V_{IH} lies in region D. Differentiate (D): $\dfrac{dV_{out}}{dV_{in}} = 1-\sqrt{\dfrac{V_{DD}-2V_t}{2V_{in}-V_{DD}}} = -1$ gives $V_{DD}-2V_t = 4(2V_{in}-V_{DD})$, so

$$V_{IH} = \frac{5V_{DD}-2V_t}{8}$$

(Kang and Leblebici obtain the same points.)

Steps to remember

Equal currents with the right region for each transistor → B: quadratic in $V_{DD}-V_{out}$, smaller root → D: quadratic in V_{out}, smaller root → C: both saturated, V_M → slope -1 in B and D gives V_{IL}, V_{IH}.

$$NM_L = V_{IL} - V_{OL}, \qquad NM_H = V_{OH} - V_{IH}$$

Example

($V_{DD}=5$ V, $V_t=1$ V, $\beta_n=\beta_p$): $V_M=2.5$ V, $V_{IL}=2.125$ V, $V_{IH}=2.875$ V, $NM_L=NM_H=2.125$ V.

Summary: the VTC is high (V_{DD}) for low inputs, falls very steeply around V_M where both transistors are saturated, and is 0 for high inputs; the transition point is set by the ratio β_p/β_n.

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3. Keyword Explanation

1. Switching threshold V_M

↩ back

The input voltage at which $V_{out} = V_{in}$; it sits in the middle of the steep part of the VTC.

2. Noise margin

↩ back

How much noise an input can carry and still be read correctly: $NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$.

4. Notes and Remarks

- With ideal long-channel transistors the VTC is exactly vertical at V_M (infinite gain) because both currents are flat in saturation; with channel-length modulation the slope is large but finite.
- A larger β_p/β_n moves V_M up (a "high-skewed" inverter); the 2020–21 paper asked about this, see 2(b) of the 2020–21 paper for extra reading.

3.(b) Find minimum delay for path $A \to B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay. Marks: 6

1. Theory

1. The delay of one gate (Core Idea)

Logical effort (Weste and Harris) writes the delay of a gate, in units of τ (the delay of an ideal inverter driving an identical one), as

$$d = f + p = g\,h + p$$

where

- g is the **logical effort**: how much more input capacitance the gate needs than an inverter to deliver the same current
- $h = C_{out}/C_{in}$ is the **electrical effort** (fanout)
- $f = gh$ is the **stage effort**
- p is the **parasitic delay** (the delay with no load)

For gates sized to the unit inverter's drive (nMOS 1, pMOS 2):

Gate	g	p
Inverter	1	1
NAND-n	$(n+2)/3$	n
NOR-n	$(2n+1)/3$	n

Table 7: Logical effort and parasitic delay

2. A path with branches

When a gate drives other gates besides the next one on our path, its current is split. The **branching effort** of that stage is

$$b = \frac{C_{on\text{-}path} + C_{off\text{-}path}}{C_{on\text{-}path}}$$

For the whole path:

$$G = \prod g_i, \quad B = \prod b_i, \quad H = \frac{C_{out}}{C_{in}}, \quad F = GBH, \quad P = \sum p_i$$

The delay is smallest when every stage has the same effort $\hat{f} = F^{1/N}$, and then

$$\hat{D} = NF^{1/N} + P$$

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The sizes follow from the load backwards: the input capacitance of each gate is $C_{in,i} = g_i\, C_{out,i}/\hat{f}$, where $C_{out,i}$ is everything that gate drives.

2. Answer

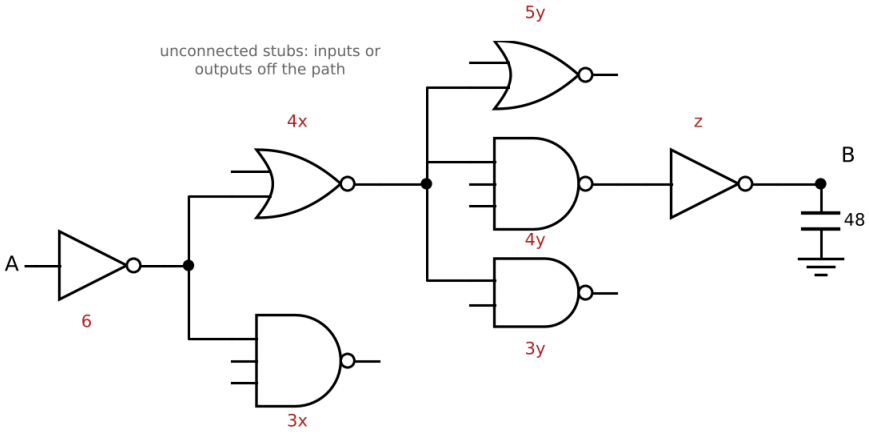


Figure 17: Repeated from the question: Figure 3.1 of the paper (redrawn)

Given (reading of Figure 3.1): input inverter with $C_{in} = 6$; it drives a NOR2 (4x, on the path) and a NAND3 (3x, off it). The 4x NOR2 drives a NOR2 (5y), a NAND3 (4y, on the path) and a NAND2 (3y). The 4y NAND3 drives the inverter z, which drives the load 48 at B. Capacitances in units of a unit transistor's gate.

Step 1: Path efforts

$$G = g_{inv}\, g_{NOR2}\, g_{NAND3}\, g_{inv} = 1 \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot 1 = \frac{25}{9} = 2.78$$

$$B = \frac{4x + 3x}{4x} \cdot \frac{5y + 4y + 3y}{4y} = \frac{7}{4} \cdot 3 = 5.25, \qquad H = \frac{48}{6} = 8$$

$$F = GBH = \frac{25}{9} \times 5.25 \times 8 = 116.67$$

Step 2: Stage effort and minimum delay

$N = 4$ stages:

$$\hat{f} = F^{1/4} = 116.67^{1/4} = 3.29$$

$$P = 1 + 2 + 3 + 1 = 7$$

$$\hat{D} = 4(3.29) + 7 = 20.15\,\tau$$

$$\boxed{\hat{D} = 20.1\,\tau \approx 4.0 \text{ FO4}}$$

Step 3: Gate sizes, from the load backwards

$$z = \frac{g_{inv} \times 48}{\hat{f}} = \frac{48}{3.29} = 14.6$$

$$4y = \frac{g_{NAND3} \times z}{\hat{f}} = \frac{(5/3)(14.6)}{3.29} = 7.41 \;\Rightarrow\; y = 1.85$$

The 4x gate drives $5y + 4y + 3y = 12y = 22.2$:

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$$4x = \frac{(5/3)(22.22)}{3.29} = 11.27 \;\Rightarrow\; x = 2.82$$

Check: the inverter drives $7x = 19.72$, so its input is $19.72/3.29 = 6.0$ ✔, the given 6.

$$\boxed{x = 2.82, \quad y = 1.85, \quad z = 14.6}$$

Gate	Input capacitance	nMOS width	pMOS width
Input inverter	6	2	4
NOR2 (4x)	11.3	2.3	9.0
NAND3 (4y)	7.4	4.4	3.0
Inverter (z)	14.6	4.9	9.7

Table 8: Sizes for the minimum delay (a NOR2 input is nMOS 1 + pMOS 4 of 5 units; a NAND3 input is nMOS 3 + pMOS 2 of 5; an inverter is 1 + 2 of 3)

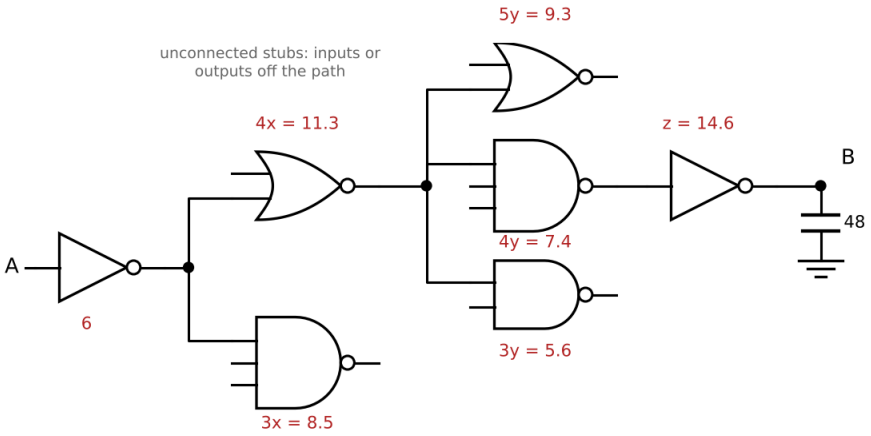


Figure 18: The path with the chosen sizes

Final Answer: $F = 116.7$, $\hat{f} = 3.29$, minimum delay $\hat{D} = 20.1\tau$ (4.0 FO4), with $x = 2.82$ (4x = 11.3, 3x = 8.5), $y = 1.85$ (5y = 9.3, 4y = 7.4, 3y = 5.6) and $z = 14.6$.

3. Keyword Explanation

1. Branching effort

↩ back

The factor by which a gate's output current is split between the path and gates off it; $b = (C_{on} + C_{off})/C_{on}$.

2. FO4 delay

↩ back

The delay of an inverter driving four copies of itself, $4 + 1 = 5\tau$.

4. Notes and Remarks

- **Every stage has effort 3.29:** inverter $1 \times 19.72/6 = 3.29$; NOR2 (4x) $\frac{5}{3} \times 22.22/11.27 = 3.29$; NAND3 (4y) $\frac{5}{3} \times 14.6/7.41 = 3.29$; inverter $48/14.6 = 3.29$.
- $\hat{f} = 3.29$ is close to the best value of about 4, so 4 stages suits this path.
- The gate types are read from the printed symbols: 4x and 5y have the curved back of a NOR; 3x and 4y have three input leads; 3y has two. The types of 3x, 5y and 3y do not matter for the answer, because only their input capacitances enter the branching effort.

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- If 4x and 4y were NAND2 gates: $G = 16/9$, $F = 74.7$, $\hat{f} = 2.94$, $P = 6$, $\hat{D} = 17.8\tau$.

3.(c) Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size. Marks: 7

1. Theory

1. How many stages? (Core Idea)

For a given path effort F, adding inverters changes the delay $\hat{D} = NF^{1/N} + P$: more stages lower each stage's effort but add parasitic delay. The best number of stages is where each stage's effort is about 4 (for $p_{inv} = 1$, the exact optimum is $\rho \approx 3.59$):

$$\hat{N} = \log_4 F$$

Round to a whole number and check the neighbours, since the delay is flat near the optimum.

2. The decoder path

A decoder sets one word line high. Each address input (after its inverter) goes to many NAND gates, so the path has a large **branching effort**, and the word line drives a whole row of register cells, so it has a large **electrical effort**. Inverters (buffers) between the NAND and the word line spread that effort over more stages.

2. Answer

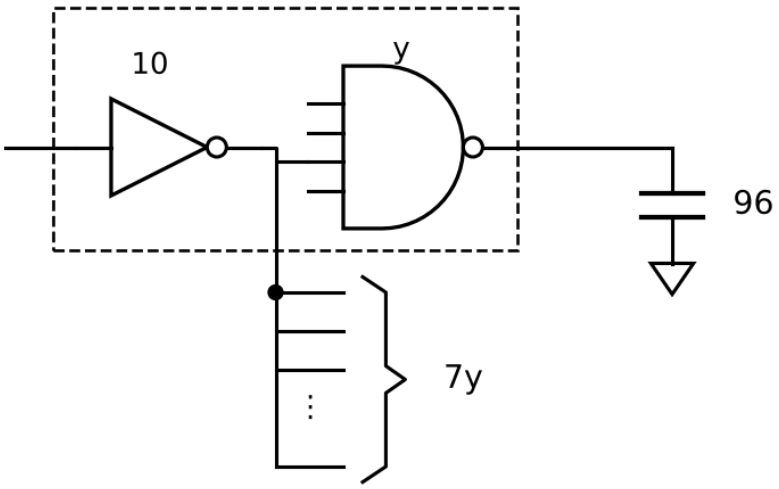


Figure 19: Repeated from the question: Figure 3.2 of the paper (redrawn)

Given: the address inverter has $C_{in} = 10$ (written on the figure) and drives 8 NAND4 gates of input capacitance y each ($B = 8$). Load: 32 bits $\times$ 3 unit transistors $= C_{out} = 96$.

Step 1: Path effort

$$G = g_{inv} \times g_{NAND4} = 1 \times \frac{4+2}{3} = 2, \qquad B = 8, \qquad H = \frac{96}{10} = 9.6$$

$$F = GBH = 2 \times 8 \times 9.6 = 153.6$$

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Step 2: Best number of stages

$$\hat{N} = \log_4 153.6 = 3.63 \;\Rightarrow\; N = 4$$

N	Inverters added	$\hat{f} = F^{1/N}$	P	$D = N\hat{f} + P$
2	0	12.39	5	29.8 τ
3	**1**	**5.36**	**6**	**22.1** τ
4	2	3.52	7	21.1 τ
5	3	2.74	8	21.7 τ

Table 9: Delay against the number of stages

Polarity decides the choice. A NAND4 output is low when its word is selected, but a word line must go high. So an **odd** number of inverters must follow the NAND4. $N = 4$ is the fastest on delay alone, but its 2 inverters leave the word line active low. The valid choices are 1 inverter ($N = 3$) and 3 inverters ($N = 5$). $N = 3$ is 2 % slower than $N = 5$, with fewer gates and less area, and it is the design in Weste and Harris.

$N = 3$: add 1 inverter between the NAND4 and the load

Step 3: Stage effort and path delay

$$\hat{f} = 153.6^{1/3} = 5.36, \qquad P = 1 + 4 + 1 = 6$$

$$\hat{D} = 3(5.36) + 6 = 22.1\,\tau$$

$$\hat{f} = 5.36, \qquad \hat{D} = 22.1\,\tau \approx 4.4 \text{ FO4}$$

Step 4: NAND gate size

From the load backwards: added inverter $96/5.36 = 17.9$; NAND4:

$$y = \frac{g_{NAND4} \times 17.9}{\hat{f}} = \frac{2 \times 17.9}{5.36} = 6.7$$

Check: the address inverter drives $8y = 53.6$, so its input is $53.6/5.36 = 10$ ✔.

$y = 6.7$ (NAND4 input capacitance: nMOS 4.5, pMOS 2.2 per input)

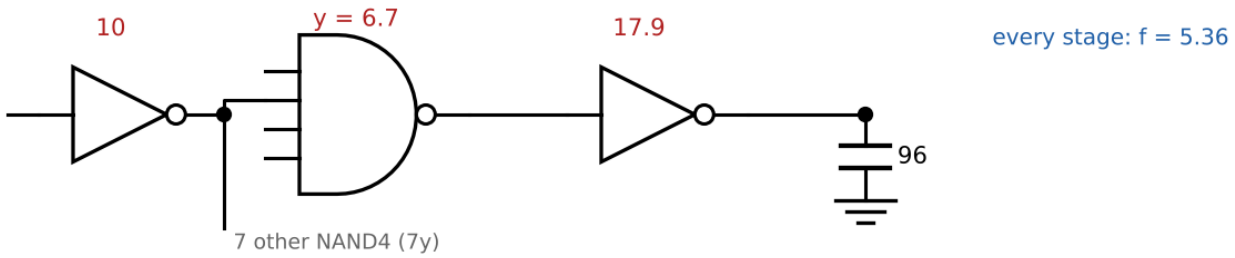


Figure 20: The decoder path: address inverter, NAND4, one added inverter, word line

Final Answer: best $N = 3$ stages (add **1 inverter**), stage effort $\hat{f} = 5.36$, path delay $\hat{D} = 22.1\tau$, NAND4 input capacitance $y = 6.7$ (added inverter 17.9).

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3. Keyword Explanation

1. Word line

↩ back

The line that selects one row (word) of a memory or register file; it drives the access transistors of all bits in that row.

4. Notes and Remarks

- $\hat{N} = 3.63$ rounds to 4, and with delay alone (ignoring polarity) $N = 4$ gives 2 inverters of 7.75 and 27.3, $\hat{f} = 3.52$, $\hat{D} = 21.1\tau$ and $y = 4.4$, but the word line is then active low. This was the earlier version of this answer.
- The input capacitance 10 of the address inverter is not in the printed text; it is taken from the figure, where "10" is written over the inverter.

3.(d) Explain input ordering delay effect. Marks: 5

1. Theory

1. Inputs of a gate are not equal (Core Idea)

In a NAND gate the nMOS transistors are in series. The transistor next to the output is the **outer** input; the one next to ground is the **inner** input. Between them is an internal node x with its own diffusion capacitance. Whether that capacitance must be discharged during the transition depends on **which input arrives last**.

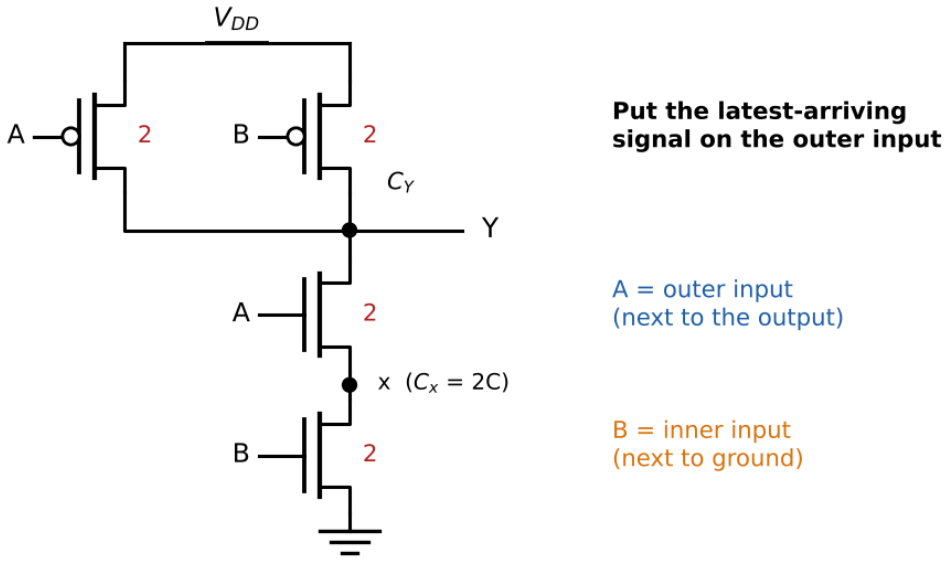


Figure 21: NAND2 with the outer input A (next to the output) and the inner input B (next to ground); x is the internal node

Reading the figure. The two nMOS (width 2 each, $R/2$) are in series; the internal node x between them has about $2C$. If B is already high, x is held at ground through the bottom transistor. If A is already high, x is connected to the output and charged to V_{DD} with it.

2. Answer

Input ordering effect: the delay of a gate depends on which input switches last. For series transistors, an input closer to the output (**outer input**) is faster than one closer to the supply rail (**inner input**), because of the internal node capacitance.

NAND2 example (unit sizing: nMOS width 2, $R/2$ each; $C_Y = 6C + 4hC$, $C_x = 2C$), output falling:

- **Outer input (A) arrives last:** B is already ON, so x is already discharged. Only C_Y discharges, through both transistors:

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$$t_{pd,A} = \left(\frac{R}{2} + \frac{R}{2}\right)(6+4h)C = (6+4h)RC \quad (p=2)$$

- **Inner input (B) arrives last:** A is already ON, so x was charged high with the output; both C_x and C_Y discharge:

$$t_{pd,B} = \frac{R}{2}(2C) + R(6+4h)C = (7+4h)RC \quad (p=7/3)$$

So the inner input has a larger parasitic delay (7/3 against 2 in units of τ); for $h=4$, $23RC$ against $22RC$. The effect grows for gates with more series transistors (NAND3, NAND4, NOR stacks).

Design rule: connect the **latest-arriving (critical) signal to the outer input**, nearest the output, and the early signals to the inner inputs, whose internal nodes are then discharged in advance.

3. Keyword Explanation

1. Internal node

↩ back

A node between series transistors that is not an input or output; its diffusion capacitance can add to the delay.

4. Notes and Remarks

- The rising transition of a NAND (parallel pMOS) is not affected by input order; for NOR gates the same effect appears in the series pMOS stack.

5. Final Concise Answer

(a) Set $I_{Dn} = |I_{Dp}|$ for each V_{in} using the I-V curves. Regions: A ($V_{in} < V_{tn}$): n off, $V_{out} = V_{DD}$; B: n sat, p linear; C: both saturated at $V_M = \dfrac{V_{tn} + r(V_{DD} - |V_{tp}|)}{1+r}$, $r = \sqrt{\beta_p/\beta_n}$, steep fall; D: n linear, p sat; E ($V_{in} > V_{DD} - |V_{tp}|$): p off, $V_{out} = 0$. Symmetric case: $V_M = V_{DD}/2$, $V_{IL} = (3V_{DD} + 2V_t)/8$, $V_{IH} = (5V_{DD} - 2V_t)/8$ (2.125 V and 2.875 V for 5 V and 1 V).

(b) $G = 25/9$, $B = 21/4$, $H = 8$, $F = 116.7$, $\hat{f} = 3.29$, $P = 7$: $\hat{D} = 20.1\tau$. Sizes: $z = 14.6$, $y = 1.85$ (4y = 7.4), $x = 2.82$ (4x = 11.3).

(c) $G = 2$, $B = 8$, $H = 9.6$, $F = 153.6$, $\hat{N} = 3.63$; polarity needs an odd number of added inverters: 3 stages (add 1 inverter), $\hat{f} = 5.36$, $\hat{D} = 22.1\tau$, NAND4 $y = 6.7$ (inverters 7.75 and 27.3).

(d) The delay depends on which series input switches last: the outer input (next to the output) is faster because the internal node is already discharged ($p = 2$ against 7/3 for a NAND2); put the latest-arriving signal on the outer input.

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4. (a) What is scaling? Explain how different parameters of transistors are scaled? Marks: 5
   (b) Define glitch? How does it increase power consumption? How it can be reduced? Marks: 6
   (c) You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at $V_{DD} = 0.9$ V. Marks: 4
   (d) Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced? Marks: 7
   (e) Define sheet resistance. How does interconnect affect the performance of a system? Marks: 3

4.(a) What is scaling? Explain how different parameters of transistors are scaled? Marks: 5

1. Theory

1. What does "scaling" mean? (Core Idea)

Every two or three years a new process generation (a new "node") makes transistors smaller. If every length shrinks by a factor S (typically $S \approx \sqrt{2} = 1.4$ per node), each transistor takes $1/S^2$ of the area, so twice as many fit on the same chip; this is how Moore's law (2(c)) was kept.

But a transistor is not only lengths: it has voltages and doping, too. How we choose to change these decides whether the smaller transistor works well. Two classic choices:

- **Constant-field (Dennard) scaling:** shrink the voltages by S as well, so the electric fields inside the device stay the same. The device behaves like a perfect miniature of the old one.
- **Constant-voltage scaling:** keep V_{DD} the same (to stay compatible with older chips); the fields then grow by S.

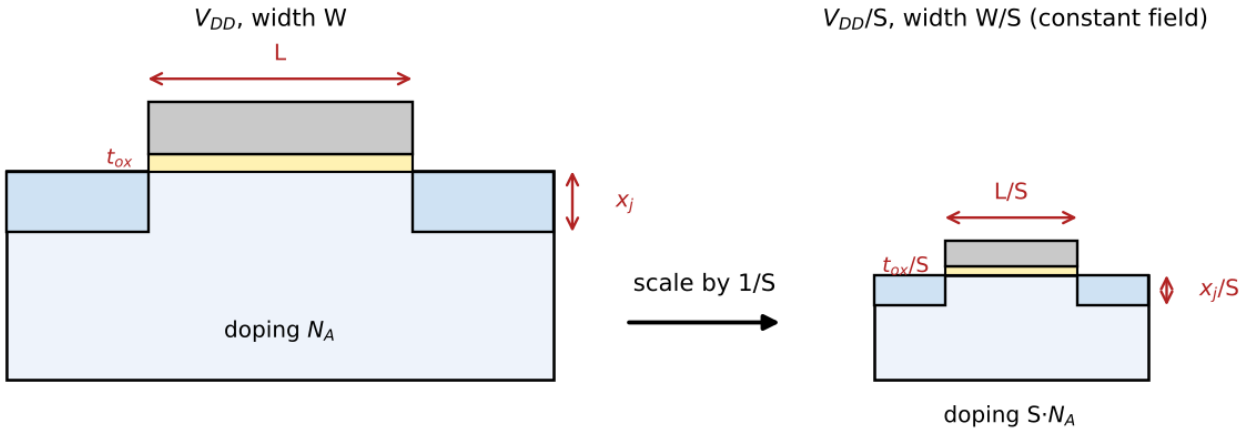


Figure 22: A transistor before and after scaling every dimension by 1/S (drawn for S = 2)

Reading the figure. Every length (channel length L, junction depth x_j, oxide thickness t_{ox}, and the width W) shrinks by $1/S$. To keep the depletion regions in proportion, the body doping is raised by S. Under constant-field scaling the supply also shrinks to V_{DD}/S.

2. Working out each parameter

The rules follow from the formulas of 2(a):

- Oxide capacitance per area $C_{ox} = \varepsilon_{ox}/t_{ox} \propto S$.
- Gain $\beta = \mu C_{ox} W/L \propto S$ (since W/L is unchanged).
- Current $I \propto \beta V^2$; gate capacitance $C \propto C_{ox} WL \propto 1/S$.

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- Gate delay $t \propto CV/I$; power per gate $P = IV$ (or $CV^2 f$).

2. Answer

Scaling is the reduction of the dimensions of a transistor (and its wires) by a factor $S > 1$ in each new technology generation, with the voltages and doping changed in a planned way, so that more, faster and lower-power transistors fit on a chip.

Parameter	Constant-field scaling	Constant-voltage scaling
Length L, width W, oxide t_{ox}, junction depth x_j	$1/S$	$1/S$
Supply V_{DD}, threshold V_t	$1/S$	1
Body doping N_A	S	S^2
C_{ox}, $\beta = \mu C_{ox} W/L$	S	S
Current per transistor $I \propto \beta V^2$	$1/S$	S
Gate capacitance $C \propto C_{ox} WL$	$1/S$	$1/S$
Gate delay CV/I	$1/S$	$1/S^2$
Power per gate IV	$1/S^2$	S
Switching energy CV^2	$1/S^3$	$1/S$
Transistors per area	S^2	S^2
Power density	1	S^3

Table 10: How the parameters scale (factor by which each changes)

Interpretation: constant-field scaling makes gates S times faster, S^2 times denser and keeps the power per area constant, an almost free improvement. Constant-voltage scaling gives even faster gates but the power density rises as S^3 and the high fields damage the devices (hot carriers, oxide breakdown), so it cannot be used for long. In practice, V_{DD} was scaled more slowly than the dimensions, and since about 2005 it has hardly scaled at all, because V_t cannot fall further without large leakage.

3. Keyword Explanation

1. Scaling factor S

↩ back

The ratio of old to new dimensions; about 1.4 per technology node (for example 90 nm → 65 nm).

4. Notes and Remarks

- Wires scale differently: a local wire's resistance per length grows by S^2 (it is thinner and narrower) while its length shrinks by S, so its RC delay stays about constant instead of falling, and global wires get slower; see 4(e).

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4.(b) Define glitch? How does it increase power consumption? How it can be reduced? Marks: 6

1. Theory

1. What is a glitch? (Core Idea)

A logic circuit computes its output from inputs that arrive through paths of different delays. While the inputs are still arriving, the output may briefly take a wrong value and then return, even though its final value does not change. Such an unwanted short pulse is a **glitch** (also called a hazard or spurious transition).

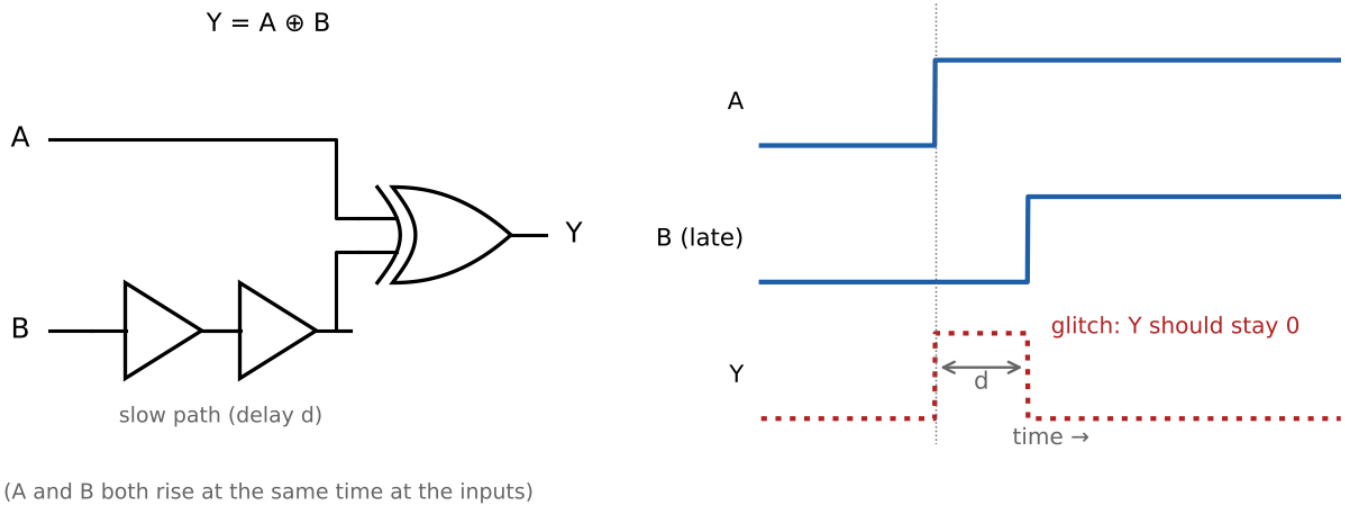


Figure 23: A glitch in an XOR gate: A and B rise together, but B arrives late through the slow path, so Y is briefly 1

Reading the figure. A and B both change from 0 to 1 at the same moment, so $Y = A \oplus B$ should stay 0. But B reaches the XOR a time d later (two buffers). During d the XOR sees A = 1, B = 0 and outputs 1. Y rises and falls again: a glitch of width d.

2. Why does a glitch cost power?

Dynamic power is spent whenever a node's capacitance is charged and discharged:

$$P_{dyn} = \alpha\, C\, V_{DD}^2\, f$$

where α is the **activity factor**, the average number of $0 \to 1$ transitions per clock cycle. A glitch is a $0 \to 1 \to 0$ pair that does no useful work, but it charges the node's capacitance once more, taking CV_{DD}^2 from the supply. It also spreads: the gates driven by the glitching node may glitch too.

2. Answer

Glitch: a short, unwanted transition (pulse) at the output of a combinational gate, caused by **unequal arrival times** of its inputs, before the output settles to its correct final value.

How it increases power:

- Each glitch is an extra charge and discharge of the node capacitance, drawing CV_{DD}^2 from the supply, so it raises the effective activity factor α in $P = \alpha C V_{DD}^2 f$.
- Glitches propagate through the following logic and can multiply: in long unbalanced paths (for example a ripple-carry adder or array multiplier) a node can toggle several times per cycle, and glitches can make up 20–70 % of the dynamic power.
- During each extra transition there is also short-circuit current while both networks are partly on.

How to reduce glitches:

1. **Balance path delays:** make the signals reaching a gate arrive together, using balanced tree structures instead of chains (for example $(A + B) + (C + D)$ instead of $((A + B) + C) + D$), or by adding delay to fast paths.
2. **Reduce logic depth:** fewer levels of logic between registers give less time for glitches to build up; insert registers (pipelining) or latches to stop glitches spreading.
3. **Restructure and reorder logic:** move high-activity or late signals close to the output, and choose logic forms with fewer reconvergent paths.
4. **Use glitch-free logic styles:** dynamic (domino) logic makes at most one transition per cycle, since its outputs only rise during evaluation; see 6(a).
5. **Gate the inputs:** hold inputs of unused blocks steady (operand isolation, clock gating), so they do not toggle at all.

3. Keyword Explanation

1. Activity factor α

↩ back

The probability that a node makes a $0 \to 1$ transition in a clock cycle; glitches increase it.

2. Reconvergent path

↩ back

Two paths that start from the same signal and meet again at a later gate, the usual cause of glitches.

4. Notes and Remarks

- A clocked flip-flop at the end of a path samples only the final value, so glitches do not cause wrong results in synchronous logic; their cost is power (and noise). On an asynchronous signal, such as a clock or a reset, a glitch can cause a wrong result.

4.(c) You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at $V_{DD} = 0.9$ V. Marks: 4

1. Theory

1. Where dynamic power comes from (Core Idea)

Each time a node rises from 0 to V_{DD}, its capacitance C takes a charge CV_{DD} from the supply, which costs energy CV_{DD}^2 (half stored, half lost in the pMOS; the stored half is lost in the nMOS when it falls). If this happens α times per cycle at clock frequency f:

$$P_{dyn} = \alpha\, C\, V_{DD}^2\, f$$

where

- α is the activity factor (0.1 for typical random logic, 1 for a clock)
- C is the total switching capacitance (gates, diffusions and wires)
- V_{DD} is the supply voltage and f the clock frequency

2. Capacitance from area

For a standard-cell design, the capacitance is roughly proportional to the area, so a process quotes an average **switching capacitance per area**; the chip's capacitance is that times the area.

2. Answer

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Given:

- $\alpha = 0.1$
- $C/A = 450\ \text{pF/mm}^2 = 450 \times 10^{-12}\ \text{F/mm}^2$, $A = 70\ \text{mm}^2$
- $f = 450\ \text{MHz} = 4.5 \times 10^8\ \text{Hz}$, $V_{DD} = 0.9\ \text{V}$

Step 1: Switching capacitance

$$C = 450\ \text{pF/mm}^2 \times 70\ \text{mm}^2 = 31\,500\ \text{pF} = 31.5\ \text{nF}$$

Step 2: Dynamic power

$$P_{dyn} = \alpha C V_{DD}^2 f = 0.1 \times 31.5 \times 10^{-9} \times (0.9)^2 \times 4.5 \times 10^8$$

$$= 0.1 \times 31.5 \times 10^{-9} \times 0.81 \times 4.5 \times 10^8 = 1.148\ \text{W}$$

$$\boxed{P_{dyn} \approx 1.15\ \text{W}}$$

Final Answer: the chip dissipates about **1.15 W** of dynamic power.

3. Notes and Remarks

- The result does not include static (leakage) power or the clock network, which has $\alpha = 1$ and is often quoted separately.

4.(d) Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced? Marks: 7

1. Theory

1. Wires are capacitors to each other (Core Idea)

Two neighbouring wires on a chip are two conductors separated by an insulator: a capacitor. In modern processes wires are taller than they are wide and packed close together, so the capacitance to the **neighbouring wire** (C_{adj}) is often larger than the capacitance to the **ground** below (C_{gnd}). When one wire switches, it pushes charge through C_{adj} onto its neighbour. This coupling is **crosstalk**. The switching wire is the **aggressor**; the one disturbed is the **victim**.

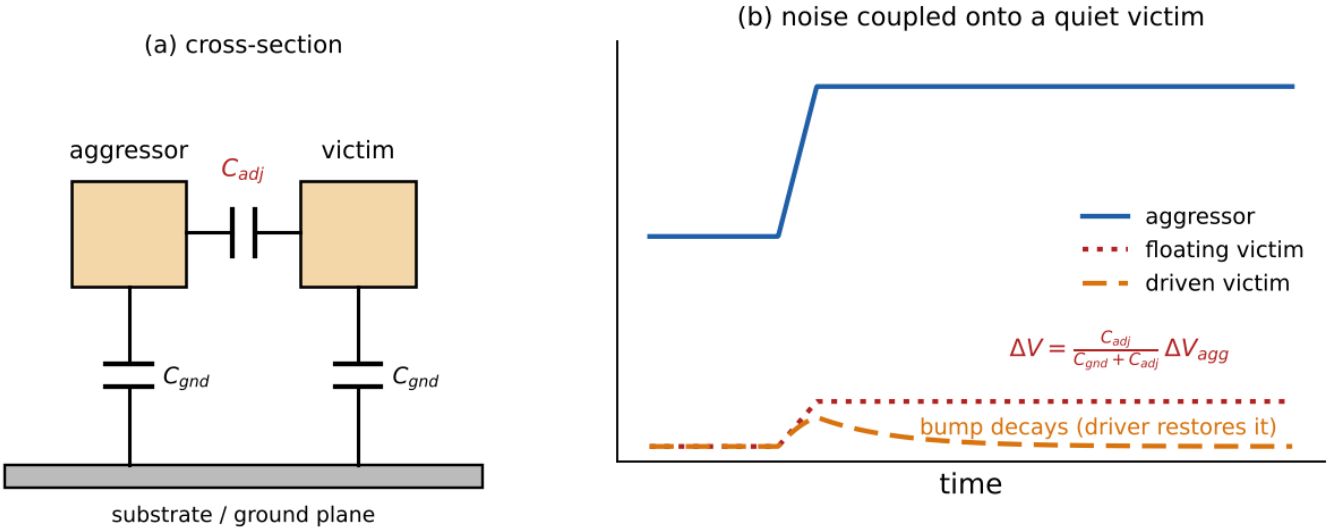


Figure 24: (a) Two neighbouring wires with their coupling and ground capacitances; (b) the noise on a quiet victim when the aggressor rises

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Reading the figure. In (a), the aggressor and the victim are coupled by C_{adj} and each has C_{gnd} to the substrate. In (b), the aggressor rises. A **floating** victim (red, no driver holding it) jumps up by a fixed fraction of the swing and stays there. A **driven** victim (orange, held low by its gate) jumps less and then returns as its driver pulls it back.

2. Answer

Crosstalk is the unwanted coupling of a signal from one wire (the aggressor) onto a neighbouring wire (the victim) through their mutual (coupling) capacitance, causing noise or a change in delay.

Noise in different cases:

Case 1: floating victim (a dynamic node or a tri-stated bus). Charge sharing through the capacitor divider:

$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_{aggressor}$$

The noise stays until the node is driven again. With $C_{adj} = C_{gnd}$, a full-swing aggressor puts $V_{DD}/2$ on the victim, which can flip a dynamic gate.

Case 2: driven victim (held by a gate with resistance R_{victim}). The victim's driver fights the noise, so the bump is smaller and decays:

$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd} + C_{adj}} \cdot \frac{1}{1+k} \Delta V_{aggressor}, \qquad k = \frac{\tau_{aggressor}}{\tau_{victim}} = \frac{R_{aggr}(C_{gnd,a} + C_{adj})}{R_{victim}(C_{gnd,v} + C_{adj})}$$

A strong victim driver (small τ_{victim}, large k) or a slow aggressor keeps the noise small.

Case 3: both wires switching (delay noise). The charge through C_{adj} depends on the voltage change *across* it, so the effective capacitance of the victim is $C_{eff} = C_{gnd} + k_M C_{adj}$:

Aggressor	Change across C_{adj}	k_M (Miller factor)	Effect on victim
Switches the same way	0	0	faster
Quiet	V_{DD}	1	normal
Switches the opposite way	$2V_{DD}$	2	slower

Table 11: Effective coupling capacitance when both wires switch

So crosstalk also makes delay depend on the data, which can cause setup or hold failures.

How to reduce crosstalk:

1. **Increase spacing** between sensitive wires (lower C_{adj}).
2. **Shield:** run a V_{DD} or ground wire between the aggressor and the victim, so the coupling goes to a fixed supply.
3. **Route neighbouring layers orthogonally** and limit the length over which two wires run in parallel.
4. **Make victims strong:** larger drivers or repeaters (staggered, so the coupling cancels); keep critical dynamic nodes short.
5. **Slow the aggressors' edges** where speed allows, and use **differential signalling** (the noise appears on both wires and cancels).

3. Keyword Explanation

1. Aggressor and victim

↩ back

The switching wire that causes the noise, and the neighbouring wire that receives it.

2. Miller coupling factor

↩ back

The factor k_M (0, 1 or 2) multiplying C_{adj} in the victim's effective capacitance, set by how the neighbour switches.

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4. Notes and Remarks

- Hand notes on the paper list "shield, space, different signal" as cures; these are items 2, 1 and 5 above.

4.(e) Define sheet resistance. How does interconnect affect the performance of a system? Marks: 3

1. Theory

1. Resistance of a thin layer (Core Idea)

A wire of resistivity ρ, length l, width w and thickness t has resistance $R = \rho l/(wt)$. On a chip all wires of one layer have the same thickness, so it is convenient to combine ρ and t into one number, the **sheet resistance** $R_\square = \rho/t$. Then

$$R = R_\square \frac{l}{w}$$

and l/w is the number of **squares** the wire contains.

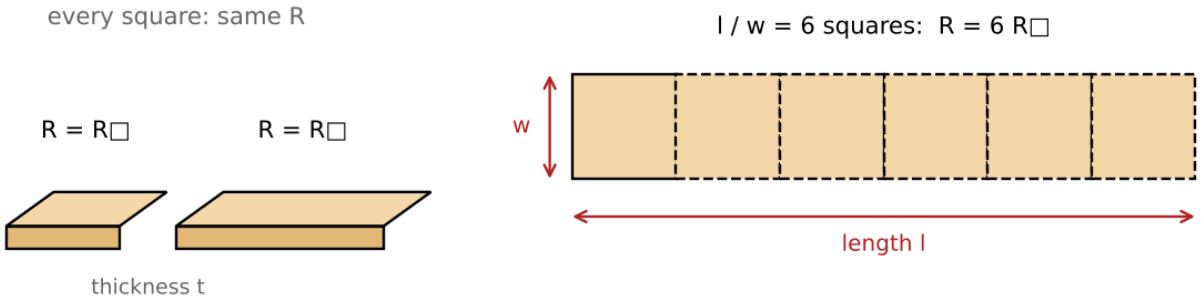


Figure 25: Sheet resistance: a square of a layer has resistance $R_\square$ whatever its size; a wire 6 squares long has 6 $R_\square$

Reading the figure. A small square and a large square of the same layer have the same resistance, because the larger square is longer (more resistance) by the same factor as it is wider (less resistance). A wire with $l/w = 6$ is six squares in series: $R = 6R_\square$.

2. Answer

Sheet resistance $R_\square$ is the resistance of a square piece of a thin conducting layer, measured between two opposite edges: $R_\square = \rho/t$ (unit $\Omega/\square$, "ohms per square"). A wire of length l and width w then has $R = R_\square\, l/w$. (Example: copper, $t = 0.5$ μm: $R_\square = 2.2 \times 10^{-8}/0.5 \times 10^{-6} = 0.044$ $\Omega/\square$; a 1 mm wire 0.25 µm wide is 4000 squares, 176 Ω.)

Effect of interconnect on performance:

- **Delay:** a wire has distributed resistance and capacitance; its delay is about $RC/2$, which grows with the **square** of its length. In deep submicron chips long wires are slower than gates and need repeaters.
- **Power:** wire capacitance is often more than half of the switched capacitance, so wires take a large share of the dynamic power.
- **Noise:** coupling capacitance causes crosstalk (4(d)), and resistance in the power grid causes IR drop.
- **Reliability and area:** high current density causes electromigration; routing takes area and limits density.

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3. Final Concise Answer

(a) Scaling shrinks all dimensions by $1/S$ per generation. Constant field: $L, W, t_{ox}, x_j, V_{DD}, V_t$ all $1/S$, $N_A \times S$; current $1/S$, capacitance $1/S$, delay $1/S$, power per gate $1/S^2$, density S^2, power density 1. Constant voltage: V fixed, $N_A \times S^2$; current S, delay $1/S^2$, power per gate S, power density S^3.

(b) A glitch is a spurious transition caused by unequal input arrival times. It charges and discharges the node capacitance without need (CV_{DD}^2 each, raising α) and propagates. Reduce by balancing paths, reducing logic depth or pipelining, restructuring logic, using dynamic logic and gating unused inputs.

(c) $C = 450\ \text{pF/mm}^2 \times 70\ \text{mm}^2 = 31.5\ \text{nF}$; $P = 0.1 \times 31.5\ \text{nF} \times 0.9^2 \times 450\ \text{MHz} = 1.15\ \text{W}$.

(d) Crosstalk: coupling through C_{adj} from an aggressor to a victim. Floating victim: $\Delta V = \frac{C_{adj}}{C_{gnd}+C_{adj}}\Delta V_{agg}$; driven victim: smaller by $1/(1+k)$ and decays; both switching: $C_{eff} = C_{gnd} + k_M C_{adj}$, $k_M = 0$, 1, 2. Cures: spacing, shielding, orthogonal routing, strong drivers or repeaters, slower edges, differential signalling.

(e) $R_\square = \rho/t$ ($\Omega/\square$), $R = R_\square l/w$. Interconnect adds RC delay (grows as length squared), dynamic power, crosstalk and IR-drop noise, electromigration and area.

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5. (a) What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution? Marks: 6

(b) The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time T_c at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures? Marks: 4

(c) Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer. Marks: 9

(d) Define stack effect. Find subthreshold leakage current for two series OFF transistor. Marks: 6

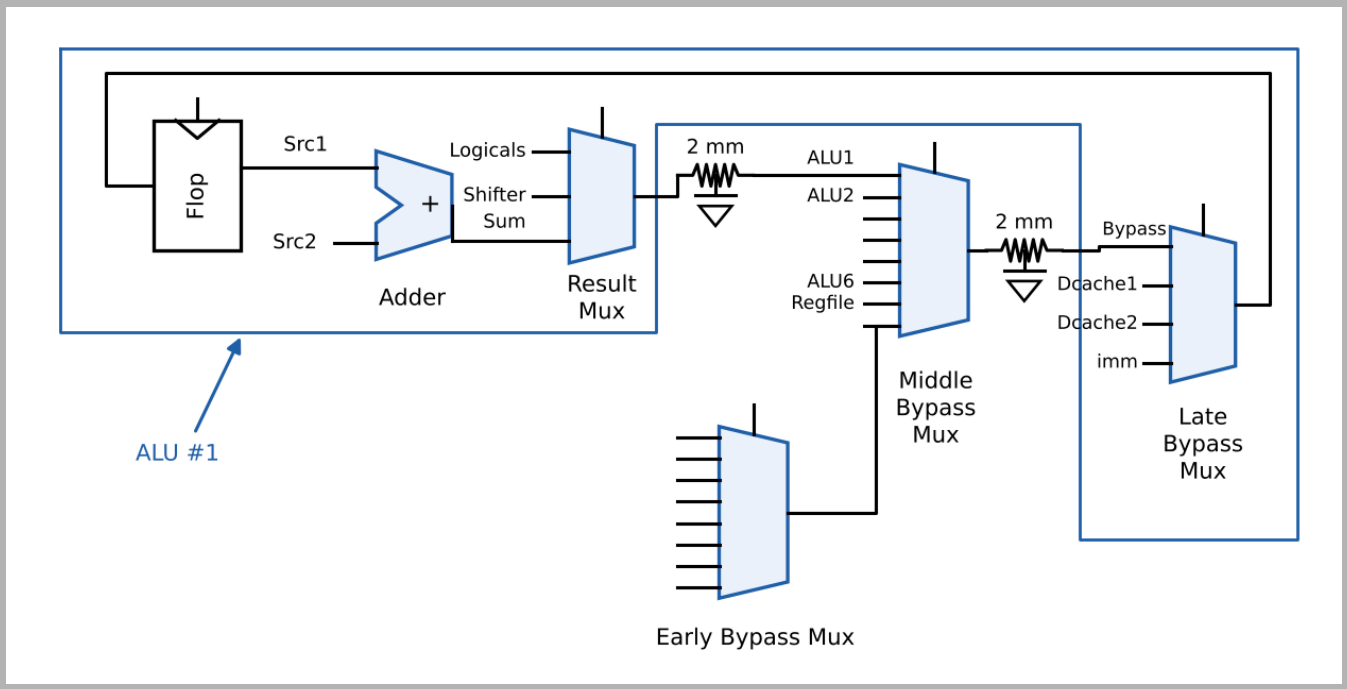


Figure 26: Figure 5.1 of the paper: the Itanium 2 ALU self-bypass path (redrawn)

Elements	Propagation delay	Contamination delay
Adder	590 ps	100 ps
Result Mux	60 ps	35 ps
Early Bypass Mux	110 ps	95 ps
Middle Bypass Mux	80 ps	55 ps
Late Bypass Mux	70 ps	45 ps
2-mm Wire	100 ps	65 ps

Table 12: Combinational Logic Delays (Table 5.1 of the paper)

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5.(a) What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution? Marks: 6

1. Theory

1. How a flip-flop system works (Core Idea)

A synchronous system is a chain: flip-flop → combinational logic → flip-flop. At each rising clock edge every flip-flop copies its input D to its output Q; the logic between them then has one clock cycle to compute the next value. Each flip-flop needs its input to be **steady** for a short time around the edge:

- **Setup time** t_{setup} before the edge, and **hold time** t_{hold} after it.

And it has two output delays:

- t_{pcq}, the **clock-to-Q propagation delay** (the latest Q changes after the edge), and
- t_{ccq}, the **clock-to-Q contamination delay** (the earliest Q starts to change).

The logic has a **propagation delay** t_{pd} (longest) and a **contamination delay** t_{cd} (shortest).

2. Two ways a path can fail

- **Too slow (max-delay, setup):** the new data does not reach FF2 before its next clock edge minus t_{setup}. Fixed by a longer clock period.
- **Too fast (min-delay, hold):** at a clock edge, FF1 launches new data; if that data races through very fast logic and reaches FF2 **within FF2's hold time after the same edge**, it corrupts the value FF2 is capturing. A longer clock period does *not* help, because both flip-flops use the same edge.

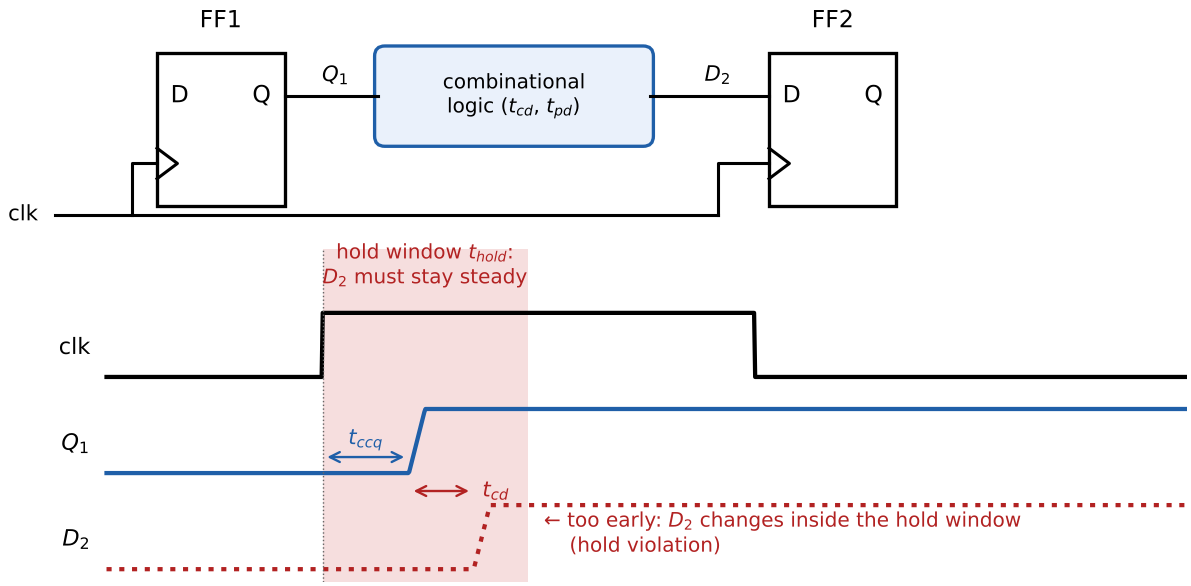


Figure 27: A flip-flop-to-flip-flop path and its min-delay timing: the new data must not reach D2 inside FF2's hold window

Reading the figure. At the clock edge, Q_1 starts changing after t_{ccq}; the change reaches D_2 after a further t_{cd} (the fastest path through the logic). FF2 needs D_2 steady until t_{hold} after the same edge (shaded). Here D_2 changes inside the shaded window: a hold violation.

2. Answer

Min-delay constraint: the requirement that the **shortest** path from one sequencing element to the next is slow enough that data launched at a clock edge does not reach the receiving element before its hold time after that same edge has passed. It is also called the **hold-time constraint**; violating it is a **race**.

Min-delay for a flip-flop system: the earliest the data at FF2 can change after the edge is $t_{ccq} + t_{cd}$. It must not change before t_{hold}:

$$t_{ccq} + t_{cd} \geq t_{hold}$$

$$\boxed{t_{cd} \geq t_{hold} - t_{ccq}}$$

With clock skew t_{skew} (FF2's clock later than FF1's): $t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$.
Key point: the constraint does not contain the clock period T_c, so a chip with a hold violation fails at *every* frequency.
Solutions if a min-delay (hold) failure occurs:

1. **Add delay to the short path:** insert buffers (pairs of inverters) or slower gates until t_{cd} meets the constraint; only short paths are padded, so the critical path is not slowed.
2. **Use flip-flops with a smaller (or negative) hold time,** or a larger t_{ccq}.
3. **Reduce clock skew** (better clock distribution), or skew the clock deliberately in the safe direction.
4. Use two-phase latches with non-overlapping clocks, which put a gap between the launch and capture.

3. Keyword Explanation

1. Contamination delay

↩ back

The shortest time from an input change until an output starts to change.

2. Race condition

↩ back

New data racing through a fast path and overwriting data still being captured.

4. Notes and Remarks

- The matching max-delay (setup) constraint is $T_c \geq t_{pcq} + t_{pd} + t_{setup}$; it is used in 5(b).

5.(b) ⚠ Note on the paper: The paper gives no flip-flop timing. Notes written on the paper give the flip-flop values $t_{setup} = 62$ ps, $t_{hold} = -10$ ps, $t_{ccq} = 75$ ps and $t_{pcq} = 90$ ps (the last is hard to read; 90 ps is the value in Weste and Harris's version of this example). These are used below.

1. Theory

1. The max-delay (setup) constraint (Core Idea)

Data launched by a flip-flop at one clock edge must reach the next flip-flop at least t_{setup} before the next edge, one period T_c later. The latest it can arrive is t_{pcq} (clock-to-Q) plus t_{pd} (the longest logic path):

$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

The smallest period that works is the right-hand side.

2. A bypass path

In a processor, the result of the ALU is needed again by the very next instruction. Rather than waiting for it to be written to the register file, **bypass multiplexers** send the result straight back to the ALU's input flip-flops. This **self-bypass loop** (adder → result mux → wire → middle and late bypass muxes, with a second wire between them → flip-flop) must fit in one cycle, so it sets the clock speed. The hold check of 5(a) is applied to the **shortest** path into the same flip-flop.

2. Answer

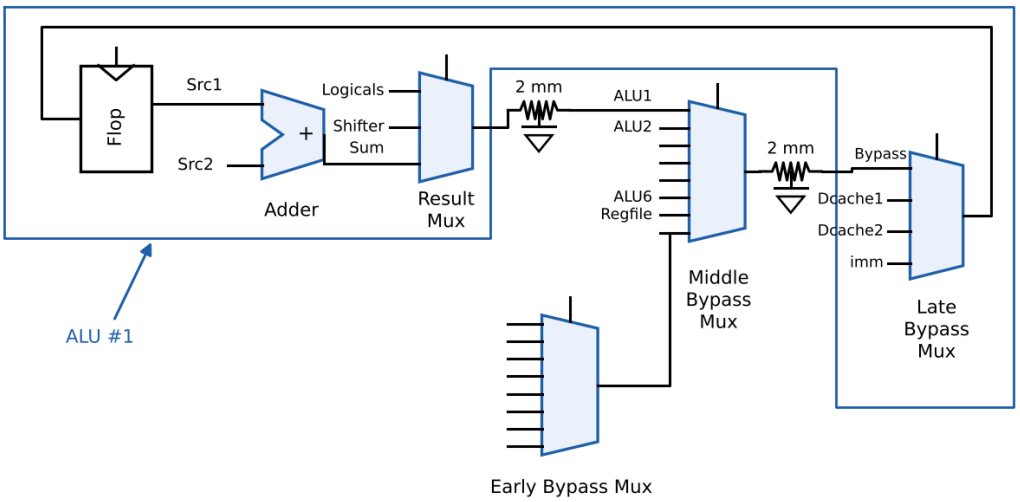


Figure 28: Repeated from the question: Figure 5.1 of the paper (redrawn)

Given: element delays in Table 5.1; flip-flops $t_{pcq} = 90$ ps, $t_{ccq} = 75$ ps, $t_{setup} = 62$ ps, $t_{hold} = -10$ ps.

Step 1: Longest path through the loop

The longest path starts at the flop and passes through the adder, the result mux, a 2-mm wire, the middle bypass mux, a second 2-mm wire and the late bypass mux. The early bypass mux feeds only one side input of the middle mux, so it is not on this path:

$$t_{pd} = 590 + 60 + 100 + 80 + 100 + 70 = 1000 \text{ ps}$$

Step 2: Minimum cycle time

$$T_c \geq t_{pcq} + t_{pd} + t_{setup} = 90 + 1000 + 62 = 1152 \text{ ps}$$

$$\boxed{T_c = 1152 \text{ ps } (f_{max} \approx 868 \text{ MHz})}$$

Step 3: Hold check

The shortest path into the flip-flop is the imm value from another flip-flop through the late bypass mux only:

$$t_{ccq} + t_{cd,\,late\,mux} = 75 + 45 = 120 \text{ ps } \geq t_{hold} = -10 \text{ ps } \checkmark$$

The data cannot change until 120 ps after the edge, while the flip-flop needs it steady only until 10 ps *before* the edge, a margin of 130 ps.

Final Answer: minimum cycle time $T_c = 1152$ ps (about 0.87 GHz). There is **no hold-time failure**: the earliest arrival (120 ps) is well after the hold time (−10 ps).

3. Notes and Remarks

- With $t_{pcq} = 98$ ps (the other reading of the note), $T_c = 1160$ ps; the hold result does not change, since it uses t_{ccq}.
- A negative hold time means the flip-flop's input may change even slightly *before* the clock edge without harm, which makes hold failures unlikely.

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5.(c) Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer. Marks: 9

1. Theory

1. Two ways to build the same function (Core Idea)

$F = AB + CD$ can be built as

- **NAND–NAND:** by De Morgan, $F = \overline{\overline{AB} \cdot \overline{CD}}$: two NAND2 gates feed one NAND2. Two stages.
- **Compound gate + inverter:** one AND-OR-INVERT gate, AOI22 $= \overline{AB + CD}$, followed by an inverter. Also two stages.

Which is faster? Logical effort (3(b)) answers it: for each design, $F = GH$, $\hat{D} = NF^{1/N} + P$.

2. Unit-resistance sizing

"Unit path resistance" means each gate is sized so its worst pull-up and pull-down match a unit inverter (nMOS 1, pMOS 2). Then:

- **NAND2:** series nMOS of width 2, parallel pMOS of width 2. One input: $2 + 2 = 4 \rightarrow g = 4/3$, $p = 2$.
- **AOI22:** the nMOS are two parallel pairs of two in series (width 2 each); the pMOS are two series pairs of two in parallel (width 4 each). One input: $2 + 4 = 6 \rightarrow g = 6/3 = 2$; the output has $2 \times 2 + 2 \times 4 = 12$ units of diffusion $\rightarrow p = 12/3 = 4$.
- **Inverter:** $g = 1$, $p = 1$.

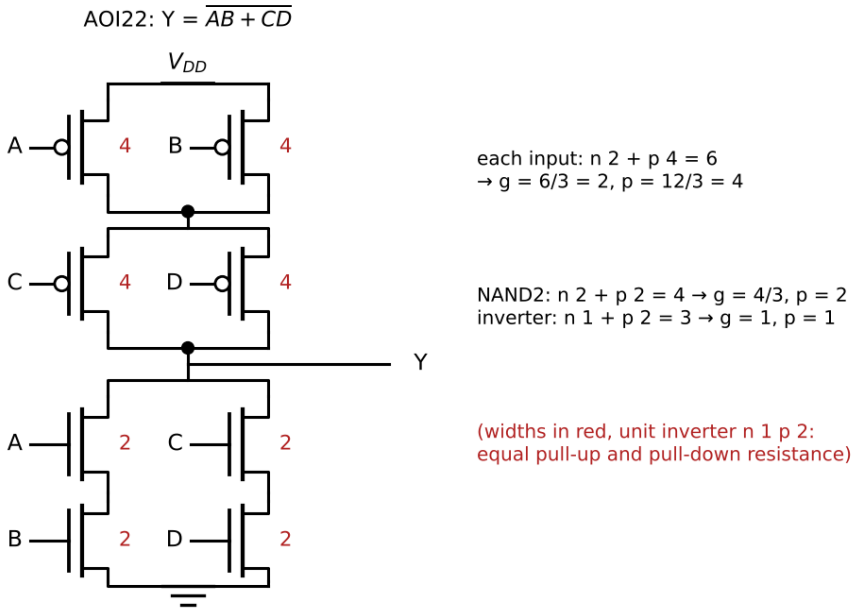


Figure 29: Unit AOI22 (widths in red) with the logical efforts of the gates used

Reading the figure. The pull-up has A and B in parallel, in series with C and D in parallel; any path from V_{DD} to Y passes two pMOS, so each is width 4. The pull-down has A in series with B, in parallel with C in series with D; each path passes two nMOS, so each is width 2. One input (say A) sees one nMOS (2) and one pMOS (4): 6 units.

2. Answer

Given: input capacitance $C_{in} = 20\lambda$ (each input), load $C_{out} = 100\lambda$: $H = 100/20 = 5$. Both designs have $N = 2$ stages and no branching.

Design (i): only NAND gates

$$G = \frac{4}{3} \cdot \frac{4}{3} = \frac{16}{9} = 1.78, \qquad F = GH = 1.78 \times 5 = 8.89$$

$$\hat{f} = \sqrt{8.89} = 2.98, \qquad P = 2 + 2 = 4$$

$$\hat{D} = 2(2.98) + 4 = 9.96\,\tau$$

$$\boxed{\hat{D}_{NAND} = 9.96\,\tau \approx 2.0 \text{ FO4}}$$

Sizes: second NAND2 input $= \dfrac{(4/3)(100)}{2.98} = 44.7\lambda$ (nMOS 22.4, pMOS 22.4); first NAND2 input $= \dfrac{(4/3)(44.7)}{2.98} = 20\lambda$ ✔ (nMOS 10, pMOS 10).

Design (ii): compound gate (AOI22 + inverter)

$$G = 2 \times 1 = 2, \qquad F = GH = 2 \times 5 = 10$$

$$\hat{f} = \sqrt{10} = 3.16, \qquad P = 4 + 1 = 5$$

$$\hat{D} = 2(3.16) + 5 = 11.32\,\tau$$

$$\boxed{\hat{D}_{AOI} = 11.3\,\tau \approx 2.3 \text{ FO4}}$$

Sizes: inverter input $= 100/3.16 = 31.6\lambda$ (nMOS 10.5, pMOS 21.1); AOI22 input $= 2(31.6)/3.16 = 20\lambda$ ✔ (nMOS 6.7, pMOS 13.3).

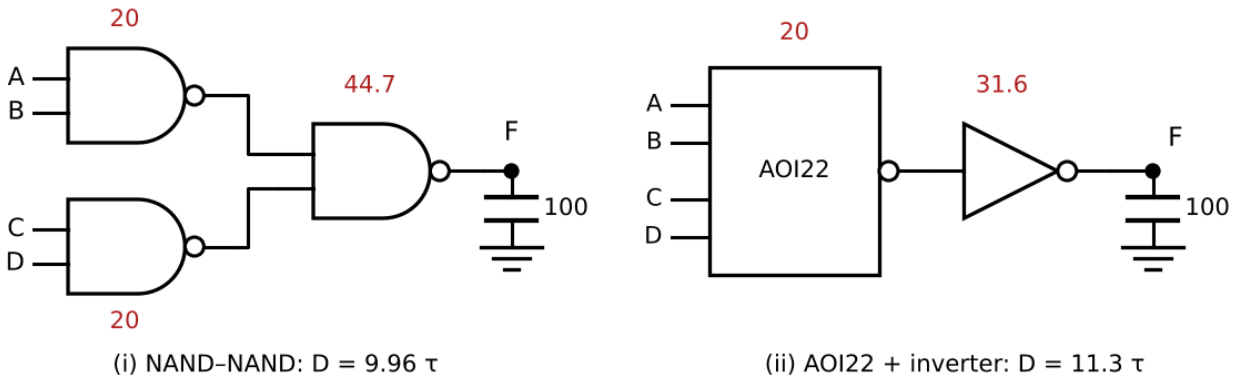


Figure 30: The two designs with their input capacitances for minimum delay

Which is faster: the NAND–NAND design, 9.96τ against 11.3τ (about 12 % faster).
Comment: the compound gate uses fewer transistors ($8 + 2 = 10$ against 12) and fewer gates, which suggests it should be faster, but its series pMOS stacks give it a larger logical effort ($g = 2$ against $4/3$) and a larger parasitic delay ($p = 4$), and it still needs an inverter for the true output. The NAND–NAND form spends its two stages on two easy gates. For a much larger load (larger H) the difference in parasitic delay matters less, but the NAND design stays ahead because its total G is smaller ($1.78 < 2$). So counting transistors or gates is not a reliable guide to speed; logical effort is.

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3. Keyword Explanation

1. Compound (complex) gate

↩ back

A single static CMOS gate that computes an AND-OR-INVERT (AOI) or OR-AND-INVERT (OAI) function in one stage.

4. Notes and Remarks

- **Every stage has the same effort:** NAND design: $\frac{4}{3} \times \frac{44.7}{20} = 2.98$ and $\frac{4}{3} \times \frac{100}{44.7} = 2.98$; AOI design: $2 \times \frac{31.6}{20} = 3.16$ and $\frac{100}{31.6} = 3.16$.

5.(d) Define stack effect. Find subthreshold leakage current for two series OFF transistor. Marks: 6

1. Theory

The basics

A MOSFET used as a switch should pass no current when off, with $V_{GS} = 0$. In fact a small **subthreshold current** flows, since a few electrons always have enough thermal energy to cross the barrier. It is small for one transistor, but a chip has billions, so this **leakage** matters. It falls by a factor of ten for each drop of a fixed amount in V_{GS}. A higher V_{DS} also lowers the barrier, which is called **DIBL**.

1. Leakage of one OFF transistor (Core Idea)

From 1(d), an OFF transistor still carries a subthreshold current, which grows exponentially with V_{GS} and (through DIBL) with V_{DS}. Ignoring body effect and taking $V_{DS} \gg v_T$:

$$I_{sub} = I'_{off}\, 10^{\frac{V_{GS}+\eta V_{DS}}{S}}$$

where S is the subthreshold slope (mV/decade), η the DIBL coefficient and I'_{off} a constant of the transistor. A single OFF nMOS ($V_{GS} = 0$, $V_{DS} = V_{DD}$) leaks $I'_{off}\, 10^{\eta V_{DD}/S}$.

2. What changes in a stack

When two OFF nMOS are in series, the same leakage current must flow through both. That forces the middle node x to rise a little above ground. This small voltage V_x does three good things to the **top** transistor: its $V_{GS} = -V_x$ is negative, its $V_{DS} = V_{DD} - V_x$ is smaller (less DIBL), and its source is above the body (body effect raises V_t). The **bottom** transistor has only $V_{DS} = V_x$, so it has almost no DIBL. Both leak far less.

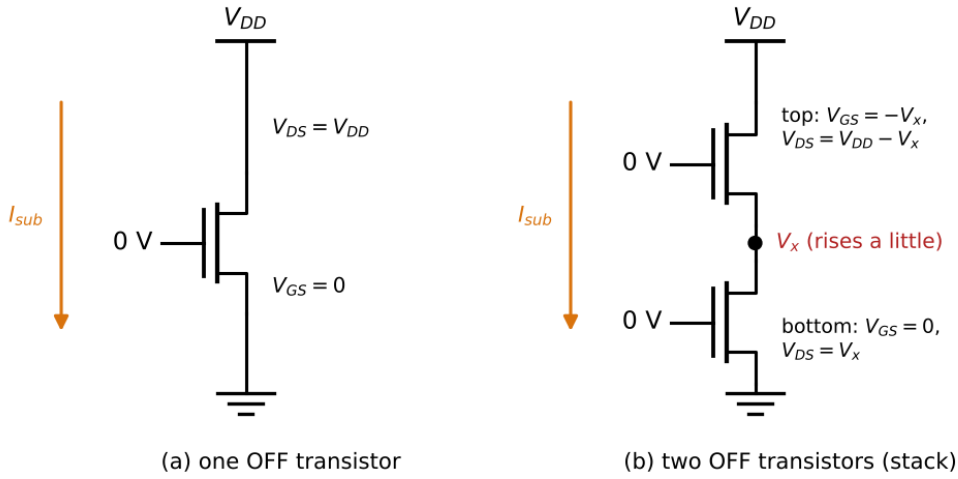


Figure 31: Subthreshold leakage through (a) one OFF transistor and (b) two OFF transistors in series, where the middle node rises to Vx

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Reading the figure. In (a) the single transistor sees the whole supply across it. In (b) the middle node V_x settles where the currents of the two transistors are equal; the top transistor's gate is then below its source, which turns it off harder.

2. Answer

Stack effect: the reduction of subthreshold leakage when two or more OFF transistors are connected in series. The intermediate node rises to a small positive voltage, which makes V_{GS} of the upper transistor negative, lowers its V_{DS} (less DIBL) and raises its threshold by body effect; the leakage of the stack is several times less than that of one transistor.
Leakage of two series OFF nMOS (both gates at 0; DIBL included, body effect neglected):

Step 1 (Basic): The subthreshold current law

Below threshold the drain current falls by ten times for every S volts (the subthreshold slope) that V_{GS} drops, and DIBL raises it by lowering the threshold by ηV_{DS}. With I'_{off} the current at $V_{GS} = 0$, $V_{DS} = 0$ threshold:

$$I_{DS} = I'_{off}\, 10^{\frac{V_{GS}+\eta V_{DS}}{S}}$$

Step 2: The current of each transistor

Let V_x be the voltage of the node between them. Bottom transistor: $V_{GS} = 0$, $V_{DS} = V_x$:

$$I_{bottom} = I'_{off}\, 10^{\frac{\eta V_x}{S}}$$

Top transistor: its source is at V_x, so $V_{GS} = -V_x$ and $V_{DS} = V_{DD} - V_x$:

$$I_{top} = I'_{off}\, 10^{\frac{-V_x+\eta(V_{DD}-V_x)}{S}}$$

Step 3: Equal currents give the middle-node voltage

The same current flows through both, so the exponents are equal:

$$\eta V_x = -V_x + \eta V_{DD} - \eta V_x \;\Rightarrow\; V_x(1+2\eta) = \eta V_{DD} \;\Rightarrow\; V_x = \frac{\eta V_{DD}}{1+2\eta}$$

Step 4: The stack leakage and the ratio

Put V_x into I_{bottom}:

$$I_{stack} = I'_{off}\, 10^{\frac{\eta V_x}{S}} = I'_{off}\, 10^{\frac{\eta^2 V_{DD}}{(1+2\eta)S}}$$

One OFF transistor alone has $V_{GS} = 0$, $V_{DS} = V_{DD}$: $I_{single} = I'_{off}\, 10^{\eta V_{DD}/S}$. Divide; the exponents subtract, $\eta V_{DD} - \dfrac{\eta^2 V_{DD}}{1+2\eta} = \dfrac{\eta V_{DD}(1+2\eta-\eta)}{1+2\eta}$:

$$\boxed{\frac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}}$$

Steps to remember

Subthreshold law $I = I'_{off} 10^{(V_{GS}+\eta V_{DS})/S}$ → bottom: $V_{GS} = 0$, $V_{DS} = V_x$; top: $V_{GS} = -V_x$, $V_{DS} = V_{DD} - V_x$ → equal currents: $V_x = \eta V_{DD}/(1+2\eta)$ → divide by the single-transistor leakage.

Example

(illustrative values: $V_{DD} = 1$ V, $\eta = 0.1$, $S = 100$ mV/decade):

$$V_x = \frac{0.1 \times 1}{1.2} = 83.3 \text{ mV}, \qquad \frac{I_{single}}{I_{stack}} = 10^{\frac{0.1\times 1\times 1.1}{1.2\times 0.1}} = 10^{0.917} = 8.3$$

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So the stack leaks about **8 times less** than a single OFF transistor; body effect (neglected here) increases the factor further, typically to 10–20.

3. Keyword Explanation

1. DIBL coefficient η

↩ back

The drop in threshold voltage per volt of drain-source voltage, $V_t = V_{t0} - \eta V_{DS}$.

4. Notes and Remarks

- The derivation drops the factor $(1 - e^{-V_{DS}/v_T})$; for the bottom transistor, $V_{DS} = V_x = 83$ mV is about $3v_T$, so the factor is 0.96 and the error is small.
- Designers use the stack effect on purpose: "stack forcing" splits one OFF transistor into two series ones on leakage-critical but timing-relaxed paths, and in standby the inputs of NAND gates are set so that their nMOS stacks are fully OFF.

5. Final Concise Answer

(a) Min-delay (hold) constraint: the shortest path must not let new data reach the next element within its hold time: $t_{cd} \geq t_{hold} - t_{ccq}$ (plus t_{skew} with skew); independent of the clock period. Fix: pad short paths with buffers, use flops with smaller hold time, reduce skew.

(b) $t_{pd} = 590 + 60 + 100 + 80 + 100 + 70 = 1000$ ps (two wires; the early bypass mux is off the path); $T_c = 90 + 1000 + 62 = 1152$ ps. Hold: $t_{ccq} + t_{cd} = 75 + 45 = 120$ ps ≥ -10 ps, so no hold failure.

(c) $H = 5$. NAND–NAND: $G = 16/9$, $F = 8.89$, $P = 4$, $\hat{D} = 9.96\tau$ (sizes 20 and 44.7). AOI22 + inverter: $G = 2$, $F = 10$, $P = 5$, $\hat{D} = 11.3\tau$ (sizes 20 and 31.6). NAND–NAND is faster: the compound gate's larger g and p outweigh its fewer transistors.

(d) Stack effect: two series OFF transistors leak much less than one. $V_x = \eta V_{DD}/(1+2\eta)$; $I_{single}/I_{stack} = 10^{\eta V_{DD}(1+\eta)/[(1+2\eta)S]}$, about 8.3 for $V_{DD} = 1$ V, $\eta = 0.1$, $S = 100$ mV/decade.

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6. (a) Explain the principle of dynamic logic circuit. Marks: 6

   (b) Draw the stick diagram of the following Boolean function: $f = (A + B).(C + D)$ with their symbolic diagram and truth table. Marks: 6

   (c) Draw logic diagram of a SRAM cell and explain its different modes of operation. Marks: 6

   (d) Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation. Marks: 7

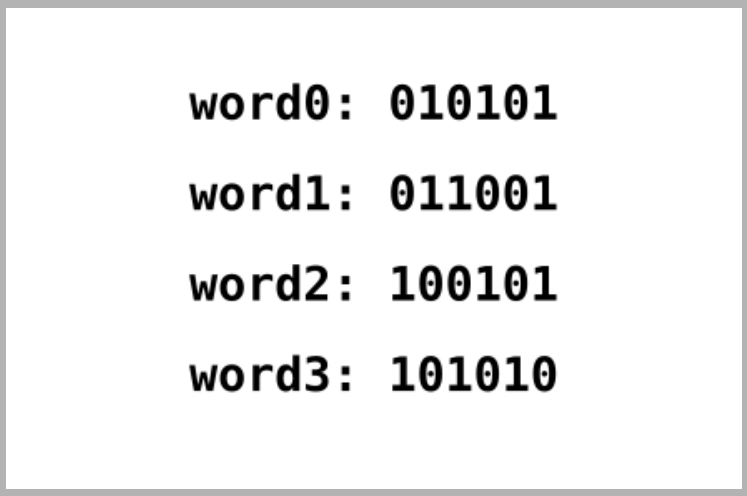


Figure 32: Figure 6.1 of the paper

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6.(a) Explain the principle of dynamic logic circuit. Marks: 6

1. Theory

1. Why not always static CMOS? (Core Idea)

A static CMOS gate has a pull-down network of nMOS and a matching pull-up network of pMOS. The pMOS network is slow (holes are slow), large, and adds input capacitance that every driving gate must charge. **Dynamic logic** keeps only the nMOS network and replaces the whole pMOS network with **one clocked pMOS**. It stores the output as charge on the output capacitance, which is why it is called dynamic: the value lasts only for a short time, like a memory that must be refreshed every cycle.

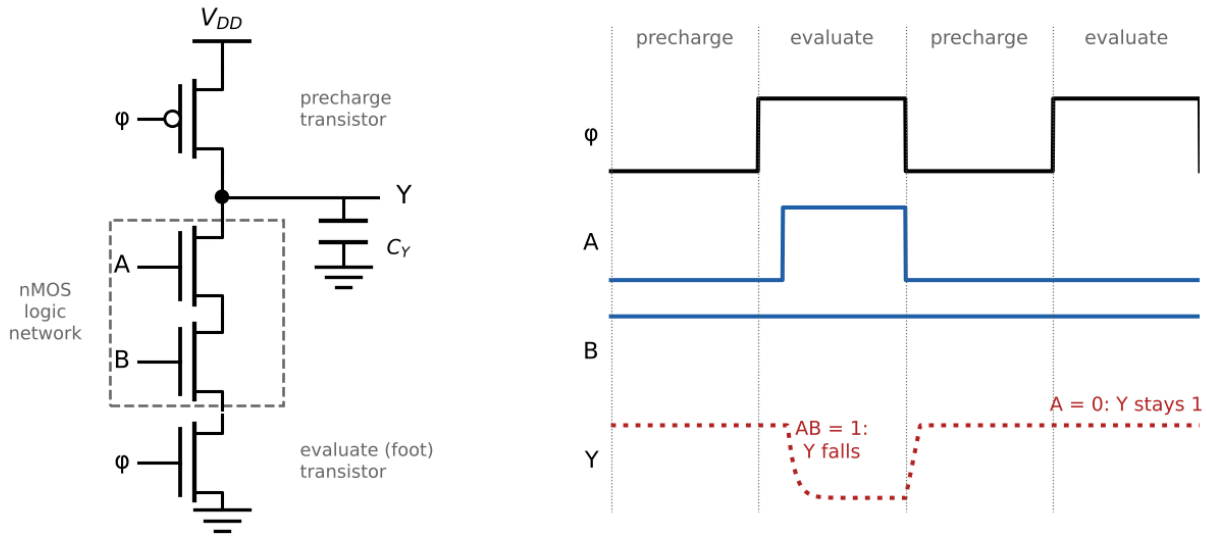


Figure 33: (a) A dynamic NAND2: clocked precharge pMOS, the nMOS logic network and the clocked evaluate (foot) nMOS; (b) its waveforms over two cycles

Reading the figure. While the clock φ is low (precharge), the pMOS charges Y to V_{DD} and the foot transistor keeps the path to ground open. When φ goes high (evaluate), the pMOS turns off and the foot turns on. In the first cycle A = B = 1, so the nMOS network conducts and Y falls to 0. In the second cycle A = 0, so there is no path to ground and Y keeps its precharged 1 (stored on C_Y).

2. Answer

Principle: a dynamic gate uses a clock φ to work in two phases on a single nMOS logic network:

1. **Precharge** ($\varphi = 0$): the pMOS precharge transistor is ON and charges the output node to V_{DD}; the nMOS evaluate (foot) transistor is OFF, so no current flows whatever the inputs are.
2. **Evaluate** ($\varphi = 1$): the precharge pMOS turns OFF and the foot turns ON. If the inputs make the nMOS network conduct, the output **discharges** to 0; otherwise it stays **high**, held only by the charge on its capacitance.

So the output is $Y = \overline{f(\text{inputs})}$, where f is the function of the nMOS network (for the figure, $Y = \overline{AB}$).

Advantages:

- Only $N + 2$ transistors for N inputs (static CMOS needs $2N$), so less area.
- Smaller input capacitance (inputs drive only nMOS): logical effort is lower and the gate is faster (about 1.5 to 2 times).
- No contention or static current (unlike pseudo-nMOS), and full output swing.

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Limitations:

- **Monotonicity:** in evaluate, the inputs may only rise ($0 \to 1$). An input that is 1 at the start and falls later has already discharged the output, which cannot be recharged until the next precharge. This is why dynamic gates cannot directly drive other dynamic gates; **domino logic** adds a static inverter after each gate to fix it.
- **Charge leakage:** the high output is floating; subthreshold leakage slowly discharges it, so the clock cannot be too slow. A weak **keeper** pMOS is added to hold it.
- **Charge sharing:** charge on the output can spread into uncharged internal nodes of the nMOS network, lowering the output voltage.
- **Noise sensitivity** (the output is not driven) and **higher clock power** (the clock drives every gate, and the output precharges every cycle, raising the activity factor).

3. Keyword Explanation

1. Foot transistor

↩ back

The clocked nMOS at the bottom of the network that blocks the path to ground during precharge; a gate without it is "unfooted".

2. Keeper

↩ back

A small pMOS, controlled by the output's inverter, that replaces charge lost by leakage and keeps a high output high.

4. Notes and Remarks

- The 2020–21 paper asked about domino logic; see 2(e) of the 2020–21 paper for extra reading.

6.(b) Draw the stick diagram of the following Boolean function: $f = (A + B).(C + D)$ with their symbolic diagram and truth table. Marks: 6

1. Theory

1. Building the gate (Core Idea)

A static CMOS gate gives an **inverted** function: its nMOS network decides when the output is 0. For $\overline{f} = \overline{(A + B)(C + D)}$, an **OR-AND-INVERT (OAI22)** gate:

- **nMOS (pull-down):** the output is 0 when $(A+B)(C+D) = 1$. OR $\to$ parallel, AND $\to$ series: (A ‖ B) in series with (C ‖ D).
- **pMOS (pull-up):** the **dual**: series becomes parallel and parallel becomes series: (A – B in series) in parallel with (C – D in series).

The function asked for, $f = (A + B)(C + D)$, is not inverted, so an **inverter** follows the OAI22.

2. Stick diagram and the Euler path

A **stick diagram** is a simplified layout: each layer is a coloured line (p-diffusion, n-diffusion, polysilicon, metal) without exact widths, and an X is a contact. A transistor appears wherever a polysilicon line crosses a diffusion line. The neatest layout uses one unbroken strip of p-diffusion and one of n-diffusion, with each input's polysilicon running vertically through *both* strips. That is possible only if the inputs can be ordered so that, in that order, the transistors form a continuous chain (an **Euler path**) in both the nMOS graph and the pMOS graph.
For this gate the order **A, B, D, C** works:

- p-strip: V_{DD} –A– p_1 –B– $\overline{f}$ –D– p_2 –C– V_{DD} (A–B in series and D–C in series, both between V_{DD} and $\overline{f}$).
- n-strip: n_1 –A– $\overline{f}$ –B– n_1 –D– GND –C– n_1 (A ‖ B between $\overline{f}$ and n_1; D ‖ C between n_1 and GND).

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2. Answer

Symbolic (transistor) diagram: OAI22 followed by an inverter; widths for unit drive (series pMOS 4, series nMOS 2; inverter n 1, p 2):

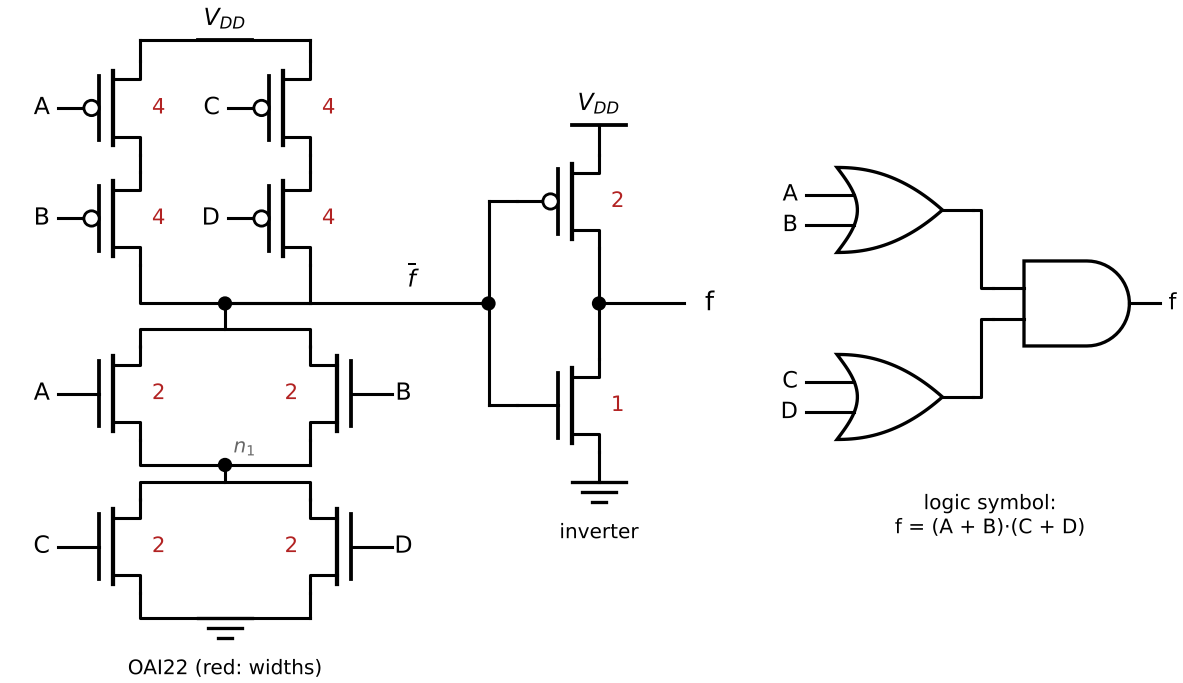


Figure 34: Symbolic diagram: OAI22 giving f-bar, then an inverter giving f, with the logic symbol of f

Stick diagram (gate order A, B, D, C on both strips):

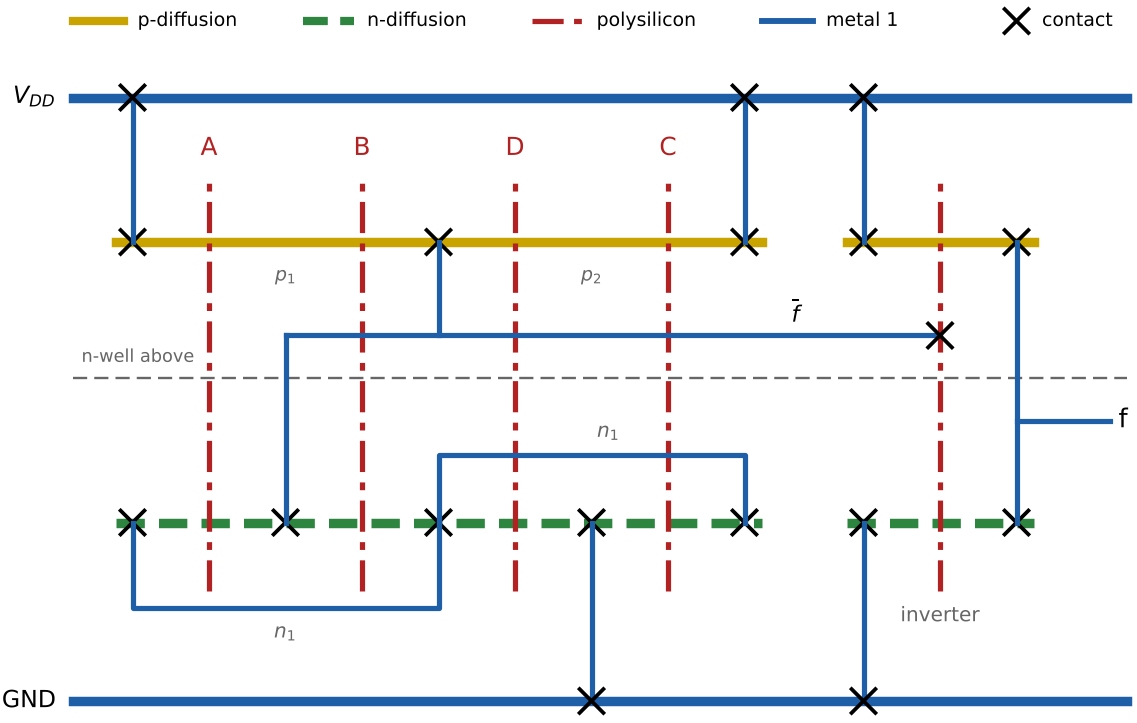


Figure 35: Stick diagram of the OAI22 and the output inverter

Reading the stick diagram. Four vertical polysilicon lines (A, B, D, C) cross both strips, making

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eight transistors; the fifth line is the inverter's gate. On the p-strip the two ends go to V_{DD} and the middle contact is $\overline{f}$. On the n-strip the contacts are n_1, $\overline{f}$, n_1, GND, n_1; the three n_1 contacts are joined by metal (below and above the strip), and GND goes down to the rail. The $\overline{f}$ metal joins the p and n contacts and runs to the inverter's polysilicon gate; the inverter's output is f.

Truth table:

A	B	C	D	A + B	C + D	f	$\overline{f}$
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	1	0	1
0	0	1	1	0	1	0	1
0	1	0	0	1	0	0	1
0	1	0	1	1	1	1	0
0	1	1	0	1	1	1	0
0	1	1	1	1	1	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	0	0	1
1	1	0	1	1	1	1	0
1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	0

Table 13: Truth table of f = (A + B)(C + D)

$f = 1$ in 9 of the 16 rows: whenever at least one of A, B *and* at least one of C, D is 1.

3. Keyword Explanation

1. Dual network

↩ back

The pMOS network obtained from the nMOS network by swapping series and parallel connections; it conducts exactly when the nMOS network does not.

2. Euler path

↩ back

A path through a graph that uses every edge (transistor) exactly once; a common Euler path in the nMOS and pMOS graphs gives a layout with unbroken diffusion strips.

4. Notes and Remarks

- A single OAI22 (without the inverter) gives $\overline{f}$; the inverter is needed because the question asks for f itself. The 2020–21 paper drew a similar complex gate; see 1(c) of the 2020–21 paper for extra reading.

6.(c) Draw logic diagram of a SRAM cell and explain its different modes of operation. Marks: 6

1. Theory

1. Storing one bit (Core Idea)

Two inverters connected in a loop, each driving the other's input, hold a value forever (as long as power is on): if Q = 1, the first inverter makes $\overline{Q} = 0$, and the second turns that back into Q = 1. This **latch** is the storage element of a **static RAM** (SRAM): static because, unlike DRAM, it needs no refresh.

To read and write the bit, two **access transistors** (nMOS) connect the two storage nodes to a pair of **bit lines** (bit and $\overline{bit}$) when the **word line** (WL) is high. One word line selects a whole row of cells;

each column has its own bit-line pair.

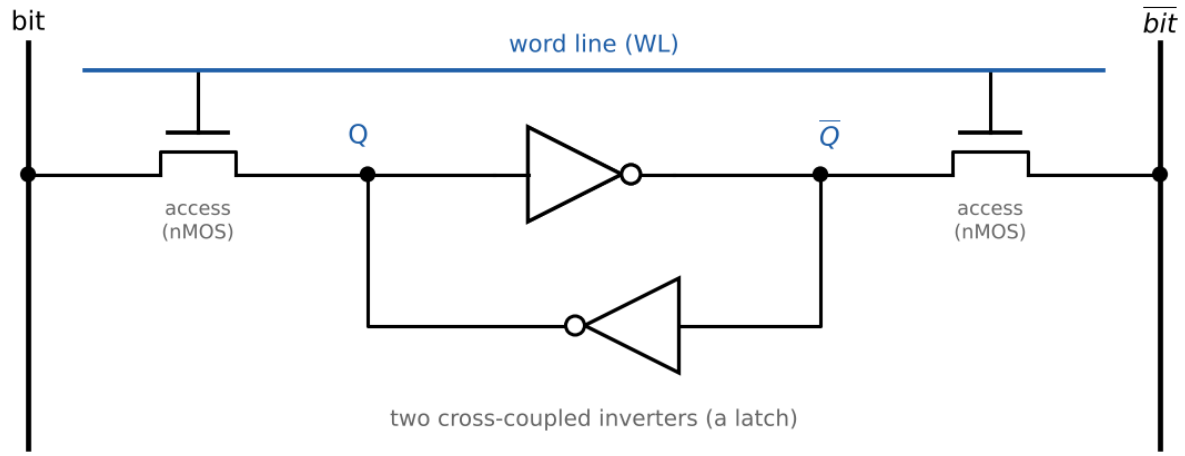


Figure 36: Logic diagram of an SRAM cell: two cross-coupled inverters and two access transistors

Reading the figure. The top inverter drives $\overline{Q}$ from Q; the bottom one drives Q from $\overline{Q}$. With WL low the access transistors are off and the loop is cut off from the bit lines. With WL high, Q connects to bit and $\overline{Q}$ to $\overline{bit}$.

2. The six transistors

Each inverter is a pMOS and an nMOS, so the cell has **6 transistors (6T)**: two pull-ups, two pull-downs and two access transistors.

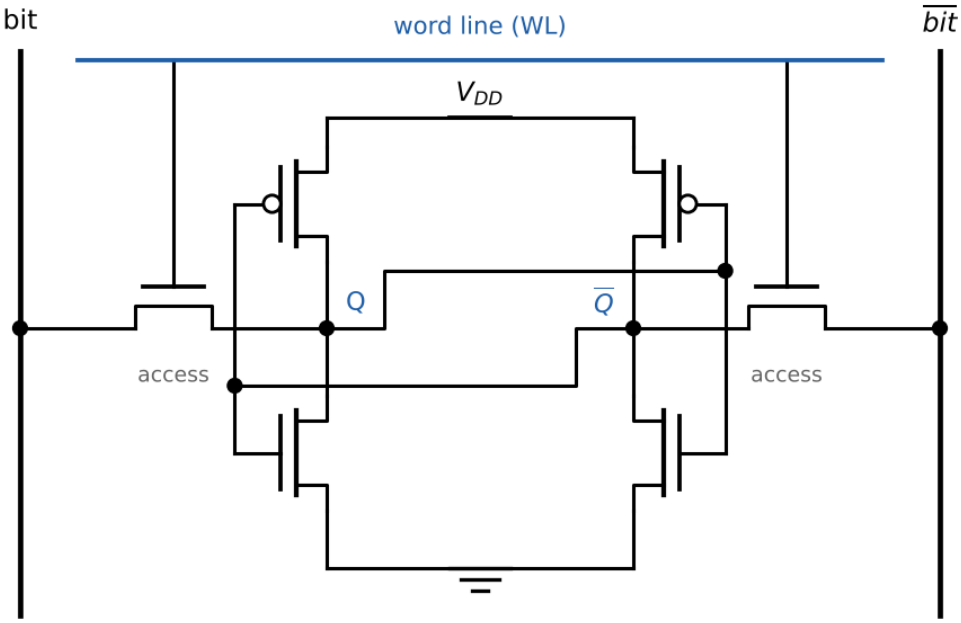


Figure 37: The 6T SRAM cell at transistor level (reused from the 2020-21 paper's solution)

2. Answer

SRAM cell (6T): two cross-coupled CMOS inverters store the bit as Q and $\overline{Q}$; two nMOS access transistors, controlled by the word line WL, connect Q to bit and $\overline{Q}$ to $\overline{bit}$ (logic diagram above).

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Modes of operation:

1. **Hold (standby):** WL = 0. The access transistors are OFF; the two inverters hold Q and $\overline{Q}$ by positive feedback for as long as power is applied. Only leakage current flows.
2. **Read:** both bit lines are first **precharged** to V_{DD}, then WL = 1. Say Q = 0: the bit line discharges through the access transistor and the ON pull-down nMOS of the inverter, while $\overline{bit}$ stays high. A **sense amplifier** detects the small difference (about 100–200 mV) and amplifies it to a full logic level. **Read stability:** the pull-down nMOS must be stronger than the access transistor (cell ratio about 1.5–2), so Q rises only a little and the cell does not flip.
3. **Write:** the write driver forces the bit lines to the new value (for example bit = 0, $\overline{bit}$ = 1 to write Q = 0), then WL = 1. The bit line at 0 pulls the node holding 1 down through the access transistor, below the other inverter's switching threshold, and the latch flips. **Writability:** the access transistor must be stronger than the pull-up pMOS (pull-up ratio below about 1), so the low bit line can overpower it.

Mode	WL	Bit lines	What happens
Hold	0	don't care	latch keeps Q
Read	1	precharged high, then floating	the side storing 0 pulls its bit line low; sense amp reads
Write	1	driven to the new data	bit lines overpower the cell; latch flips

Table 14: SRAM cell modes

3. Keyword Explanation

1. Sense amplifier

↩ back

A circuit that detects a small voltage difference between bit and $\overline{bit}$ and amplifies it to a full logic level, so the read does not wait for a full swing.

4. Notes and Remarks

- The 2020–21 paper asked for a 2×2 SRAM array; see 5(a) of the 2020–21 paper for extra reading.

6.(d) Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation. Marks: 7

1. Theory

1. What a ROM is (Core Idea)

A **read-only memory** stores fixed data, chosen when the chip is made. Each **word line** (a row) selects one stored word; each **bit line** (a column) gives one bit of it. A bit is stored by the **presence or absence of a transistor** at the crossing of a word line and a bit line, so a ROM is very dense.

2. The pseudo-nMOS NOR ROM

- Each bit line has a **pMOS pull-up with its gate grounded**: always ON, weak, so the bit line is high unless something pulls it down (this is a pseudo-nMOS gate).
- At a crossing that stores a transistor, an nMOS with its gate on the word line connects the bit line to ground.
- An address decoder raises exactly one word line. Every nMOS on that row turns on and pulls its bit line low; bit lines without a transistor on that row stay high.
- Each bit line is therefore a **NOR** of the word lines that have transistors on it. An **inverter** at the bottom of each bit line turns "pulled low" into a 1, so **a transistor stores a 1** at the output.

The **dot diagram** is the shorthand for such an array: a grid of word lines and bit lines with a dot where there is a transistor.

2. Answer

Data (Figure 6.1): 4 words × 6 bits; bit columns $D_5 \ldots D_0$ from left to right:

Word	D_5	D_4	D_3	D_2	D_1	D_0
word0	0	1	0	1	0	1
word1	0	1	1	0	0	1
word2	1	0	0	1	0	1
word3	1	0	1	0	1	0

Table 15: Data to store

Design:

1. **Address decoder:** 4 words need 2 address bits, A_1A_0; a 2:4 decoder drives word0 to word3 (exactly one high).
2. **Array:** 6 bit lines, each with a pseudo-nMOS pull-up (pMOS, gate at ground) to V_{DD}, and an output inverter giving $D_5 \ldots D_0$.
3. **Programming:** an nMOS (gate = word line, drain = bit line, source = GND) at every position where the stored bit is **1**; nothing where it is 0. That gives $3 + 3 + 3 + 3 = 12$ transistors.

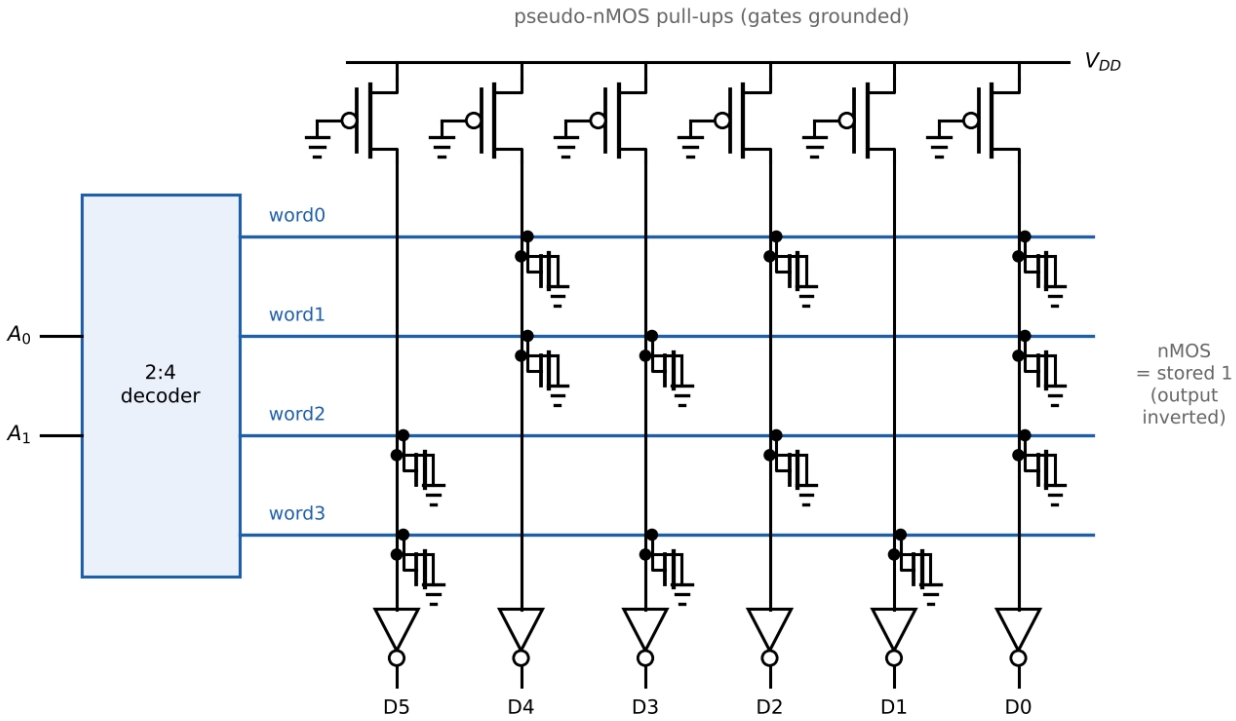


Figure 38: The 4-word × 6-bit pseudo-nMOS ROM: decoder, word lines, grounded-gate pMOS pull-ups, an nMOS for each stored 1, and output inverters

Operation (read word1 = 011001): the decoder raises word1. The nMOS on word1 sit on D_4, D_3 and D_0; they pull those bit lines low, and the inverters output 1. D_5, D_2 and D_1 have no transistor on word1, so their pull-ups keep them high and their inverters output 0. Output: 0 1 1 0 0 1 ✔.

Dot diagram (a dot = a transistor = a stored 1):

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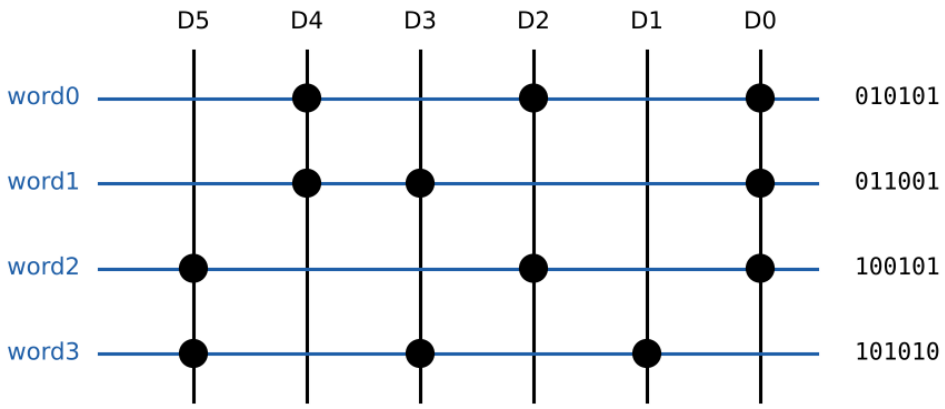


Figure 39: Dot diagram of the ROM

3. Keyword Explanation

1. Pseudo-nMOS

↩ back

A logic style with an nMOS pull-down network and a single always-on (grounded-gate) pMOS load; it is small and fast but draws static current while the output is low.

4. Notes and Remarks

- **Without output inverters**, a transistor stores a 0 (the bit line reads low); then the transistors go wherever the data has a 0, and the array would have $24 - 12 = 12$ transistors as well for this data.
- The pull-ups draw static current while a selected word line holds a bit line low, so pseudo-nMOS ROMs are often replaced by dynamic (precharged) bit lines in low-power designs.

5. Final Concise Answer

(a) Dynamic logic: one clocked pMOS precharges the output to V_{DD} while $\varphi = 0$ (foot nMOS off); when $\varphi = 1$ the nMOS network evaluates and discharges the output if it conducts, else the output stays high on its capacitance. Fewer transistors ($N+2$), lower input capacitance, faster; needs monotonically rising inputs (hence domino), suffers leakage (keeper), charge sharing and higher clock power.

(b) $\overline{f}$ from an OAI22: nMOS (A ∥ B) in series with (C ∥ D), pMOS (A–B) ∥ (C–D); then an inverter gives f. Euler order A, B, D, C on both strips; $f = 1$ in 9 of 16 rows (at least one of A, B and one of C, D).

(c) 6T SRAM: cross-coupled inverters with two access nMOS on WL to bit/$\overline{bit}$. Hold: WL = 0, latch keeps data. Read: bit lines precharged, WL = 1, the 0 side discharges its bit line, sense amplifier reads (pull-down stronger than access). Write: bit lines driven, WL = 1, they overpower the cell (access stronger than pull-up).

(d) 2:4 decoder, 4 word lines, 6 bit lines with grounded-gate pMOS pull-ups and output inverters; an nMOS at each stored 1: word0 on D_4, D_2, D_0; word1 on D_4, D_3, D_0; word2 on D_5, D_2, D_0; word3 on D_5, D_3, D_1 (12 transistors); the dot diagram puts a dot at each of these.

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FORMULA SHEET

EEE455

VLSI I

Formula Sheet

Session 2019–20 paper

Every formula behind the 4th Year 1st Semester Final Examination-2023

TOPICS

1 Fabrication — 9 formulas

2 MOS transistor — 9 formulas

3 CMOS inverter — 4 formulas

4 Delay and logical effort — 12 formulas

5 Scaling, power and interconnect — 8 formulas

6 Sequential timing — 4 formulas

7 Logic, memory — 7 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	F = fluorine radical, the main etchant	1(a)
↑ the name you search for	↑ the formula, exactly as derived	↑ every new symbol with its unit	↑ link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

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1 Fabrication 4

2 MOS transistor 4

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4 Delay and logical effort 6

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6 Sequential timing 8

7 Logic, memory 8

8 Key formulas 9

9 Symbol index 10

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Conventions: nMOS quantities carry the subscript n, pMOS p. Widths and capacitances in logical effort are in units of the unit inverter's nMOS (unit inverter: nMOS 1, pMOS 2). Delays are in units of $\tau = 3RC$; 1 FO4 $= 5\tau$. References point to the solutions of the 2019–20 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Fabrication:

Title	Equation	Notation	Ref.
Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	F = fluorine radical, the main etchant	1(a)
★ **Silicon etch**	$Si + 4F \rightarrow SiF_4 \uparrow$	SiF_4 = volatile product, pumped away	1(a)
Oxide etch (dry)	$SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$	slower than Si; needs ion help	1(a)
Oxide etch (wet)	$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$	buffered HF (with NH_4F)	1(a)
★ **Adding O_2**	$O + CF_3 \rightarrow COF_2 + F$	more free F: Si rate rises to a peak near 12 % O_2	1(a)
★ **Adding H_2**	$H + F \rightarrow HF$	less free F: Si rate falls, SiO_2:Si selectivity rises	1(a)
Selectivity	$S = \dfrac{r_{film}}{r_{under}}$	S = selectivity r_{film}, r_{under} = etch rates of the film and of the layer under it (nm/min)	1(a)
★ **RCA clean**	SC-1: NH_4OH:H_2O_2:H_2O = 1:1:5; HF 1:50; SC-2: HCl:H_2O_2:H_2O = 1:1:6	SC-1: organics, particles; HF: oxide; SC-2: metals (75–80 °C)	1(b)
Lithography resolution	$W_{min} = k_1 \dfrac{\lambda}{NA}$	W_{min} = smallest printable feature (m) k_1 = process factor λ = wavelength (m) NA = numerical aperture	1(c)

Table 1: Fabrication

2. MOS transistor:

Title	Equation	Notation	Ref.
Oxide capacitance	$C_{ox} = \dfrac{\varepsilon_{ox}}{t_{ox}}$	C_{ox} = capacitance per area (F/m^2) ε_{ox} = permittivity of the oxide (F/m) t_{ox} = oxide thickness (m)	2(a)
Gain factor	$\beta = \mu C_{ox} \dfrac{W}{L}$	β = gain factor (A/V^2) μ = mobility ($m^2/V \cdot s$) W, L = width, length (m)	2(a)
★ **Current, linear**	$I_{DS} = \beta \left(V_{GS} - V_t - \dfrac{V_{DS}}{2} \right) V_{DS}$	I_{DS} = drain current (A) V_{GS} = gate-to-source voltage (V) V_t = threshold voltage (V) V_{DS} = drain-to-source voltage (V) valid for $V_{DS} < V_{GS} - V_t$	2(a)
★ **Current, saturation**	$I_{DS} = \dfrac{\beta}{2}(V_{GS} - V_t)^2$	valid for $V_{DS} \geq V_{GS} - V_t$; cutoff: $I_{DS} = 0$	2(a)

Title	Equation	Notation	Ref.
★ **DIBL**	$V_t = V_{t0} - \eta V_{DS}$	V_{t0} = threshold at $V_{DS} = 0$ (V) η = DIBL coefficient (about 0.1)	1(d)
★ **Subthreshold current**	$I_{DS} = I_{ds0}\, e^{\frac{V_{GS}-V_t+\eta V_{DS}}{nv_T}} \left(1 - e^{-V_{DS}/v_T}\right)$	I_{ds0} = current at threshold (A) n = slope factor (1.3–1.5) v_T = thermal voltage, kT/q (25.9 mV at 300 K)	1(d)
Subthreshold slope	$S = nv_T \ln 10$	S = subthreshold slope (mV/decade) $S \geq 60$ mV/decade at 300 K	1(d)
Stack node voltage	$V_x = \dfrac{\eta V_{DD}}{1+2\eta}$	V_x = middle node of two OFF nMOS (V) V_{DD} = supply voltage (V)	5(d)
★ **Stack effect**	$\dfrac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}$	I_{single}, I_{stack} = leakage of one OFF transistor and of two in series (A) about 8 for $V_{DD} = 1$ V, $\eta = 0.1$, $S = 100$ mV/dec	5(d)

Table 2: MOS transistor

3. CMOS inverter:

Title	Equation	Notation	Ref.
Operating point	$I_{Dn} = \lvert I_{Dp}\rvert$	I_{Dn}, I_{Dp} = nMOS and pMOS drain currents (A) V_{in} = input voltage (V) regions A–E as V_{in} rises	3(a)
★ **Switching threshold**	$V_M = \dfrac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1+r},\ r = \sqrt{\beta_p/\beta_n}$	V_M = switching threshold (V) V_{tn}, V_{tp} = nMOS and pMOS thresholds (V) V_{DD} = supply voltage (V) $r = \sqrt{\beta_p/\beta_n}$ β_p, β_n = gain factors (A/V^2) $V_M = V_{DD}/2$ when $\beta_n = \beta_p$	3(a)
Unity-gain inputs	$V_{IL} = \dfrac{3V_{DD} + 2V_t}{8},\ V_{IH} = \dfrac{5V_{DD} - 2V_t}{8}$	V_{IL}, V_{IH} = input low and high limits (V) V_t = threshold of either transistor (V) symmetric inverter	3(a)
Noise margins	$NM_L = V_{IL} - V_{OL},\ NM_H = V_{OH} - V_{IH}$	NM_L, NM_H = low and high noise margins (V) V_{OL}, V_{OH} = output low and high levels, 0 and V_{DD} (V)	3(a)

Table 3: CMOS inverter

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4. Delay and logical effort:

Title	Equation	Notation	Ref.
★ **Elmore delay**	$t_{pd} = \sum_i R_i C_i$	t_{pd} = propagation delay (s) R_i = resistance shared with the path to node i (Ω) C_i = capacitance at node i (F)	2(b)
NAND3 rising, worst	$t_{pdr} = (9 + 5h)RC$	t_{pdr} = rising delay (s) h = number of NAND3 loads R = unit transistor resistance (Ω) C = unit gate capacitance (F)	2(b)
NAND3 falling, worst	$t_{pdf} = (12 + 5h)RC$	t_{pdf} = falling delay (s) bottom input last	2(b)
NAND3 contamination	$t_{cdr} = \left(3 + \frac{5h}{3}\right) RC,\ t_{cdf} = (9 + 5h)RC$	t_{cdr}, t_{cdf} = rising and falling contamination delays (s) $h = 4$: 29, 32, 9.67, 29 RC	2(b)
★ **Gate delay**	$d = gh + p$	d = delay in units of τ g = logical effort h = electrical effort, C_{out}/C_{in} p = parasitic delay	3(b)
Logical efforts	inv $g = 1$; NAND-n $\frac{n+2}{3}$; NOR-n $\frac{2n+1}{3}$; AOI22 2	n = number of inputs p: inv 1, NAND/NOR-n n, AOI22 4	3(b)
Branching effort	$b = \dfrac{C_{on} + C_{off}}{C_{on}}$	b = branching effort C_{on} = load on the path (F) C_{off} = load off the path (F)	3(b)
★ **Path effort**	$F = GBH,\ G = \prod g_i,\ B = \prod b_i,\ H = \dfrac{C_{out}}{C_{in}}$	F = path effort g_i, b_i = logical and branching efforts of stage i G = path logical effort, $\prod g_i$ B = path branching effort, $\prod b_i$ H = path electrical effort C_{out}, C_{in} = load of the path and input capacitance of its first stage (F) $P = \sum p_i$	3(b)
★ **Minimum delay**	$\hat{f} = F^{1/N},\ \hat{D} = NF^{1/N} + P$	$\hat{f}$ = best stage effort $\hat{D}$ = least path delay N = number of stages P = path parasitic delay, $\sum p_i$	3(b)
Gate size	$C_{in,i} = \dfrac{g_i C_{out,i}}{\hat{f}}$	$C_{in,i}$ = input capacitance of stage i (F) g_i = logical effort of stage i $C_{out,i}$ = load of stage i (F) work back from the load	3(b)
★ **Best stages**	$\hat{N} = \log_4 F$	$\hat{N}$ = best number of stages each stage effort near 4	3(c)
Input order (NAND2)	outer $p = 2$, inner $p = 7/3$	latest signal to the outer input	3(d)

Table 4: Delay and logical effort

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5. Scaling, power and interconnect:

Title	Equation	Notation	Ref.
★ **Constant-field scaling**	$L, W, t_{ox}, V_{DD}, V_t \times \frac{1}{S};\ N_A \times S$	L, W = channel length and width t_{ox} = oxide thickness V_{DD} = supply voltage V_t = threshold voltage S = scaling factor (> 1) N_A = substrate doping delay $1/S$, power per gate $1/S^2$, density S^2, power density 1	4(a)
Constant-voltage scaling	$L, W, t_{ox} \times \frac{1}{S};\ V$ fixed; $N_A \times S^2$	V = supply voltage delay $1/S^2$, power per gate S, power density S^3	4(a)
★ **Dynamic power**	$P_{dyn} = \alpha C V_{DD}^2 f$	P_{dyn} = dynamic power (W) α = activity factor C = switched capacitance (F) f = clock (Hz)	4(c)
Capacitance from area	$C = (C/A) \times A$	A = chip area (mm^2) 450 pF/mm^2 × 70 mm^2 = 31.5 nF	4(c)
★ **Crosstalk, floating victim**	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_a$	ΔV_v = noise on the victim (V) ΔV_a = swing of the aggressor (V) C_{adj} = coupling capacitance (F) C_{gnd} = capacitance to ground (F)	4(d)
Crosstalk, driven victim	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \cdot \frac{1}{1+k} \Delta V_a,\ k = \frac{\tau_a}{\tau_v}$	$k = \tau_a/\tau_v$ τ_a, τ_v = RC time constants of the aggressor and victim drivers (s)	4(d)
Miller coupling	$C_{eff} = C_{gnd} + k_M C_{adj}$	C_{eff} = effective capacitance (F) k_M = 0 (same), 1 (quiet), 2 (opposite)	4(d)
★ **Sheet resistance**	$R_\square = \frac{\rho}{t},\ R = R_\square \frac{l}{w}$	$R_\square$ = sheet resistance (Ω/square) ρ = resistivity ($\Omega \cdot$ m) t = thickness (m) R = wire resistance (Ω) l = length (m) w = width (m)	4(e)

Table 5: Scaling, power and interconnect

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6. Sequential timing:

Title	Equation	Notation	Ref.
★ **Max-delay (setup)**	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$	T_c = clock period (s) t_{pcq} = clock-to-Q propagation (s) t_{pd} = longest logic delay (s) t_{setup} = setup time (s)	5(b)
★ **Min-delay (hold)**	$t_{cd} \geq t_{hold} - t_{ccq}$	t_{cd} = shortest logic delay (s) t_{hold} = hold time (s) t_{ccq} = clock-to-Q contamination (s)	5(a)
Hold with skew	$t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$	t_{skew} = clock skew (s) independent of T_c	5(a)
Itanium 2 bypass	$T_c = 90 + 1000 + 62 = 1152$ ps	hold: $75 + 45 = 120 \geq -10$ ps	5(b)

Table 6: Sequential timing

7. Logic, memory:

Title	Equation	Notation	Ref.
NAND–NAND form	$AB + CD = \overline{\overline{AB} \cdot \overline{CD}}$	A, B, C, D = inputs $\hat{D} = 9.96\tau$ for $H = 5$	5(c)
Compound gate	AOI22 $= \overline{AB + CD}$	+ inverter: $\hat{D} = 11.3\tau$	5(c)
★ **OAI22**	$\overline{f} = \overline{(A + B)(C + D)}$	$f = (A + B)(C + D)$ nMOS $(A \parallel B)$–$(C \parallel D)$; pMOS dual; inverter gives f	6(b)
Dynamic gate output	$Y = \overline{f}$ in evaluate, $Y = 1$ in precharge	Y = output inputs must rise monotonically	6(a)
CVSL outputs	$Y = f$, $\overline{Y} = \overline{f}$	nMOS networks f and $\overline{f}$, cross-coupled pMOS	2(d)
SRAM sizing	pull-down > access > pull-up	read stability; writability	6(c)
★ **NOR ROM**	bit line $= \overline{\sum \text{word lines with a transistor}}$	with output inverter: transistor = 1	6(d)

Table 7: Logic, memory

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Silicon etch	$\mathrm{Si} + 4\mathrm{F} \rightarrow \mathrm{SiF}_4 \uparrow$
Adding O_2	$\mathrm{O} + \mathrm{CF}_3 \rightarrow \mathrm{COF}_2 + \mathrm{F}$
Adding H_2	$\mathrm{H} + \mathrm{F} \rightarrow \mathrm{HF}$
RCA clean	SC-1: $\mathrm{NH_4OH{:}H_2O_2{:}H_2O}$ = 1:1:5; HF 1:50; SC-2: $\mathrm{HCl{:}H_2O_2{:}H_2O}$ = 1:1:6
Current, linear	$I_{DS} = \beta \left(V_{GS} - V_t - \frac{V_{DS}}{2} \right) V_{DS}$
Current, saturation	$I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2$
DIBL	$V_t = V_{t0} - \eta V_{DS}$
Subthreshold current	$I_{DS} = I_{ds0}\, e^{\frac{V_{GS} - V_t + \eta V_{DS}}{n v_T}} \left(1 - e^{-V_{DS}/v_T}\right)$
Stack effect	$\frac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}$
Switching threshold	$V_M = \frac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1+r},\ r = \sqrt{\beta_p/\beta_n}$
Elmore delay	$t_{pd} = \sum_i R_i C_i$
Gate delay	$d = gh + p$
Path effort	$F = GBH,\ G = \prod g_i,\ B = \prod b_i,\ H = \frac{C_{out}}{C_{in}}$
Minimum delay	$\hat{f} = F^{1/N},\ \hat{D} = NF^{1/N} + P$
Best stages	$\hat{N} = \log_4 F$
Constant-field scaling	$L, W, t_{ox}, V_{DD}, V_t \times \frac{1}{S};\ N_A \times S$
Dynamic power	$P_{dyn} = \alpha C V_{DD}^2 f$
Crosstalk, floating victim	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_a$
Sheet resistance	$R_\square = \frac{\rho}{t},\ R = R_\square \frac{l}{w}$
Max-delay (setup)	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$
Min-delay (hold)	$t_{cd} \geq t_{hold} - t_{ccq}$
OAI22	$\overline{f} = \overline{(A+B)(C+D)}$
NOR ROM	bit line = $\overline{\sum \text{word lines with a transistor}}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
F	fluorine radical, the main etchant	Plasma dissociation
SiF_4	volatile product, pumped away	Silicon etch
S	selectivity	Selectivity
r_{film}, r_{under}	etch rates of the film and of the layer under it (nm/min)	Selectivity
W_{min}	smallest printable feature (m)	Lithography resolution
k_1	process factor	Lithography resolution
λ	wavelength (m)	Lithography resolution
NA	numerical aperture	Lithography resolution
C_{ox}	capacitance per area (F/m^2)	Oxide capacitance
ε_{ox}	permittivity of the oxide (F/m)	Oxide capacitance
t_{ox}	oxide thickness (m)	Oxide capacitance
β	gain factor (A/V^2)	Gain factor
μ	mobility ($m^2/V \cdot s$)	Gain factor
W, L	width, length (m)	Gain factor
I_{DS}	drain current (A)	Current, linear
V_{GS}	gate-to-source voltage (V)	Current, linear
V_t	threshold voltage (V)	Current, linear
V_{DS}	drain-to-source voltage (V)	Current, linear
V_{t0}	threshold at $V_{DS} = 0$ (V)	DIBL
η	DIBL coefficient (about 0.1)	DIBL
I_{ds0}	current at threshold (A)	Subthreshold current
n	slope factor (1.3–1.5)	Subthreshold current
v_T	thermal voltage, kT/q (25.9 mV at 300 K)	Subthreshold current
S	subthreshold slope (mV/decade)	Subthreshold slope
V_x	middle node of two OFF nMOS (V)	Stack node voltage
V_{DD}	supply voltage (V)	Stack node voltage
I_{single}, I_{stack}	leakage of one OFF transistor and of two in series (A)	Stack effect
I_{Dn}, I_{Dp}	nMOS and pMOS drain currents (A)	Operating point
V_{in}	input voltage (V)	Operating point
V_M	switching threshold (V)	Switching threshold
V_{tn}, V_{tp}	nMOS and pMOS thresholds (V)	Switching threshold
r	$\sqrt{\beta_p/\beta_n}$	Switching threshold
β_p, β_n	gain factors (A/V^2)	Switching threshold
V_{IL}, V_{IH}	input low and high limits (V)	Unity-gain inputs
V_t	threshold of either transistor (V)	Unity-gain inputs
NM_L, NM_H	low and high noise margins (V)	Noise margins

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Symbol	Meaning	First used in
V_{OL}, V_{OH}	output low and high levels, 0 and V_{DD} (V)	Noise margins
t_{pd}	propagation delay (s)	Elmore delay
R_i	resistance shared with the path to node i (Ω)	Elmore delay
C_i	capacitance at node i (F)	Elmore delay
t_{pdr}	rising delay (s)	NAND3 rising, worst
h	number of NAND3 loads	NAND3 rising, worst
R	unit transistor resistance (Ω)	NAND3 rising, worst
C	unit gate capacitance (F)	NAND3 rising, worst
t_{pdf}	falling delay (s)	NAND3 falling, worst
t_{cdr}, t_{cdf}	rising and falling contamination delays (s)	NAND3 contamination
d	delay in units of τ	Gate delay
g	logical effort	Gate delay
h	electrical effort, C_{out}/C_{in}	Gate delay
p	parasitic delay	Gate delay
n	number of inputs	Logical efforts
b	branching effort	Branching effort
C_{on}	load on the path (F)	Branching effort
C_{off}	load off the path (F)	Branching effort
F	path effort	Path effort
g_i, b_i	logical and branching efforts of stage i	Path effort
G	path logical effort, $\prod g_i$	Path effort
B	path branching effort, $\prod b_i$	Path effort
H	path electrical effort	Path effort
C_{out}, C_{in}	load of the path and input capacitance of its first stage (F)	Path effort
$\hat{f}$	best stage effort	Minimum delay
$\hat{D}$	least path delay	Minimum delay
N	number of stages	Minimum delay
P	path parasitic delay, $\sum p_i$	Minimum delay
$C_{in,i}$	input capacitance of stage i (F)	Gate size
g_i	logical effort of stage i	Gate size
$C_{out,i}$	load of stage i (F)	Gate size
$\hat{N}$	best number of stages	Best stages
L, W	channel length and width	Constant-field scaling
S	scaling factor (> 1)	Constant-field scaling
N_A	substrate doping	Constant-field scaling
V	supply voltage	Constant-voltage scaling
P_{dyn}	dynamic power (W)	Dynamic power
α	activity factor	Dynamic power
C	switched capacitance (F)	Dynamic power

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Symbol	Meaning	First used in
f	clock (Hz)	Dynamic power
A	chip area (mm^2)	Capacitance from area
450 pF/mm^2 × 70 mm^2	31.5 nF	Capacitance from area
ΔV_v	noise on the victim (V)	Crosstalk, floating victim
ΔV_a	swing of the aggressor (V)	Crosstalk, floating victim
C_{adj}	coupling capacitance (F)	Crosstalk, floating victim
C_{gnd}	capacitance to ground (F)	Crosstalk, floating victim
k	τ_a/τ_v	Crosstalk, driven victim
τ_a, τ_v	RC time constants of the aggressor and victim drivers (s)	Crosstalk, driven victim
C_{eff}	effective capacitance (F)	Miller coupling
k_M	0 (same), 1 (quiet), 2 (opposite)	Miller coupling
$R_\square$	sheet resistance (Ω/square)	Sheet resistance
ρ	resistivity ($\Omega \cdot$ m)	Sheet resistance
t	thickness (m)	Sheet resistance
R	wire resistance (Ω)	Sheet resistance
l	length (m)	Sheet resistance
w	width (m)	Sheet resistance
T_c	clock period (s)	Max-delay (setup)
t_{pcq}	clock-to-Q propagation (s)	Max-delay (setup)
t_{pd}	longest logic delay (s)	Max-delay (setup)
t_{setup}	setup time (s)	Max-delay (setup)
t_{cd}	shortest logic delay (s)	Min-delay (hold)
t_{hold}	hold time (s)	Min-delay (hold)
t_{ccq}	clock-to-Q contamination (s)	Min-delay (hold)
t_{skew}	clock skew (s)	Hold with skew
A, B, C, D	inputs	NAND–NAND form
Y	output	Dynamic gate output
with output inverter: transistor	1	NOR ROM

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